Geometric Class Field Theory

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Abstract

We calculate the ramification of the symmetric product $\mathcal{F}^{[\deg \mathfrak{m}]}$ of a local system $\mathcal{F}$ on a curve $C \setminus \mathfrak{m}$. Assuming the ramification of $\mathcal{F}$ is bounded by $\mathfrak{m}$, we find that the symmetric product $\mathcal{F}^{[\deg \mathfrak{m}]}$ is at most tamely ramified at the generic point of the exceptional divisor $E_{\mathfrak{m}}$ of the blowup of $C^{(\deg \mathfrak{m})}$ at $\mathfrak{m}$. As a primary application, we utilize this result to prove Geometric Class Field Theory. Our approach builds upon the geometric framework for the unramified case originally established by Deligne for the rank-one Langlands correspondence, following the subsequent extensions to the ramified case developed by Takeuchi and Guignard.

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Introduction

In this thesis, we give an elementary proof of a certain imporant geometric theorem occuring in Deligne's approach to geometric class field theory. We (usually) work over a perfect field k, C is a projective smooth geometrically connected curve over k, with genus g. One of the main geometric ingredients in the approach, is showing why a local system $\mathcal{F}$ with ramification bounded by a modulus $\mathfrak{m} = \sum n_i P_i$ on $U = C \setminus \mathfrak{m}$ descends via the Abel-Jacobi $\Phi : U \to \mathrm{Pic}_{C,\mathfrak{m}}$ to $\mathrm{Pic}_{C,\mathfrak{m}}$. The approach, innovated by Deligne, relies on analyzing the symmetric powers $\mathcal{F}^{[d]}$ of $\mathcal{F}$ on the symmetric powers $U^{(d)}$ of U, and showing that for sufficiently large d, $\mathcal{F}^{[d]}$ descends to $\mathrm{Pic}^d_{C,\mathfrak{m}}$ via the degree d Abel-Jacobi map $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$. The geometric-fibers of Φ_d (for $d \geq \deg \mathfrak{m} + 2g - 1$) over any point are isomorphic to

$$\begin{cases} \mathbb{A}^{d-\deg \mathfrak{m}-g+1}_{k^{sep}} & \text{if } \mathfrak{m} > 0 \\ \mathbb{P}^{d-g}_{k^{sep}} & \text{if } \mathfrak{m} = 0 \end{cases}$$

Where g is the genus of the curve C. The unramified case ($\mathfrak{m} = 0$) is relatively simple, as the Abel-Jacobi map is proper, surjective with geometrically connected fibers, which follows from the fact that it is a fibration in projective spaces. Thus, by using the homotopy exact sequence for the étale fundamental group, one gets an isomorphism between the étale fundamental group of $U^{(d)}(= C^{(d)})$ and that of $\mathrm{Pic}^d_{C,\mathfrak{m}}(= \mathrm{Pic}^d_C)$.

The ramified case ($\mathfrak{m} > 0$) is more subtle, as the Abel-Jacobi map is not proper anymore, and one needs to analyze the ramification of $\mathcal{F}^{[d]}$ "along the boundary" of $U^{(d)}$ in $C^{(d)}$.

Previous work has generalized Deligne's approach to the ramified case, most notably by Guignard [Gui19] and Takeuchi [Tak19]. Their approaches differ. To descend, Guignard proves that the restriction of $\mathcal{F}^{[d]}$ to any line in the fiber of the degree d Abel-Jacobi map is a constant étale sheaf. He achieves this by demonstrating that the restriction is at most tamely ramified and invoking the triviality of the tame fundamental group of $\mathbb{A}^1_k$. His analysis relies on local geometric class field theory. It is also worth noting that Guignard's method generalizes to relative curves over arbitrary base schemes. Takeuchi, on the other hand, constructs a compactification of $U^{(d)}$ by blowing up $C^{(d)}$ along certain well-chosen centers. This compactification, denoted by $\tilde{C}^{(d)}_{\mathfrak{m}}$, has $U^{(d)}$ as an open subscheme with a codimension 1 closed subscheme H as complement. He then shows that the Abel-Jacobi map extends to a proper morphism from $\tilde{C}^{(d)}_{\mathfrak{m}}$ to $\mathrm{Pic}^d_{C,\mathfrak{m}}$, which is a fibration in projective spaces. Thus, by the homotopy exact sequence for the étale fundamental group, one gets an isomorphism between the étale fundamental group of $\tilde{C}^{(d)}_{\mathfrak{m}}$ and that of $\mathrm{Pic}^d_{C,\mathfrak{m}}$. To conclude the descent, Takeuchi analyzes the ramification of $\mathcal{F}^{[d]}$ along the boundary H of $\tilde{C}^{(d)}_{\mathfrak{m}}$, showing that it is tamely ramified there, which suffices. His methods relies on the theory of refined Swan conductors.

For an account of these approaches, see [Gui19] and [Tak19]. For a full approach following Deligne's method in the unramified case, and the tamely ramified case see [Ten15], and [Tót11].

In this thesis, we calculate the ramification of $\mathcal{F}^{[d]}$ directly to show that it is at most tame at the generic point of H, avoiding the use of Swan conductors. To prove Geometric Class Field Theory, we utilize Takeuchi's construction of the compactification $\tilde{C}^{(d)}_{\mathfrak{m}}$ of $U^{(d)}$ via the blowing up of $C^{(d)}$.

In the rest of the introduction, we state the main theorem of geometric class field theory Theorem 1, and its reductions to Theorem 2 and Theorem 3, which we prove in this thesis.

Let k be a perfect field, and let C be a projective smooth geometrically connected curve over k, with genus g. Geometric class field theory gives a geometric description of abelian coverings of C by relating it to isogenies of the generalized Picard schemes.

Fix a modulus $\mathfrak{m}$, i.e. an effective Cartier divisor of C and let U be its complement in C. The pairs $(\mathcal{L}, \alpha)$, where $\mathcal{L}$ is an invertible $\mathcal{O}_C$-module and α is a rigidification of $\mathcal{L}$ along $\mathfrak{m}$, are parametrized by a k-group scheme $\mathrm{Pic}_{C,\mathfrak{m}}$, called the rigidified Picard scheme. The Abel-Jacobi morphism

$$\Phi : U \to \mathrm{Pic}_{C,\mathfrak{m}}$$

is the morphism which sends a section x of U to the pair $(\mathcal{O}(x), 1)$. The fundamental result of GCFT can be formulated as:

Theorem 1 (Geometric Class Field Theory)**.** *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on U, with ramification bounded by $\mathfrak{m}$. Then, there exists a unique (up to isomorphism) multiplicative*[1] *étale sheaf of Λ-modules $\mathcal{G}$ on $\mathrm{Pic}_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}$ by Φ is isomorphic to $\mathcal{F}$.*

Let d be a positive integer. We denote by $U^{(d)}$ the d-th symmetric power of U over k. For an étale sheaf $\mathcal{F}$ on $U_{\text{ét}}$, we denote by $\mathcal{F}^{[d]}$ the d-th symmetric power of $\mathcal{F}$ on $U^{(d)}$ The degree d Abel-Jacobi morphism is defined as the map

$$\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$$

which sends a section $x_1 + \cdots + x_d$ of $U^{(d)}$ to the pair $(\mathcal{O}(x_1 + \cdots + x_d), 1)$.

A method of descent shows that to prove Theorem 1, it suffices to prove the following reduced version[2]:

Theorem 2. *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{\text{ét}}$, with ramification bounded by $\mathfrak{m}$. Then, for sufficiently large integer d, there exists a unique (up to isomorphism) étale sheaf of Λ-modules $\mathcal{G}_d$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, locally free of rank 1, such that the pullback of $\mathcal{G}_d$ by Φ_d is isomorphic to $\mathcal{F}^{[d]}$.*

To prove Theorem 2 we follow the work of [Tak19] (and similiary done in [Tót11]), analyzing the ramification of $\mathcal{F}^{[d]}$ after blowing up $C^{(d)}$. We analyze this ramification using elementary methods. We prove the following Theorem:

Theorem 3. *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on $U_{\text{ét}}$, with ramification bounded by $\mathfrak{m} = \sum n_i P_i$. Let $\mathfrak{n} = n_i P_i \subset \mathfrak{m}$ be a sub modulus of $\mathfrak{m}$. Then $\mathcal{F}^{[\deg \mathfrak{n}]}$ is tamely ramified at the generic point $\eta_{\mathfrak{n}}$ of the exceptional divisor $E_{\mathfrak{n}}$ of the blowup $X_{\mathfrak{n}}$ of $C^{(\deg \mathfrak{n})}$ at the closed point $\mathfrak{n}$.*

The thesis is organized as follows:

[1]The notion of a multiplicative locally free Λ-module of rank 1 is due to [Gui19] and corresponds to isogenies $G \to \mathrm{Pic}_{C,\mathfrak{m}}$ with constant kernel $\Lambda^\times$. This concept corresponds to multiplicative characters of $H^1(\mathrm{Pic}_{C,\mathfrak{m}}, \mathbb{Q}/\mathbb{Z})$ in the formulation of [Tak19], and generalizes Hecke eigensheaves in the context of [Ten15].

[2]See the last page of [Gui19], Section 8.3 of [Ten15], or the proof of Theorem 1.2 in [Tak19] for details on this reduction.

Chapter 1 - Ramification After Blowup Is Tame is devoted to the proof of Theorem 3, along with several corollaries that will be instrumental in the proof of Theorem 2. It opens with a preliminaries section covering the foundational material upon which this work is based, generally without providing proofs.

Chapter 2 - Geometric Class Field Theory presents the proof of Theorem 2.

Appendix - A Wrong Path documents an earlier attempted approach to Theorem 3, and the difficulties that led us to abandon it.

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Chapter 1

Ramification After Blowup Is Tame

This chapter is devoted to the proof of Theorem 3. We begin with the necessary preliminaries, and then turn to the proof itself.

1.1 Preliminaries

This section establishes the foundational definitions and theorems necessary for the remainder of this work. We focus specifically on the theory of G-torsors, which are central to our study due to their correspondence with locally free sheaves of rank 1. Subsequently, we review the relevant background on ramification theory from the existing literature. Finally, we conclude with several algebraic geometric remarks and notational conventions that will be employed implicitly throughout this thesis.

1.1.1 Scheme Quotients in general

basic definitions, categorail qoutient, when it exists, etc, properties, universal properties, etc... prop 14 beolw

1.1.2 Torsors

In what follows, we largely adhere to the treatment of torsors and group objects found in published notes of Alex Youcis [You20]. Let $\mathscr{C} = (\mathcal{C}, J)$ be a site and let $\mathcal{E} = Sh(\mathscr{C})$ be the associated topos. Let $\mathcal{G}$ be a group object in $\mathcal{E}$. We denote by $\mathcal{G}\mathcal{E}$ the category of objects in $\mathcal{E}$ endowed with a left $\mathcal{G}$-action.

Definition 4. A **$\mathcal{G}$-torsor** in $\mathcal{E}$ is an object $\mathcal{P}$ of $\mathcal{G}\mathcal{E}$ satisfying the following conditions:

1. The structural morphism $\mathcal{P} \to 1$ is an epimorphism in $\mathcal{E}$ (i.e., $\mathcal{P}$ is locally non-empty).
2. The map $\mathcal{G} \times \mathcal{P} \to \mathcal{P} \times \mathcal{P}$ defined by $(g, p) \mapsto (g \cdot p, p)$ is an isomorphism in $\mathcal{E}$ (i.e., $\mathcal{G}$ acts simply transitively on $\mathcal{P}$).

Since $\mathcal{E}$ is the topos of sheaves on a site $\mathscr{C}$, the definition can be reformulated in terms of covers. A $\mathcal{G}$-sheaf $\mathcal{P}$ is a $\mathcal{G}$-torsor if:

1. For every object $X \in \mathcal{C}$, there exists a covering $\{U_i \to X\} \in J$ such that $\mathcal{P}(U_i) \neq \emptyset$ for all i.
2. For any $X \in \mathcal{C}$ where $\mathcal{P}(X)$ is non-empty, the action of $\mathcal{G}(X)$ on $\mathcal{P}(X)$ is simply transitively.

A fundamental property of torsors is their local triviality: a $\mathcal{G}$-sheaf $\mathcal{P}$ is a $\mathcal{G}$-torsor if and only if it is locally isomorphic to the trivial torsor. Specifically, for every $X \in \mathcal{C}$, there must exist a cover $\{U_i \to X\}$ such that the restriction $\mathcal{P}|_{U_i}$ is isomorphic, as a $\mathcal{G}|_{U_i}$-sheaf, to $\mathcal{G}|_{U_i}$ acting on itself by left multiplication.

A **morphism of $\mathcal{G}$-torsors** $f : \mathcal{P}_1 \to \mathcal{P}_2$ is a morphism of sheaves that is equivariant with respect to the $\mathcal{G}$-action.

Proposition 5. *Every morphism of $\mathcal{G}$-torsors is an isomorphism. Consequently, the category of $\mathcal{G}$-torsors in $\mathcal{E}$ forms a groupoid.*

Proof. Locally on a cover trivializing both torsors, such a morphism is a $\mathcal{G}$-equivariant endomorphism of the trivial torsor $\mathcal{G}$, i.e. right translation by a section of $\mathcal{G}$, hence an isomorphism; and being an isomorphism may be checked on a cover. □

Definition 6. We denote the groupoid of $\mathcal{G}$-torsors in $\mathcal{E}$ by $\mathbf{Tors}(\mathcal{E}, \mathcal{G})$. The set of isomorphism classes of $\mathcal{G}$-torsors is denoted by $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$.

Torsors exhibit functoriality with respect to the group object:

Definition 7. Let $\varphi : \mathcal{G}_1 \to \mathcal{G}_2$ be a morphism of group sheaves on $\mathscr{C}$, and let $\mathcal{P}$ be a $\mathcal{G}_1$-torsor. We define the **contracted product** $\mathcal{G}_2 \times^{\mathcal{G}_1} \mathcal{P}$ as the quotient sheaf $\mathcal{G}_1 \backslash (\mathcal{G}_2 \times \mathcal{P})$, where $\mathcal{G}_1$ acts on the product by:

$$g_1 \cdot (g_2, p) = (g_2 \varphi(g_1)^{-1}, g_1 \cdot p)$$

The contracted product inherits a natural left $\mathcal{G}_2$-action given on local sections by $h \cdot [g_2, p] = [hg_2, p]$, which endows it with the structure of a $\mathcal{G}_2$-torsor.

This construction yields a functor:

$$\varphi_* : \mathbf{Tors}(\mathcal{E}, \mathcal{G}_1) \to \mathbf{Tors}(\mathcal{E}, \mathcal{G}_2), \quad \mathcal{P} \mapsto \mathcal{G}_2 \times^{\mathcal{G}_1} \mathcal{P}$$

On the level of isomorphism classes, φ_* induces a map of pointed sets $\mathrm{Tors}(\mathcal{E}, \mathcal{G}_1) \to \mathrm{Tors}(\mathcal{E}, \mathcal{G}_2)$, sending the class of the trivial $\mathcal{G}_1$-torsor to the class of the trivial $\mathcal{G}_2$-torsor.

When $\mathcal{G}$ is a **sheaf of abelian groups** (an *abelian sheaf*), the pointed set $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$ inherits the structure of an abelian group:

Definition 8. Assume $\mathcal{G}$ is abelian sheaf. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be objects of $\mathbf{Tors}(\mathcal{E}, \mathcal{G})$. We define the **product** $\mathcal{P}_1 \otimes^{\mathcal{G}} \mathcal{P}_2$ to be the quotient sheaf $(\mathcal{P}_1 \times \mathcal{P}_2)/\mathcal{G}$, where $\mathcal{G}$ acts on T-points by having:

$$g \cdot (f_1, f_2) := (gf_1, g^{-1} f_2)$$

The group object $\mathcal{G}$ then acts on the resulting quotient via its action on the presheaf quotient, which is given on classes by:

$$g \cdot [(f_1, f_2)] = [(gf_1, f_2)] = [(f_1, gf_2)]$$

where the square brackets denote the class in the quotient set.[1]

[1]Equivantly, the sum is obtained as the contracted product of the $\mathcal{G} \times \mathcal{G}$-torsor $\mathcal{P}_1 \times \mathcal{P}_2$ along the multiplication map $m : \mathcal{G} \times \mathcal{G} \to \mathcal{G}$.

Let $[\mathcal{P}_1]$ and $[\mathcal{P}_2]$ be the isomorphism classes of $\mathcal{P}_1$ and $\mathcal{P}_2$ in $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$. We define the sum $[\mathcal{P}_1] + [\mathcal{P}_2]$ to be the class $[\mathcal{P}_1 \otimes^{\mathcal{G}} \mathcal{P}_2]$. This structure turns $\mathrm{Tors}(\mathcal{E}, \mathcal{G})$ into an abelian group, where the identity is the class of the trivial torsor and the inverse is obtained by the opposite action.

Let Λ be a commutative ring object in the topos $\mathcal{E}$. We denote by $\mathbf{Pic}(\mathcal{E}, \Lambda)$ the groupoid of locally free Λ-modules of rank 1 in $\mathcal{E}$ (i.e., invertible modules and their isomorphisms). For the multiplicative group object $G = \Lambda^\times$, there exists a classical correspondence between such Λ-modules and G-torsors. Indeed, both categories are symmetric monoidal categories, $\mathbf{Pic}(\mathcal{E}, \Lambda)$ under the tensor product $\otimes_\Lambda$, with unit object Λ. And $\mathbf{Tors}(\mathcal{E}, G)$ under the product $\otimes^G$, with unit object G acting on itself by translation.

Throughout this thesis, we shall primarily adopt the language of G-torsors, with the understanding that the associated Λ-module structure is implicitly inferred. The following proposition formalizes this dictionary:

Proposition 9. *The associated module functor $\Phi : \mathcal{P} \mapsto \mathcal{P} \otimes^G \Lambda$ induces a canonical equivalence of monoidal categories:*

$$\Phi : (\mathbf{Tors}(\mathcal{E}, G), \otimes^G, G) \xrightarrow{\sim} (\mathbf{Pic}(\mathcal{E}, \Lambda), \otimes_\Lambda, \Lambda)$$

with quasi-inverse functor given by $\Psi : \mathcal{L} \mapsto \underline{\mathrm{Isom}}_\Lambda(\Lambda, \mathcal{L})$.

Lastly, for any object $\mathcal{X} \in \mathcal{E}$, there is a canonical identification between the slice category $(\mathcal{G}\mathcal{E})/\mathcal{X}$ and the category of group objects $\mathcal{G}(\mathcal{E}/\mathcal{X})$, where $\mathcal{X}$ is viewed as having the trivial $\mathcal{G}$-action.

We denote by $\mathbf{Tors}(\mathcal{X}, \mathcal{G})$ the category of G-torsors over $\mathcal{X}$ in $\mathcal{G}\mathcal{E}_{/\mathcal{X}}$.

1.1.3 Torsors over the Étale's Sites

Fix a scheme X, and let G be a smooth affine X-group scheme. Denote by $X_{\text{Ét}}$ and $X_{\text{ét}}$ the big and small étale sites on X, respectively. The functor of points of G, which we denote by $G(\cdot) = \mathrm{Hom}_X(-, G)$, defines a group sheaf on $X_{\text{ét}}$.

Definition 10. A *principal G-bundle* (or *principal homogeneous space*) for G is a scheme $f : Y \to X$ equipped with a left G-action $\rho : G \times_X Y \to Y$ satisfying:

1. The morphism $G \times_X Y \to Y \times_X Y$ sending $(g, y) \mapsto (gy, y)$ is an isomorphism of X-schemes.
2. There exists an étale covering $\{U_i \to X\}$ such that $Y_{U_i} \cong G_{U_i}$ as G-schemes, where G acts on itself by left translation.

Remark. Since $G \to X$ is smooth and Y is étale-locally isomorphic to G, the morphism $f : Y \to X$ is smooth and affine. Note that Y is generally not étale over X unless G is an étale group scheme (e.g., a finite constant group). Similarly, if G is finite then $Y \to X$ is finite.

Similarly to the case of G-torsors of a topos, we define a morphism of principal G-bundles to be a morphism of X-schemes commuting with the G-action. We now have:

Theorem 11. *The morphism sending $Y \mapsto \mathrm{Hom}_X(-, Y)$ is an equivalence of categories from the category of principal G-bundles to the category of G-torsors on $X_{Ét}$ Similarly, the morphism sending Y to $\mathrm{Hom}_X(-, Y)$ to the category of G-torsors on $X_{ét}$ is an equivalence.*

Corollary 12. *There is a natural equivalence $\mathbf{Tors}(X_{ét}, G) \cong \mathbf{Tors}(X_{Ét}, G)$ inducing a bijection of pointed sets $\mathrm{Tors}(X_{ét}, G) \xrightarrow{\cong} \mathrm{Tors}(X_{Ét}, G)$ which is an isomorphism of abelian groups if G is* *abelian. And every G-torsor, in each of the above sites can be realized as a scheme, which is a prinicipal G-bundle.*

Constant Finite Group Torsors

Let G be a finite group, within this section, we denote the associated constant group scheme over X by $\underline{G}$ to maintain a clear distinction between the group as a set and the group as a scheme. In subsequent sections, we shall follow the standard practice of identifying G with its constant group scheme and omit the underline for brevity.

$\underline{G}$ is given by $\coprod_{g \in G} X$ with the action shuffeling the X's according to multiplication.

By following the definitions, one sees that if X be a connected scheme and $f : Y \to X$ is a finite Galois cover with Galois group G. Then, $f : Y \to X$ is a principal $\underline{G}$-bundle. (Recall that a *finite Galois cover* is a finite étale surjection $Y \to X$ with Y connected and such that $G = \mathrm{Aut}(Y/X)$ acts transitively on the geometric points of Y lying over any geometric point of X).

On the otherhand if $f : Y \to X$ is a principal $\underline{G}$-bundle with Y connected, then Y is a finite Galois cover with automorphism group G.

However, not all $\underline{G}$-torsors are connected. If $H \subset G$ is a proper subgroup then any connected finite étale cover $f : Y \to X$ with Galois group H gives rise to a non-connected $\underline{G}$-torsor by looking at the induced $\underline{G}$-torsor $\varphi_*(Y)$ under the inclusion $\varphi : \underline{H} \to \underline{G}$. On the otherhand, if we fix a geometric point $\overline{x} \to X$, then to give an homomorphism $\rho \in \mathrm{Hom}_{\mathrm{cont}}(\pi_1^{\mathrm{et}}(X, x), G)$ is equivilant to give a connected pointed Galois cover $(Y, \overline{y}) \to (X, \overline{x})$ with Galois group $H = \rho(\pi_1^{\mathrm{et}}(X, x)) \subset G$. Thus, pushing forward to $\underline{G}$ we get a prinicipal $\underline{G}$-bundle. The choice of a different geometric point $\overline{x'} \to X$ differ the homomorphism by an inner automorphism, thus we have:

Theorem 13. *Let X be a connected scheme and $\overline{x}$ a geometric point of X. Suppose in addition that G is a finite abstract group. Define a map*

$$\mathrm{Hom}_{\mathrm{cont}}(\pi_1^{\mathrm{et}}(X, \overline{x}), G)/\mathrm{Inn}(G) \to \mathrm{Tors}(X_{\acute{E}t}, \underline{G}) \tag{1.1}$$

by sending a homomorphism $\rho : \pi_1^{\mathrm{et}}(X, \overline{x}) \to G$ to the principal $\underline{G}$-bundle $\varphi_*(Y)$ where Y is the principal $\underline{\rho(\pi_1^{\mathrm{et}}(X, \overline{x}))}$-bundle obtained above and φ is the inclusion $\rho(\pi_1^{\mathrm{et}}(X, \overline{x})) \hookrightarrow G$. Then, the map is a bijection of pointed sets where the trivial homomorphisms (which is the only element of its $\mathrm{Inn}(G)$-orbit) is the distinguished element of the left hand side.

If G is abelian then $\mathrm{Inn}(G)$ is trivial, and we obtain:

Corollary 14. *Let G be a finite abelian group, X a connected scheme, and $\overline{x}$ a geometric point of X. Then, the map from Theorem 13 induces an isomorphism of abelian groups*

$$\mathrm{Hom}_{\mathrm{cont.}}(\pi_1^{\acute{e}t}(X, \overline{x}), G) \xrightarrow{\cong} \mathrm{Tors}(X_{\acute{E}t}, \underline{G}) \tag{1.2}$$

Decomposition of Torsors

because the contracted product is also P/H, we should add this!!! say this. etc... maybe have a section about qoutient of schemes. In the case of finite groups, the functoriality of the contracted product has a concrete geometric interpretation as a tower of covers. Let G be a finite abelian group and $H \subseteq G$ a subgroup. Let $\pi : G \to G/H$ be the natural quotient map.

Proposition 15. *Let $\mathcal{P} \to X$ be a G-torsor in $X_{\acute{e}t}$. There exists a factorization of the morphism $\mathcal{P} \to X$:*

$$\mathcal{P} \xrightarrow{\phi} \mathcal{P}_{G/H} \xrightarrow{\psi} X$$

that is unique up to a unique isomorphism commuting with the projections, where:

1. *$\mathcal{P}_{G/H} := \mathcal{P} \times^G (G/H)$ is a G/H-torsor over X.*
2. *$\mathcal{P}$ is an H-torsor over the scheme $\mathcal{P}_{G/H}$.*

Proof. The first assertion follows directly from the definition of the contracted product functor $\pi_* : \mathbf{Tors}(X, G) \to \mathbf{Tors}(X, G/H)$. To see the second assertion, we note that the map $\phi : \mathcal{P} \to \mathcal{P}_{G/H}$ is surjective. Locally on X, we may choose a cover $U \to X$ such that $\mathcal{P}|_U \cong G \times U$. Over this cover, the map ϕ is identified with the product of the quotient map and the identity:

$$\pi \times \mathrm{id}_U : G \times U \to (G/H) \times U$$

Since $G \to G/H$ is a trivial H-torsor (the fiber over any element $\bar{g} \in G/H$ is a coset gH, which is an H-set isomorphic to H acting on itself), $\mathcal{P}$ is locally an H-torsor over $\mathcal{P}_{G/H}$. By the descent of torsors, $\mathcal{P}$ is an H-torsor over $\mathcal{P}_{G/H}$ globally.

For uniqueness, suppose $\mathcal{P} \to \mathcal{Q} \to X$ is another such factorization. Because $\mathcal{P} \to \mathcal{Q}$ is an H-torsor, $\mathcal{Q}$ represents the categorical quotient $\mathcal{P}/H$ in $X_{\text{ét}}$. Since $\mathcal{P}_{G/H}$ also represents this quotient, the universal property of quotients yields a unique isomorphism $\mathcal{Q} \xrightarrow{\sim} \mathcal{P}_{G/H}$ making all associated diagrams commute.

□

1.1.4 Symmetric Powers of Schemes and Torsors

This section reviews the construction of quotients for schemes and torsors under finite group actions, specifically focusing on symmetric powers. To ensure these quotients exist as schemes, we utilize the framework of admissible actions from [SGA1]. Our treatment here closely follows the exposition in [Gui19] The definitions and results presented below are adapted from their work, the difference being that we ask Γ to act on the left, to make the later construction of symmetric powers more natural. This foundation provides the necessary criteria for admissibility and base change required to define the symmetric powers of a scheme X and a G-torsor $\mathcal{P}$ over X.

Let S be a scheme.

Definition 16 (([SGA1], V.1.7).)**.**

- Let T be an object of a category $\mathcal{C}$ endowed with a right action of a group Γ. We say that **the quotient** T/Γ **exists** in $\mathcal{C}$ if the covariant functor

$$\mathcal{C} \to \mathrm{Sets}$$
$$U \mapsto \mathrm{Hom}_{\mathcal{C}}(T, U)^{\Gamma}$$

  is representable by an object of $\mathcal{C}$.

- Let T be an S-scheme. An action of a finite group Γ on T is **admissible** if there exists an affine Γ-invariant morphism $f : T \to T'$ such that the canonical morphism $\mathcal{O}_{T'} \to f_*\mathcal{O}_T$ induces an isomorphism from $\mathcal{O}_{T'}$ to $(f_*\mathcal{O}_T)^{\Gamma}$.

Proposition 17. *The following holds:*

1. ***([SGA1] V.1.3).*** *Let T be an S-scheme endowed with an admissible right action of a finite group Γ. If $f : T \to T'$ is an affine Γ-invariant morphism such that the canonical morphism $\mathcal{O}_{T'} \to f_*\mathcal{O}_T$ induces an isomorphism from $\mathcal{O}_{T'}$ to $(f_*\mathcal{O}_T)^\Gamma$, then the quotient T/Γ exists and is isomorphic to T'.*

2. ***([SGA1], V.1.8).*** *Let T be an S-scheme endowed with a right action of a finite group Γ. Then, the action of Γ on T is admissible if and only if T is covered by Γ-invariant affine open subsets.*

3. ***([SGA1], V.1.9).*** *Let T be an S-scheme endowed with an admissible right action of a finite group Γ, and let S' be a flat S-scheme. Then, the action of Γ on the S'-scheme $T \times_S S'$ is admissible, and the canonical morphism*

$$(T \times_S S')/\Gamma \to (T/\Gamma) \times_S S'$$

is an isomorphism.

Now let X be an S-scheme and let $d \geq 0$ be an integer. The group S_d of permutations of $[\![1, d]\!]$ acts on the right on the S-scheme $X^{\times_S d} = X \times_S \cdots \times_S X$ by the formula

$$(x_i)_{i \in [\![1,d]\!]} \cdot \sigma = (x_{\sigma(i)})_{i \in [\![1,d]\!]}.$$

Proposition 18 ([Gui19] Proposition 2.27)**.** *If X is a scheme, Zariski locally quasi-projective over S, then the right action of the symmetric group S_d on the d-fold fiber product $X^{\times_S d}$ is admissible. Consequently, the quotient $\mathrm{Sym}^d_S(X) = X^{\times_S d}/S_d$ exists as a scheme over S.*

Definition 19. Under the hypotheses of the proposition above, we define the **relative symmetric product** of X over S of degree d as the quotient

$$\mathrm{Sym}^d_S(X) \coloneqq X^{\times_S d}/S_d.$$

When the base scheme S is clear from the context, we shall denote this quotient by $X^{(d)}$.

Guingard shows that when $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ then $\mathrm{Sym}^d_S(X)$ is representable by an affine S-scheme (See [Gui19] Remark 2.28).

Proposition 20 ([Gui19] Proposition 2.28)**.** *If X is flat and Zariski-locally quasi-projective over S, then $Sym^d_S(X)$ is flat over S. Moreover, for any S-scheme S', the canonical morphism*

$$Sym^d_{S'}(X \times_S S') \to Sym^d_S(X) \times_S S'$$

is an isomorphism.

Symmetric Powers of Torsors

Let G be a finite abelian group, let $\mathcal{P}$ be a G-torsor over an S-scheme X in $S_{\text{Ét}}$.

Proposition 21 ([[SGA1]], IX.5.8)**.** *Assume that $\mathcal{P}$ and X are endowed with right actions from a finite group Γ such that the morphism $\mathcal{P} \to X$ is Γ-equivariant, and that the following properties hold:*

(a) The right Γ-action on $\mathcal{P}$ commutes with the left G-action.

(b) The right Γ-action on X is admissible, and the quotient morphism $X \to X/\Gamma$ is finite.

(c) For any geometric point $\bar{x}$ of X, the action of the stabilizer $\Gamma_{\bar{x}}$ of $\bar{x}$ in Γ on the fiber $\mathcal{P}_{\bar{x}}$ of $\mathcal{P}$ at $\bar{x}$ is trivial.

Then the action of Γ on $\mathcal{P}$ is admissible, and $\mathcal{P}/\Gamma$ is a G-torsor over X/Γ in $S_{\acute{E}t}$.

By Theorem 11, $\mathcal{P}$ is representable by a finite étale X-scheme. For each $i \in [\![1, d]\!]$ let $p_i : X^{\times_S d} \to X$ be the projection on i-th factor, and let us consider the G-torsor

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} = K_G \backslash \mathcal{P}^{\times_S d}$$

over $X^{\times_S d}$, where $K_G \subseteq G^d$ is the kernel of the multiplication morphism $G^d \to G$ (the dependence on d is suppressed in the notation). [2] The object $K_G \backslash \mathcal{P}^{\times_S d}$ of $S_{\text{Ét}}$ is too representable by an S-scheme which is finite étale over $X^{\times_S d}$. The group S_d acts on the right on $K_G \backslash \mathcal{P}^{\times_S d}$ by the formula

$$(p_i)_{i\in[\![1,d]\!]} \cdot \sigma = (p_{\sigma(i)})_{i\in[\![1,d]\!]}.$$

This action of S_d commutes with the left action of G on $K_G \backslash \mathcal{P}^{\times_S d}$.

Proposition 22 ([Gui19] Proposition 2.32.). *If X is Zariski-locally quasi-projective on S, then the right action of S_d on $K_G \backslash \mathcal{P}^{\times_S d}$ is admissible, so that the quotient $\mathcal{P}^{[d]}$ of $K_G \backslash \mathcal{P}^{\times_S d}$ by S_d exists as a scheme over S. Moreover, the canonical morphism $\mathcal{P}^{[d]} \to \mathrm{Sym}^d_S(X)$ is a G-torsor, and the morphism*

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} \to r^{-1}\mathcal{P}^{[d]}$$

where $r : X^{\times_S d} \to \mathrm{Sym}^d_S(X)$ is the canonical projection, is an isomorphism of G-torsors over $X^{\times_S d}$.

When G is abelian, and letting the S_d-action be a left action (via the inverse), then the condition $\prod_{i=1}^d g_i = 1$ defining K_G is invariant under permuting the coordinates. Therefore, the left action of S_d normalizes the action of K_G. The group generated by these two actions is the semidirect product $\Xi = K_G \rtimes S_d$, which is a finite group acting on the left on $\mathcal{P}^d$. Then the double qoutient $(K_G \backslash \mathcal{P}^{\times_S d})/S_d \cong S_d \backslash (K_G \backslash \mathcal{P}^{\times_S d})$ add here, explain why the qoutineing by the double qoutient exists, should the action be free or something? is the same as the quotient $\Xi \backslash \mathcal{P}^{\times_S d}$, where Ξ acts on $\mathcal{P}^{\times_S d}$ by the formula:

$$((g_1, \ldots, g_d), \sigma) \star (p_1, \ldots, p_d) = (g_1 p_{\sigma^{-1}(1)}, g_2 p_{\sigma^{-1}(2)}, \ldots, g_d p_{\sigma^{-1}(d)})$$

Corollary 23. *Under the assumptions of the above discussion, there is a natural canonical morphism*

$$q : S_d \backslash \mathcal{P}^{\times_S d} \to \Xi \backslash \mathcal{P}^{\times_S d}$$

or equivlantly

$$q : S_d \backslash \mathcal{P}^{\times_S d} \to (K_G \backslash \mathcal{P}^{\times_S d})/S_d$$

between the symmetric power of $\mathcal{P}$ as a scheme and the symmetric power of $\mathcal{P}$ as a G-torsor. We make it a convention to denote the symmetric power of $\mathcal{P}$ as a scheme by $\mathcal{P}^{(d)}$, and the symmetric power of $\mathcal{P}$ as a G-torsor by $\mathcal{P}^{[d]}$. So that the above qoutient map is also written as: $q : \mathcal{P}^{(d)} \to \mathcal{P}^{[d]}$.

[2] Equivantly, the sum is obtained as the contracted product of the G^d-torsor $p_1^{-1}\mathcal{P}_1 \times \cdots \times p_d^{-1}\mathcal{P}_d$ along the multiplication map $m : G^d \to G$.

Proof. S_d is a subgroup of Ξ, hence any map that is Ξ-invariant is automatically S_d-invariant. $\square$

Remark. Note that the above discussion does not require that $\mathcal{P}$ is a G-torsor over X, only that it is a scheme with a left G-action.

Quotient of Torsors Towers

We now study how the symmetric power construction of Proposition 22 interacts with the canonical decomposition of Proposition 15. Throughout this subsection, G denotes a finite abelian group, $H \subseteq G$ a subgroup, and $\mathcal{P} \to X$ a G-torsor over an S-scheme X that is Zariski-locally quasi-projective over S. Recall that $\mathcal{P} \to X$ factors canonically as

$$\mathcal{P} \xrightarrow{\phi} \mathcal{P}_{G/H} \xrightarrow{\psi} X,$$

where $\mathcal{P}_{G/H} = \mathcal{P} \times^G (G/H)$ is a G/H-torsor over X and ϕ is an H-torsor.

In addition to the kernel $K_G = \ker(G^d \to G)$ of the summation morphism appearing in Proposition 22, we shall need the analogous *summation kernels*

$$K_H := \ker(H^d \to H), \qquad K_{G/H} := \ker((G/H)^d \to G/H).$$

Applying the snake lemma to the commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & H^d \arrow[r] \arrow[d, two heads] & G^d \arrow[r] \arrow[d, two heads] & (G/H)^d \arrow[r] \arrow[d, two heads] & 0 \\ 0 \arrow[r] & H \arrow[r] & G \arrow[r] & G/H \arrow[r] & 0, \end{tikzcd}$$

in which the vertical maps are the surjective summation morphisms, yields a short exact sequence

$$0 \to K_H \to K_G \to K_{G/H} \to 0. \tag{1.3}$$

The induced isomorphism $K_G/K_H \cong K_{G/H}$ will play a central role below.

We define four quotient schemes

$$\begin{aligned} \mathcal{P}^{[d]_G} &:= (K_G \rtimes S_d)\backslash \mathcal{P}^{\times_S d}, \\ \mathcal{P}^{[d]_H} &:= (K_H \rtimes S_d)\backslash \mathcal{P}^{\times_S d}, \\ (\mathcal{P}_{G/H})^{(d)} &:= (H^d \rtimes S_d)\backslash \mathcal{P}^{\times_S d}, \\ (\mathcal{P}_{G/H})^{[d]} &:= ((H^d + K_G) \rtimes S_d)\backslash \mathcal{P}^{\times_S d}, \end{aligned}$$

where S_d acts on $\mathcal{P}^{\times_S d}$ as in Corollary 23 and the subgroups $K_H \subseteq K_G \subseteq H^d + K_G$ and $H^d \subseteq H^d + K_G$ of G^d act diagonally on $\mathcal{P}^{\times_S d}$ via the G-action on $\mathcal{P}$. The first space is the symmetric G-power of $\mathcal{P}$ as in Proposition 22; the remaining three are, respectively, the symmetric H-power of $\mathcal{P}$ relative to ϕ, the symmetric power of the scheme $\mathcal{P}_{G/H}$, and the symmetric G/H-power of $\mathcal{P}_{G/H}$. **The latter two identifications follow from $(\mathcal{P}_{G/H})^{\times_S d} \cong H^d \backslash \mathcal{P}^{\times_S d}$ together with the fact that the preimage of $K_{G/H} \subseteq (G/H)^d$ under $G^d \twoheadrightarrow (G/H)^d$ equals $H^d + K_G$.** the last isomorphism like that, is from the correspondence theorem, claude generated at the end of the article, add it somewhere

Proposition 24. *The four quotient schemes above exist and fit into a canonical commutative diagram*

$$\begin{tikzcd}
\mathcal{P}^{[d]_H} \arrow[r, "H"] \arrow[d] & (\mathcal{P}_{G/H})^{(d)} \arrow[d, "q"] \\
\mathcal{P}^{[d]_G} \arrow[r, "H"'] \arrow[dr, bend right, "G"'] & (\mathcal{P}_{G/H})^{[d]} \arrow[d, "G/H"] \\
 & X^{(d)},
\end{tikzcd}$$

in which each labelled arrow is a torsor under the indicated constant group. The right-hand vertical q is the canonical projection of Corollary 23, applied to the G/H-torsor $\mathcal{P}_{G/H} \to X$; in general it is not *a torsor under any constant group scheme (see the remark below). The unlabelled left-hand vertical $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ is the canonical projection induced by the inclusion of S_d-stable subgroups $K_H \rtimes S_d \subseteq K_G \rtimes S_d$ in $G^d \rtimes S_d$; it is the base change of q along the structural map $\mathcal{P}^{[d]_G} \to (\mathcal{P}_{G/H})^{[d]}$ via the Cartesian property below, and likewise fails to be a torsor under a constant group scheme in general.*

Moreover, the upper square is Cartesian:

$$\mathcal{P}^{[d]_H} \cong \mathcal{P}^{[d]_G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)}.$$

Proof. The four labelled arrows are torsors under the indicated constant groups by direct application of Proposition 22 and Proposition 15:

- Applied to the G-torsor $\mathcal{P} \to X$, Proposition 22 gives the G-torsor $\mathcal{P}^{[d]_G} \to X^{(d)}$ (the diagonal arrow).
- Applied to the H-torsor $\phi : \mathcal{P} \to \mathcal{P}_{G/H}$ (taking $\mathcal{P}_{G/H}$ in place of the base X), Proposition 22 gives the H-torsor $\mathcal{P}^{[d]_H} \to (\mathcal{P}_{G/H})^{(d)}$ (the top arrow).
- Applied to the G/H-torsor $\psi : \mathcal{P}_{G/H} \to X$, Proposition 22 gives the G/H-torsor $(\mathcal{P}_{G/H})^{[d]} \to X^{(d)}$ (the bottom-right vertical), and Corollary 23 produces the canonical morphism $q : (\mathcal{P}_{G/H})^{(d)} \to (\mathcal{P}_{G/H})^{[d]}$.

The factorization $\mathcal{P}^{[d]_G} \xrightarrow{H} (\mathcal{P}_{G/H})^{[d]} \xrightarrow{G/H} X^{(d)}$ of the diagonal arrow into an H-torsor followed by a G/H-torsor is the canonical decomposition of Proposition 15, applied to the G-torsor $\mathcal{P}^{[d]_G} \to X^{(d)}$ and the subgroup $H \subseteq G$. Indeed, both $\mathcal{P}^{[d]_G} \times^G (G/H)$ and $(\mathcal{P}_{G/H})^{[d]}$ are realized as the admissible quotient $((H^d + K_G) \rtimes S_d)\backslash \mathcal{P}^{\times_S d}$. here maybe add more info, like isomorphism theorem for torsors and qoutients, or something like that, to make it more clear, from which it follows

To prove the upper square is Cartesian, set $F := \mathcal{P}^{[d]_G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)}$. Base-changing the H-torsor $\mathcal{P}^{[d]_G} \to (\mathcal{P}_{G/H})^{[d]}$ along q exhibits F as an H-torsor over $(\mathcal{P}_{G/H})^{(d)}$. The universal property of the fiber product furnishes a canonical morphism $\mathcal{P}^{[d]_H} \to F$ over $(\mathcal{P}_{G/H})^{(d)}$, which is H-equivariant since the H-actions on both source and target are induced by the diagonal H-action on $\mathcal{P}^{\times_S d}$ add earlier in the section of torsors that in fact base changing an H-torsor along a morphism gives another H-torsor is a torsor. Any morphism of H-torsors is an isomorphism, so

$\mathcal{P}^{[d]_H} \cong F$ and the upper square is Cartesian. The left vertical $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ is then identified, via this Cartesian square, with the base change of q along $\mathcal{P}^{[d]_G} \to (\mathcal{P}_{G/H})^{[d]}$. $\square$

Remark. The two vertical morphisms $q : (\mathcal{P}_{G/H})^{(d)} \to (\mathcal{P}_{G/H})^{[d]}$ and $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ are not, in general, torsors under any constant group scheme. The obstruction is purely group-theoretic: in $K_G \rtimes S_d$ the subgroup $K_H \rtimes S_d$ is not normal, because the conjugation action of S_d on K_G by coordinate permutation is non-trivial — for $k \in K_G$ and $\sigma \in S_d$, the components of $k - \sigma \cdot k$ are $k_i - k_{\sigma^{-1}(i)} \in G$, which need not lie in H. Equivalently, $H^d \rtimes S_d$ is not normal in $(H^d + K_G) \rtimes S_d$. So the naive set-theoretic quotients $(K_G \rtimes S_d)/(K_H \rtimes S_d)$ and $((H^d + K_G) \rtimes S_d)/(H^d \rtimes S_d)$ are not groups, even though $K_G/K_H \cong K_{G/H}$ and $(H^d + K_G)/H^d \cong K_{G/H}$ as abstract groups.

Compatibility with Addition

The construction of symmetric powers is compatible with a natural addition law. For any integers $d_1, d_2 \geq 0$, there exists a canonical **addition morphism**

$$+_{d_1,d_2} : X^{(d_1)} \times_S X^{(d_2)} \to X^{(d_1+d_2)}$$

induced by the equivariant isomorphism $X^{\times_S d_1} \times_S X^{\times_S d_2} \cong X^{\times_S (d_1+d_2)}$ with respect to the inclusion of the product of symmetric groups $S_{d_1} \times S_{d_2} \subseteq S_{d_1+d_2}$.

Proposition 25. *Let $\mathcal{P}$ be a G-torsor over X. There is a canonical isomorphism of G-torsors over $X^{(d_1)} \times_S X^{(d_2)}$:*

$$+_{d_1,d_2}^{-1}\left(\mathcal{P}^{[d_1+d_2]}\right) \cong r_1^{-1}\mathcal{P}^{[d_1]} \otimes^G r_2^{-1}\mathcal{P}^{[d_2]}.$$

Proof. Let $\pi = r_{d_1} \times r_{d_2}$. By the commutativity of the following diagram

$$\begin{tikzcd} X^{\times_S d_1} \times_S X^{\times_S d_2} \arrow[r, "\sim"] \arrow[d, "\pi"'] & X^{\times_S (d_1+d_2)} \arrow[d, "r_{d_1+d_2}"] \\ X^{(d_1)} \times_S X^{(d_2)} \arrow[r, "+_{d_1,d_2}"] & X^{(d_1+d_2)} \end{tikzcd}$$

and applying Proposition 22, we obtain canonical isomorphisms:

$$\pi^{-1}\left(+_{d_1,d_2}^{-1}\mathcal{P}^{[d_1+d_2]}\right) \cong r_{d_1+d_2}^{-1}\mathcal{P}^{[d_1+d_2]} \cong \bigotimes_{i=1}^{d_1+d_2}{}^{G}\, p_i^{-1}\mathcal{P}$$

and

$$\pi^{-1}(r_1^{-1}\mathcal{P}^{[d_1]} \otimes^G r_2^{-1}\mathcal{P}^{[d_2]}) \cong \left(\bigotimes_{i=1}^{d_1}{}^{G}\, p_i^{-1}\mathcal{P}\right) \otimes^G \left(\bigotimes_{j=d_1+1}^{d_1+d_2}{}^{G}\, p_j^{-1}\mathcal{P}\right).$$

Since both pullbacks identify with

$$\bigotimes_{k=1}^{d_1+d_2}{}^{G}\, p_k^{-1}\mathcal{P}$$

in an $S_{d_1} \times S_{d_2}$-equivariant manner, the isomorphism descends to the quotient $X^{(d_1)} \times_S X^{(d_2)}$ by Proposition 20. $\square$

1.1.5 Algebraic Preliminaries on Ramification

This section reviews the ramification of discrete valuations, beginning with the general theory of *discrete valuation rings* and their integral closures in finite separable extensions. We then specialize to complete rings in Galois extensions to describe the ramification filtration via lower and upper numbering, following the treatments in [Stacks, Tag 0EXQ] and [Ser79].

Definition 26 ([Stacks, Tag 09E4])**.** We say that $A \to B$ or $A \subset B$ is an extension of discrete valuation rings if A and B are discrete valuation rings and $A \to B$ is injective and local. In particular, if π_A and π_B are uniformizers of A and B, then $\pi_A = u\pi_B^e$ for some $e \geq 1$ and unit u of B. The integer e does not depend on the choice of the uniformizers as it is also the unique integer ≥ 1 such that

$$\mathfrak{m}_A B = \mathfrak{m}_B^e$$

The integer e is called the *ramification index* of B over A. We say that B is *weakly unramified* over A if $e = 1$. If the extension of residue fields $\kappa_A = A/\mathfrak{m}_A \subset \kappa_B = B/\mathfrak{m}_B$ is finite, then we set $f = [\kappa_B : \kappa_A]$ and we call it the *residual degree* or residue degree of the extension $A \subset B$.

Note that we do not require the extension of fraction fields to be finite.

Now let A be a discrete valuation ring with fraction field K, let L/K be a finite separable field extension and let $B \subset L$ be the integral closure of A in L. Then the ring extension $A \subset B$ is finite, hence B is Noetherian. The dimension of B is 1, hence B is a Dedekind domain. Let $\mathfrak{m}_1, \dots, \mathfrak{m}_n$ be the maximal ideals of B (i.e., the primes lying over $\mathfrak{m}_A$). We obtain extensions of discrete valuation rings

$$A \subset B_{\mathfrak{m}_i}$$

and hence ramification indices e_i and residue degrees f_i. We have

$$[L : K] = \sum\nolimits_{i=1,\dots,n} e_i f_i$$

Note that if A is henselian (e.g. A is complete) then $n = 1$.

Definition 27 ([Stacks, Tag 09E9])**.** Let A be a discrete valuation ring with fraction field K. Let L/K be a finite separable extension. With B and $\mathfrak{m}_i$, $i = 1, \dots, n$ as above, we say the extension L/K is

1. unramified with respect to A if $e_i = 1$ and the extension $\kappa(\mathfrak{m}_i)/\kappa_A$ is separable for all i,
2. tamely ramified with respect to A if either the characteristic of κ_A is 0 or the characteristic of κ_A is $p > 0$, the field extensions $\kappa(\mathfrak{m}_i)/\kappa_A$ are separable, and the ramification indices e_i are prime to p, and
3. totally ramified with respect to A if $n = 1$ and the residue field extension $\kappa(\mathfrak{m}_1)/\kappa_A$ is trivial.

If the discrete valuation ring A is clear from context, then we sometimes say L/K is unramified, totally ramified, or tamely ramified for short.

Now and for the rest of the section assume A is a *complete discrete valuation ring* with uniformizer π and residue field κ, which we assume to be perfect. When A and κ are of the same characteristic

$p > 0$, then A contains a coefficient field $k \cong \kappa$ and a well known structure theorem holds: $A = k[[\pi]] \cong k[[t]]$. Let K be the fraction field of A, then $K = k((\pi))$.

For the remainder of this section, assume A is a *complete discrete valuation ring* with uniformizer π and a perfect residue field κ. In the equal characteristic case, where $\mathrm{char}(A) = \mathrm{char}(\kappa) = p > 0$, the Cohen Structure Theorem implies that A contains a coefficient field $k \cong \kappa$, such that $A \cong k[[\pi]] \cong k[[t]]$. Consequently, the fraction field is given by $K = k((\pi))$. Let L/K be a finite Galois extension with Galois group G. The integral closure B of A in L is itself a complete discrete valuation ring. Since the residue field κ is perfect, the extension of residue fields is separable, and there exists a uniformizer $\Pi \in B$ such that $B = A[\Pi]$.

The *lower numbering ramification filtration* of G, denoted by $(G_i)_{i\geq -1}$, is defined as:

$$G_i = \{\sigma \in G \mid v_B(\sigma(x) - x) \geq i + 1 \text{ for all } x \in B\}$$

where v_B is the valuation on L associated with B. In this indexing, $G_{-1} = G$ and G_0 corresponds to the *inertia group* of the extension L/K. Note that L/K is unramified if and only if $G_0 = \{1\}$, and it is tamely ramified if and only if $G_1 = \{1\}$.

A standard result in the theory of local fields(complete local fields with a discrete valuation and perfect residue field) ensures that the condition in the definition of G_i need only be checked for the uniformizer Π of B. Specifically, if we define the function

$$i_K^L(\sigma) = v_B(\sigma(\Pi) - \Pi)$$

for $\sigma \in G$, then $G_i = \{\sigma \in G \mid i_K^L(\sigma) \geq i + 1\}$. The subgroups G_i are normal in G and become trivial for sufficiently large i.

Consider a tower of fields $K \subset E \subset L$, and let $H = \mathrm{Gal}(L/E) \subset G$. The lower numbering is compatible with subgroups:

$$G_i \cap H = H_i \quad \text{for all } i \geq -1$$

which follows from the identity $i_E^L = i_K^L|_H$. However, the lower numbering does not behave as simply with respect to quotients. If H is normal in G, the image of G_i in G/H is generally not $(G/H)_i$. Instead, we have $G_iH/H = (G/H)_j$, where the index j is given by the formula:

$$j = \frac{1}{e_{L/E}} \sum_{\tau\in H} \min(i_K^L(\tau), i + 1) - 1$$

In the literature, one reindexes the ramification groups by defining the Herbrand function $\phi_{L/K} : [-1, \infty) \to [-1, \infty)$:

$$\phi_{L/K}(i) = \frac{1}{e_{L/K}} \sum_{\sigma\in G} \min(i_K^L(\sigma), i + 1) - 1 = \int_0^i \frac{1}{[G_0 : G_t]} dt$$

This function is continuous, strictly increasing, and piecewise linear, making it a bijection. It satisfies the composition law $\phi_{L/K} = \phi_{E/K} \circ \phi_{L/E}$ for a tower of extensions $K \subset E \subset L$. Using this function, we define the *upper numbering ramification groups* as:

$$G^i = G_{\phi_{L/K}^{-1}(i)}$$

The primary advantage of this reindexing is its compatibility with quotients: for any normal subgroup $H \subset G$, we have:

$$G^iH/H = (G/H)^i$$

for all $i \geq -1$.

Proposition 28 ([Ser79], IV Corollary 3). *the quotients G_i/G_{i+1}, $i \geq 1$, are abelian groups, and are direct products of cyclic groups of order p. The group G_1 is a p-group.*

Proposition 29 ([Ser79], IV Corollary 4). *The inertia group G_0 has the following property: It is the semi-direct product of a cyclic group of order prime to p with a normal subgroup whose order is a power of p.*

Kummer and Artin-Schreier Theories

For cyclic extensions, Kummer and Artin-Schreier theories provide explicit ways to calculate ramification. These results show how the valuation of a defining element $a \in K$ determines the ramification index and the jumps in the upper numbering filtration. Throughout this section, let K be a discrete valuation field with a perfect residue field κ of characteristic $p > 0$.

Theorem 30 (Ramification in Kummer Extensions, [Koc97], Proposition 1.83). *Assume K contains the n-th roots of unity μ_n. Let L/K be the extension given by the equation $X^n = a$ for some $a \in K^\times$ and denote by G its Galois group. Then we have:*

1. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in the residue field κ is an n-th power, the extension L/K is trivial.*
2. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in the residue field κ is not an n-th power, the extension L/K is cyclic and unramified.*
3. *If $v_K(a) \notin n\mathbb{Z}$, the extension L/K is cyclic and ramified. Specifically, if $\gcd(|v_K(a)|, n) = 1$, the extension is totally ramified of degree n. Otherwise it has ramification index $\frac{n}{|gcd(v_K(a)|,n)}$*

Conversely, Kummer theory ensures that every cyclic extension of degree n, prime to p of a field that contains n-th roots of unity, is of the above form. Moreover, in the above we can always take $a \in \mathcal{O}_K$

Note that in the case of total ramification the extesntion is tamely ramified.

Theorem 31 (Ramification in Artin-Schreier Extensions, [Tho05]). *Let $\wp(x) = x^p - x$ be the Artin-Schreier opeartor. Let L/K be the extension given by the equation $X^p - X = a$ for some $a \in K$ and denote by G its Galois group. Then we have:*

1. *If $v_K(a) > 0$ or if $v_K(a) = 0$ and $a \in \wp(K)$, the extension L/K is trivial.*
2. *If $v_K(a) = 0$ and if $a \notin \wp(K)$, the extension L/K is cyclic of degree p and unramified.*
3. *If $v_K(a) = -m < 0$ with $m \in \mathbb{Z}_{>0}$ and if m is prime to p, the extension L/K is cyclic of degree p again and totally ramified. Moreover, its ramification groups are given by:*

$$G = G^{(-1)} = \cdots = G^{(m)} \quad \text{and} \quad G^{(m+1)} = 1.$$

Conversely, Artin-Schreier theory ensures that every cyclic extension of degree p takes this form. Moreover, in the above and under the isomorphism $K \cong k((t))$, one can always take a of the form $ct^{-m} + a_{-m+1}t^{-m+1} + \cdots a_{-1}t^{-1} + a_0$. If k is algebraically closed then there is a change of variables such that $a = u^{-m}$.

1.1.6 Witt Vectors and Artin–Schreier–Witt Theory

Artin–Schreier theory, together with the ramification formula of Theorem 31, gives complete control over the cyclic extensions of degree p of a local field of characteristic p. To treat the groups $G = \mathbb{Z}/p^r\mathbb{Z}$ we need the analogous control over cyclic extensions of degree p^r, which is provided by Witt's generalization: the field K is replaced by the ring $W_r(K)$ of p-typical Witt vectors of length r, the operator $\wp(u) = u^p - u$ by its Witt-vector analogue $\wp = F - \mathrm{id}$, and the classical ramification-break formula by the Schmid–Witt formula. This subsection collects the definitions and the elementary lemmas that we use; our references for the general theory are [Ser79, II, §6] and [Rab14].

Throughout, p is a fixed prime, and $r \geq 1$ is an integer. All the Witt vectors appearing in this thesis are p-typical and of finite length.

The ring of Witt vectors

For $n \geq 0$, the n-th **Witt polynomial** in the variables $X_0, \dots, X_n$ is

$$w_n(X_0, \dots, X_n) \;=\; \sum_{j=0}^{n} p^j X_j^{p^{n-j}} \;=\; X_0^{p^n} + pX_1^{p^{n-1}} + \cdots + p^n X_n.$$

Theorem 32 (Witt; [Ser79, II, §6, Theorem 6], [Rab14, §1]). *There exists a unique family of polynomials $S_n, P_n \in \mathbb{Z}[X_0, \dots, X_n; Y_0, \dots, Y_n]$, $n \geq 0$, such that*

$$w_n(S_0, \dots, S_n) = w_n(X_0, \dots, X_n) + w_n(Y_0, \dots, Y_n), \qquad w_n(P_0, \dots, P_n) = w_n(X_0, \dots, X_n) \cdot w_n(Y_0, \dots, Y_n)$$

for all n. For any commutative ring A, the operations

$$\vec{a} \oplus \vec{b} := \big(S_n(\vec{a}; \vec{b})\big)_{0 \leq n \leq r-1}, \qquad \vec{a} \odot \vec{b} := \big(P_n(\vec{a}; \vec{b})\big)_{0 \leq n \leq r-1}$$

*make the set A^r into a commutative ring $W_r(A)$, the ring of **p-typical Witt vectors of length r over** A, with zero $(0, \dots, 0)$ and unit $(1, 0, \dots, 0)$. Additive inverses are likewise given by universal integral polynomials; we write $\ominus$ for Witt subtraction. The **ghost map***

$$w \colon W_r(A) \longrightarrow A^r, \qquad \vec{a} \longmapsto \big(w_0(\vec{a}), w_1(\vec{a}), \dots, w_{r-1}(\vec{a})\big)$$

is a ring homomorphism to the product ring A^r; it is injective when p is a non-zero-divisor in A, and bijective when $p \in A^\times$.

We index the components of a Witt vector $\vec{a} = (a_0, \dots, a_{r-1})$ by $0, \dots, r-1$. For $r = 1$ the polynomials reduce to $S_0 = X_0 + Y_0$ and $P_0 = X_0 Y_0$, so $W_1(A) = A$ as a ring.

Since the ring operations are given by universal polynomials with integer coefficients, W_r is a functor: a ring homomorphism $\sigma \colon A \to B$ induces the ring homomorphism

$$W_r(\sigma) \colon W_r(A) \to W_r(B), \qquad W_r(\sigma)(a_0, \dots, a_{r-1}) = (\sigma a_0, \dots, \sigma a_{r-1}),$$

acting *componentwise*. Two consequences of this trivial-looking observation are used repeatedly and deserve emphasis: a ring automorphism σ of A commutes with $\oplus$ and $\ominus$ on $W_r(A)$, and a Witt vector $\vec{a} \in W_r(A)$ satisfies $W_r(\sigma)(\vec{a}) = \vec{a}$ if and only if *each component* a_n is σ-invariant. In particular, for a group Γ acting on A by ring automorphisms,

$$W_r(A)^\Gamma \;=\; W_r\big(A^\Gamma\big) \qquad \text{inside } W_r(A). \tag{1.4}$$

The uniqueness statement in Theorem 32 is an instance of a general principle that we invoke several times, so we record it explicitly.

Lemma 33 (ghost principle). *Let Φ and Ψ be two operations on Witt vectors given by universal polynomials with integer coefficients in the components of their arguments. If $w \circ \Phi = w \circ \Psi$ holds over the polynomial ring $\mathbb{Z}[X_0, X_1, \dots]$ on the components of the arguments, then $\Phi = \Psi$ over every commutative ring.*

Proof. Over $\mathbb{Z}[X_0, X_1, \dots]$ the prime p is a non-zero-divisor, so the ghost map is injective by Theorem 32 and $\Phi = \Psi$ there; this is an identity of integral polynomials, which persists under every base change $\mathbb{Z}[X_0, X_1, \dots] \to A$. □

The only structural feature of the addition polynomials S_n that we shall need is their "triangular" shape:

Proposition 34 (shape of Witt addition). *For every $n \geq 0$ there is a polynomial $C_n \in \mathbb{Z}[X_0, \dots, X_{n-1}; Y_0, \dots, Y_{n-}$ involving only the variables of index* strictly less *than n, such that*

$$(\vec{a} \oplus \vec{b})_n \;=\; a_n + b_n + C_n(a_0, \dots, a_{n-1}; b_0, \dots, b_{n-1}). \tag{1.5}$$

*We refer to C_n as the **carry polynomial**. The analogous statement holds for $\ominus$.*

Proof. By induction on n, using the recursion defining S_n:

$$p^n S_n \;=\; \sum_{j \leq n} p^j \left(X_j^{p^{n-j}} + Y_j^{p^{n-j}} \right) \;-\; \sum_{j<n} p^j S_j^{p^{n-j}}.$$

The terms with $j = n$ contribute $p^n(X_n + Y_n)$, and all remaining terms involve only variables and polynomials S_j of index $j < n$, which by induction lie in $\mathbb{Z}[X_{<n}; Y_{<n}]$. Since S_n has integer coefficients (Theorem 32), $C_n := S_n - X_n - Y_n$ is an integral polynomial in the variables of index $< n$. The statement for $\ominus$ follows by the same argument applied to the inversion polynomials. □

Frobenius, Verschiebung and truncation

From now on, A denotes an $\mathbb{F}_p$-algebra; this is the only case we use. Three operators act on the rings $W_r(A)$:

Definition 35. Let A be an $\mathbb{F}_p$-algebra.

1. The **Frobenius** $F\colon W_r(A) \to W_r(A)$ is $F := W_r(\varphi_A)$, where $\varphi_A\colon a \mapsto a^p$ is the absolute Frobenius of A. Explicitly, F acts componentwise: $F(a_0, \dots, a_{r-1}) = (a_0^p, \dots, a_{r-1}^p)$. By functoriality, F is a ring endomorphism of $W_r(A)$.

2. The **Verschiebung** is the shift

$$V\colon W_{r-1}(A) \to W_r(A), \qquad V(a_0, \dots, a_{r-2}) = (0, a_0, \dots, a_{r-2}).$$

   Iterating, $V^j\colon W_{r-j}(A) \to W_r(A)$; in particular $V^{r-1}\colon A = W_1(A) \to W_r(A)$ sends $c \mapsto (0, \dots, 0, c)$.

3. The **truncation** is

$$t\colon W_r(A) \to W_{r-1}(A), \qquad t(a_0, \dots, a_{r-1}) = (a_0, \dots, a_{r-2}).$$

Remark. For a general (not necessarily $\mathbb{F}_p$-) algebra A, the Frobenius is defined by universal polynomials characterized by $w_n \circ F = w_{n+1}$, and the componentwise formula above holds precisely in characteristic p; see [Ser79, II, §6] or [Rab14, §2]. We will not need the general case.

We record from the general theory ([Ser79, II, §6], [Rab14, §2]) how multiplication by p interacts with these operators.

Proposition 36. *Let A be an $\mathbb{F}_p$-algebra. Then:*

1. *t is a natural ring homomorphism commuting with F, and its kernel is*
$$\ker\big(t\colon W_r(A) \to W_{r-1}(A)\big) \;=\; V^{r-1}A \;=\; \{(0,\dots,0,c) : c \in A\}.$$
2. *V is additive, and $FV = VF$.*
3. *Multiplication by p on $W_r(A)$ equals $V \circ F = F \circ V$ (composed with the appropriate truncation); explicitly,*
$$p \cdot (a_0,\dots,a_{r-1}) \;=\; (0,\, a_0^p,\, \dots,\, a_{r-2}^p).$$
Iterating, $p^j \cdot \vec{a} = (0,\dots,0,a_0^{p^j},\dots,a_{r-1-j}^{p^j})$; in particular $p^r = 0$ on $W_r(A)$, and
$$p^{r-1} \cdot (a_0,\dots,a_{r-1}) \;=\; \big(0,\dots,0,\, a_0^{p^{r-1}}\big). \tag{1.6}$$

Proof. (1) That t is a ring homomorphism follows from the recursive definition of the S_n, P_n: the first $r-1$ components of a sum or product depend only on the first $r-1$ components of the arguments (Proposition 34). Naturality and the commutation with the (componentwise) F are clear, and the kernel is computed componentwise. (2) Additivity of V follows from Lemma 33 and the ghost identity $w_n(V\vec{a}) = p\, w_{n-1}(\vec{a})$ (with $w_{-1} := 0$); the commutation $FV = VF$ is immediate since F is componentwise. (3) The identity $FV = p$ holds over arbitrary rings for the general Frobenius, by Lemma 33 applied to $w_n(FV\vec{a}) = w_{n+1}(V\vec{a}) = p\, w_n(\vec{a}) = w_n(p\vec{a})$; in characteristic p the general Frobenius is componentwise, giving the displayed formula. See [Ser79, II, §6, Proposition 8 ff.] for details. ☐

Example 37. $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$ canonically. Indeed, by (1.6) the unit $\underline{1} = (1,0,\dots,0)$ satisfies $p^j \cdot \underline{1} = (0,\dots,0,1,0,\dots,0)$ (1 in position j), so $\underline{1}$ has additive order p^r; since $|W_r(\mathbb{F}_p)| = p^r$, the additive group is cyclic of order p^r generated by the unit, and $m \mapsto m \cdot \underline{1}$ is a ring isomorphism $\mathbb{Z}/p^r\mathbb{Z} \xrightarrow{\cong} W_r(\mathbb{F}_p)$. We write $\underline{m} \in W_r(\mathbb{F}_p)$ for the image of $m \in \mathbb{Z}/p^r\mathbb{Z}$. (Similarly, the full ring of Witt vectors satisfies $W(\mathbb{F}_p) \cong \mathbb{Z}_p$; we shall not need this.)

Definition 38. The **Artin–Schreier–Witt operator** on $W_r(A)$, A an $\mathbb{F}_p$-algebra, is
$$\wp \;:=\; F - \mathrm{id}, \qquad \wp\,\vec{u} \;=\; F\vec{u} \ominus \vec{u}.$$
For $r = 1$ this is the classical operator $\wp(u) = u^p - u$ of Theorem 31.

Since F is a ring endomorphism, $\wp$ is an endomorphism of the additive group $(W_r(A), \oplus)$: $\wp(\vec{u} \oplus \vec{v}) = \wp\vec{u} \oplus \wp\vec{v}$. Moreover $\wp$ commutes with $W_r(\sigma)$ for every ring homomorphism σ (both F and the ring operations are natural), and with V and t by Proposition 36.

The following bookkeeping lemma is elementary but does the essential work of reducing statements about W_r to statements about W_{r-1} and about A itself: the truncation t and the inclusion V^{r-1} of its kernel slice W_r into lower-length pieces, compatibly with $\wp$, and *addition of an element of the kernel is carry-free.*

Lemma 39 (Witt bookkeeping). *Let A be an $\mathbb{F}_p$-algebra.*

(i) $t\colon W_r(A) \to W_{r-1}(A)$ is a ring homomorphism commuting with F, hence with $\wp$; its kernel is $V^{r-1}A$.

(ii) $V^{r-1}\colon A \to W_r(A)$ is additive, and $\wp \circ V^{r-1} = V^{r-1} \circ \wp$, where on the left $\wp$ is the operator on $W_r(A)$ and on the right the classical $\wp(c) = c^p - c$ on A.

(iii) (carry-free top-level addition) For all $\vec{a} \in W_r(A)$ and $c \in A$,

$$\vec{a} \oplus V^{r-1}(c) \;=\; (a_0, \ldots, a_{r-2},\, a_{r-1} + c).$$

Proof. (i) and (ii) restate parts of Proposition 36 (for (ii): V^{r-1} is additive and commutes with the componentwise F, hence with $\wp = F - \mathrm{id}$). (iii) By Lemma 33 it suffices to verify the identity over $\mathbb{Z}[a_0, \ldots, a_{r-1}, c]$, where the ghost map is injective. Put $\vec{b} := (0, \ldots, 0, c)$ and $\vec{s} := \vec{a} \oplus \vec{b}$. For $n < r-1$ we have $w_n(\vec{b}) = 0$, so $w_n(\vec{s}) = w_n(\vec{a})$, and by induction on n (solving for s_n, using that p is a non-zero-divisor) $s_n = a_n$ for $n \le r-2$. For $n = r-1$, $w_{r-1}(\vec{b}) = p^{r-1}c$ gives $p^{r-1}s_{r-1} = p^{r-1}a_{r-1} + p^{r-1}c$, i.e. $s_{r-1} = a_{r-1} + c$. ☐

Remark. Lemma 39(iii) fails for the addition of vectors supported in lower positions: in general the carry polynomials C_n of (1.5) are non-zero, and Witt addition is *not* componentwise. In particular, operations performed componentwise on a Witt vector — such as discarding the regular parts of Laurent-series components — do *not* in general respect Witt addition, and will be carefully avoided.

The Artin–Schreier–Witt isogeny and its torsors

The componentwise ring operations make $\mathbb{A}^r_{\mathbb{F}_p}$ a ring scheme, the **Witt vector group scheme** $\mathbb{W}_r$: its underlying scheme is $\mathbb{A}^r_{\mathbb{F}_p} = \operatorname{Spec} \mathbb{F}_p[T_0, \ldots, T_{r-1}]$, with the group law given by the addition polynomials S_n, so that $\mathbb{W}_r(A) = W_r(A)$ functorially. The operator $\wp = F - \mathrm{id}$ is an endomorphism of the group scheme $\mathbb{W}_r$, and it is the Witt-vector instance of a *Lang isogeny*: by Example 37 and the lemma below, it sits in a short exact sequence of group schemes over $\mathbb{F}_p$ for the étale topology

$$0 \longrightarrow \mathbb{Z}/p^r\mathbb{Z} \cong W_r(\mathbb{F}_p) \longrightarrow \mathbb{W}_r \xrightarrow{\;\wp\;} \mathbb{W}_r \longrightarrow 0. \tag{1.7}$$

The corresponding covers admit the following completely explicit description, which is the form in which we use them. For $\vec{g} \in W_r(A)$ we write $A[\vec{X}]/(\wp\vec{X} \ominus \vec{g})$, $\vec{X} = (X_0, \ldots, X_{r-1})$, for the quotient of $A[X_0, \ldots, X_{r-1}]$ by the ideal generated by the r components of the Witt vector $\wp\vec{X} \ominus \vec{g} \in W_r(A[X_0, \ldots, X_{r-1}])$.

Lemma 40 (ASW covers are étale torsors). *Let A be an $\mathbb{F}_p$-algebra and $\vec{g} \in W_r(A)$. Let*

$$B \;:=\; A[\vec{X}]/\big(\wp\vec{X} \ominus \vec{g}\big).$$

Then B is finite free over A with basis $\big\{\prod_n X_n^{a_n} : 0 \le a_n \le p-1\big\}$, and étale over A. The translation action of $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$,

$$m\colon \vec{X} \longmapsto \vec{X} \oplus \underline{m},$$

makes $\operatorname{Spec} B$ *a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over* $\operatorname{Spec} A$.

Proof. By (1.5) and the componentwise formula for F, the n-th defining equation has the triangular form

$$X_n^p - X_n - g_n + C'_n(X_0, \dots, X_{n-1};\, g_0, \dots, g_{n-1}) \;=\; 0$$

for universal integral polynomials C'_n. Triangularity gives the free basis; the Jacobian matrix of the system is lower triangular with diagonal entries $pX_n^{p-1} - 1 = -1$, a unit, so B is étale over A. Finally, $\operatorname{Spec} B$ is the pullback along $\vec{g}\colon \operatorname{Spec} A \to \mathbb{W}_r$ of the covering $\wp\colon \mathbb{W}_r \to \mathbb{W}_r$; by the triangular form just established (over $A = \mathbb{F}_p[T_0, \dots, T_{r-1}]$, $\vec{g} = (T_0, \dots, T_{r-1})$), $\wp$ is a surjective étale homomorphism of group schemes, and its kernel is $\{\vec{u} : F\vec{u} = \vec{u}\} = W_r(\mathbb{F}_p)$ since F is componentwise. A surjective étale group homomorphism is a torsor under its kernel, and torsors pull back to torsors; hence $\operatorname{Spec} B$ is a $W_r(\mathbb{F}_p)$-torsor over $\operatorname{Spec} A$. $\square$

Artin–Schreier–Witt theory for fields

Over a field, the sequence (1.7) computes all cyclic extensions of p-power degree, exactly as classical Artin–Schreier theory does for degree p.

Theorem 41 (Artin–Schreier–Witt theory; [Tho05, §3–4])**.** *Let $M \supseteq \mathbb{F}_p$ be a field, M^s a separable closure, and $\Gamma = \operatorname{Gal}(M^s/M)$. Then $\wp$ is surjective on $W_r(M^s)$ with kernel $W_r(\mathbb{F}_p)$, and there is a canonical isomorphism of abelian groups*

$$\delta\colon\; W_r(M)/\wp\, W_r(M) \;\xrightarrow{\;\cong\;}\; \operatorname{Hom}_{\text{cont}}\big(\Gamma,\, \mathbb{Z}/p^r\mathbb{Z}\big) \;=\; H^1\big(M, \mathbb{Z}/p^r\mathbb{Z}\big), \qquad [\vec{g}] \longmapsto \big(\sigma \mapsto \sigma(\vec{u}) \ominus \vec{u}\big),$$

where $\vec{u} \in W_r(M^s)$ is any solution of $\wp\vec{u} = \vec{g}$. Under the correspondence between characters and torsors (Corollary 14), the class $[\vec{g}]$ corresponds to the $\mathbb{Z}/p^r\mathbb{Z}$-torsor

$$\operatorname{Spec}\; M[\vec{X}]/\big(\wp\vec{X} \ominus \vec{g}\big)$$

of Lemma 40. Moreover, for a class of order p^s in $W_r(M)/\wp W_r(M)$, the fixed field $M(\vec{u}) := M(u_0, \dots, u_{r-1})$ of $\ker \delta[\vec{g}]$ is a cyclic extension of M of degree p^s; the torsor is connected if and only if $[\vec{g}]$ has order p^r, in which case $\operatorname{Spec} M[\vec{X}]/(\wp\vec{X} \ominus \vec{g}) \cong \operatorname{Spec} M(\vec{u})$ is (the spectrum of) a cyclic field extension of degree p^r. Every cyclic extension of M of degree dividing p^r arises this way.

Sketch of proof. Surjectivity of $\wp$ on $W_r(M^s)$ and the computation of its kernel follow from the triangular equations in the proof of Lemma 40: each component of a preimage is obtained by solving a separable equation $X_n^p - X_n = (\dots)$, and $F\vec{u} = \vec{u}$ holds if and only if $u_n^p = u_n$ for all n, i.e. $\vec{u} \in W_r(\mathbb{F}_p)$. Given $\vec{g}$ and a solution $\wp\vec{u} = \vec{g}$, for $\sigma \in \Gamma$ we have $\wp(\sigma\vec{u} \ominus \vec{u}) = \sigma\vec{g} \ominus \vec{g} = 0$ (using (1.4) and that $W_r(\sigma)$ commutes with $\wp$), so $\sigma\vec{u} \ominus \vec{u} \in W_r(\mathbb{F}_p)$; one checks it is a continuous homomorphism in σ (the kernel is fixed pointwise by Γ), independent of the choices modulo nothing and of the representative $\vec{g}$ modulo $\wp W_r(M)$. Injectivity: if $\delta[\vec{g}] = 0$ then $\vec{u}$ is Γ-invariant, hence lies in $W_r(M)$ by (1.4), so $\vec{g} = \wp\vec{u} \in \wp W_r(M)$. Surjectivity of δ is the additive Hilbert 90: $H^1(\Gamma, W_r(M^s)) = 0$, by induction on r along the exact sequence $0 \to M^s \xrightarrow{V^{r-1}} W_r(M^s) \xrightarrow{t} W_{r-1}(M^s) \to 0$ of Γ-modules (Lemma 39(i)) and the classical case $H^1(\Gamma, M^s) = 0$. The remaining statements follow from Theorem 13 applied to the character $\delta[\vec{g}]$, whose image has the same order as the class $[\vec{g}]$. For details see [Tho05, §3–4] or [Rab14]. $\square$

The order of a class is partially visible in its 0-th component:

Corollary 42. *Let $M \supseteq \mathbb{F}_p$ be a field and $\vec{g} \in W_r(M)$ with $g_0 = 0$. Then $p^{r-1}[\vec{g}] = 0$ in $W_r(M)/\wp W_r(M)$; equivalently, the cyclic extension attached to $[\vec{g}]$ has degree at most p^{r-1}. Consequently, if $[\vec{g}]$ has order p^r (e.g. if $\vec{g}$ defines a connected torsor), then* every *representative of the class has non-zero* 0*-th component.*

Proof. By (1.6), $p^{r-1}\vec{g} = (0, \ldots, 0, g_0^{p^{r-1}}) = 0$ already in $W_r(M)$. □

Remark. When $M = k((x))$ with k algebraically closed, every finite separable extension L/M is totally ramified: the residue field extension is trivial (as k is algebraically closed) and $\mathcal{O}_L$ is a complete discrete valuation ring, so $[L : M] = ef = e$. Thus a class of order p^r in $W_r(k((x)))/\wp$ corresponds to a cyclic, *totally ramified* extension of degree p^r.

Isobaricity of the Witt addition polynomials

The following homogeneity property of the universal addition polynomials is what makes degree estimates on Witt sums possible: even though the carry polynomials C_n of (1.5) are complicated, they are *isobaric* for the weighting in which the j-th component has weight p^j.

Lemma 43 (isobaricity of Witt addition)**.** *Assign to the variables X_j and Y_j the weight p^j. Then the n-th addition polynomial S_n is isobaric of weight p^n: every monomial of S_n has total weight exactly p^n. The same holds for each component of a d-fold iterated Witt sum $\bigoplus_{i=1}^{d} \vec{a}_i$, viewed as a universal polynomial in the components $a_{i,j}$ of weight p^j.*

Proof. The n-th ghost polynomial $w_n = \sum_{j \le n} p^j X_j^{p^{n-j}}$ is isobaric of weight p^n for this weighting. The recursion

$$p^n S_n = \sum_{j \le n} p^j \big(X_j^{p^{n-j}} + Y_j^{p^{n-j}}\big) - \sum_{j<n} p^j S_j^{p^{n-j}}$$

preserves isobaricity of weight p^n by induction on n: each S_j is isobaric of weight p^j, so $S_j^{p^{n-j}}$ is isobaric of weight p^n. Isobaricity persists under iterated substitution of Witt sums into Witt sums (substituting an isobaric polynomial of weight p^j for a variable of weight p^j preserves weights), giving the statement for d-fold sums. □

Ramification breaks: the Schmid–Witt formula

Finally we state the generalization of the ramification formula of Theorem 31 to Artin–Schreier–Witt extensions. As in the classical case, the formula reads off the largest upper-numbering ramification break from the pole orders of a defining Witt vector — provided the representative is normalized so that no interference between the levels can occur. The relevant normalization is the following.

Definition 44. Let $K = k((x))$. For $g \in K$ put $m(g) := \max\big(0, -v_x(g)\big)$ (the pole order of g, or 0 if g is regular). A vector $\vec{g} = (g_0, \ldots, g_{r-1}) \in W_r(K)$ is **reduced** if for every j, either $m(g_j) = 0$ or $p \nmid m(g_j)$.

Theorem 45 (Schmid–Witt break formula)**.** *Let $K = k((x))$ with k perfect of characteristic p, and let L/K be a cyclic totally ramified extension of degree p^r whose Artin–Schreier–Witt class (Theorem 41) is represented by a* reduced *vector $\vec{g} \in W_r(K)$, with $m_j := m(g_j)$. Then the largest upper-numbering ramification break of L/K equals*

$$u \;=\; \max_{0 \le j \le r-1} \, p^{\,r-1-j}\, m_j.$$

This is due, for finite residue fields, to Kanesaka–Sekiguchi [KS79] and Thomas [Tho05] (building on Schmid [Sch37] for $r = 1$), and for arbitrary perfect residue fields to Elder–Keating [EK25]. We use it as stated, in one direction only: *evaluating the formula on one reduced representative computes* u; no minimization over representatives is needed.

Remark. Reducedness is exactly what makes the formula exact. If two levels $j < j'$ with $m_j, m_{j'} \neq 0$ satisfied $p^{r-1-j} m_j = p^{r-1-j'} m_{j'}$, then $m_{j'} = p^{j'-j} m_j$ would be divisible by p, contradicting Definition 44; hence the maximum is attained at a unique level, and no cancellation of breaks between levels can occur.

Remark. For $r = 1$, a reduced vector is a single element $g \in K$ whose pole order m_0 is zero or prime to p, and Theorem 45 recovers Theorem 31: the largest (indeed the only) upper break is m_0. The existence of reduced representatives — and, over $K = k((x))$, of reduced representatives whose components are polynomials in x^{-1} without constant term — is not part of the general theory quoted here; it will be proved where it is used, by induction on r through the truncation maps of Lemma 39.

1.1.7 Algebraic Geometry

In this section we group together some general theorems in algebraic geometry that we will be employing throughtout the text. All schemes are assumed to be locally of finite type.

Theorem 46. *Let $f : X \to Y$ be a finite flat map between integral schemes, of finite type over a field k. if $Z \subset X$ is a prime divisor with generic point η_Z, then $f(Z) \subset X$ is a prime divisor with generic point $\eta_{f(Z)}$ satisfying $f(\eta_Z) = \eta_{f(Z)}$*

Proof. f is finite hence proper hence closed so $f(Z)$ is closed subset of Y, it is irreducible as the image of an irreducible. Since $Z = \overline{\{\eta_Z\}}$ we get:

$$\{f(\eta_Z)\} \subset f(Z) = f(\overline{\{\eta_Z\}}) \subseteq \overline{f(\{\eta_Z\})} = \overline{\{f(\eta_Z)\}}$$

And since $f(Z)$ is closed we get $f(Z) = \overline{\{f(\eta_Z)\}}$.

For flat map of integral schemes we have for every $x \in X$, $y = f(x)$ the dimension formula:

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X_y,x})$$

And since $\dim(\mathcal{O}_{X_y,x}) = 0$ we get $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y})$ concluding that $f(Z)$ is a prime divisor as well. $\square$

A known theorem states that:

Theorem 47. *Let X, Y be two integral schemes over a field k. If X is geometrically integral then $X \times_k Y$ is integral. If both X, Y are geometrically integral, then $X \times_k Y$ is geometrically integral.*

Theorem 48. *Let C be smooth projective curve geometrically connected over a field k. Then:*

1. *$C^{(d)}$ is smooth*
2. *For every d, $C^{(d)}$ is integral.*
3. *For every d, $C^{(d)}$ is geometrically integral.*
4. *The product of every finite number of $C^{(d)}$ is geometrically integral.*

Proof. 1. Let t_i be a local parameter for C at P_i. The local ring of the product C^d at the point $(P_1, \ldots, P_d)$ is isomorphic $k[[t_1, t_2, \ldots, t_d]]$ and the local ring of the qoutient at the divisor $D = \sum P_i$ is $k[[t_1, \ldots, t_d]]^{S_d}$ which is isomorphic to $k[[t_1, \ldots, t_d]]^{S_d} \cong k[[s_1, \ldots, s_d]]$ where the s_i are the symmetric polynomials, hence this ring is regular local ring.

2. C is irreducible hence C^d is irreducible hence $C^{(d)}$ is irreducible. Since $C^{(d)}$ is smooth it is reduced.

3. By [Stacks, Tag 0366], C is geometrically integral, so it follows from the above.

4. Theorem 47

□

Blowups

Theorem 49 ([Stacks, Tag 0805])**.** *Let $X_1 \to X_2$ be a flat morphism of schemes. Let $Z_2 \subset X_2$ be a closed subscheme. Let Z_1 be the inverse image of Z_2 in X_1. Let X_i' be the blowup of Z_i in X_i. Then there exists a cartesian diagram*

$$\begin{tikzcd} X_1' \arrow[r] \arrow[d] & X_2' \arrow[d] \\ X_1 \arrow[r] & X_2 \end{tikzcd}$$

of schemes.

Theorem 50. *If X is integral then $Bl_Z(X)$ is integral.*

If X is a smooth curve, we have the following result regarding its symmetric power:

Proposition 51. *Let $f : X \to S$ be a smooth morphism of relative dimension 1. Suppose f is flat and Zariski-locally quasi-projective. Then the relative symmetric power $X_S^{(d)} = Sym_S^d(X)$ is smooth over S of relative dimension d.*

Proof. Since $f : X \to S$ is smooth, the d-fold product X_S^d is smooth over S. By Proposition 20, the quotient $X_S^{(d)}$ is flat over S. Smoothness is a fiberwise property for flat morphisms of finite presentation (cf. [Stacks, Tag 01V8]); thus, it suffices to verify the smoothness of the fibers. For every $s \in S$, the fiber of the symmetric power is given by:

$$(\mathrm{Sym}_S^d(X))_s \cong \mathrm{Sym}_{\kappa(s)}^d(X_s)$$

Consequently, we may reduce to the case where $S = \mathrm{Spec}(k)$ for a field k. Since smoothness is preserved under base change to the algebraic closure, we may further assume $k = \bar{k}$. Let $z = \sum_{i=1}^r d_i \cdot x_i \in X^{(d)}$ be a point represented by an effective cycle of degree d, where the points $x_i \in X(k)$ are distinct and $\sum d_i = d$. Choose pairwise disjoint Zariski-open subsets $U_i \subset X$ such that $x_i \in U_i$. Let $W \subset X^{(d)}$ be the open subset consisting of cycles whose support is contained in $\bigcup U_i$ and which meet each U_i with degree exactly d_i. There is a canonical isomorphism:

$$W \cong \prod_{i=1}^r U_i^{(d_i)}$$

Thus, it suffices to show that each $U_i^{(d_i)}$ is smooth of dimension d_i at the point $d_i \cdot x_i$.

We may therefore restrict our attention to the "worst" case: the diagonal point $z = d \cdot x$. Let $R = \mathcal{O}_{X,x}$ be the local ring of the curve at x. Since X is a smooth curve over k, R is a regular local ring of dimension 1, i.e., a discrete valuation ring. Its completion is $\hat{R} \cong k[[t]]$. The completed local ring of the product X^d at the point $(x, \dots, x)$ is:

$$\hat{\mathcal{O}}_{X^d,(x,\dots,x)} \cong \hat{R} \hat{\otimes}_k \dots \hat{\otimes}_k \hat{R} \cong k[[t_1, \dots, t_d]]$$

where t_i is the uniformizer for the i-th copy of X. The symmetric group S_d acts on this power series ring by permuting the variables t_i. By the *Fundamental Theorem of Symmetric Polynomials*, the ring of invariants is:

$$\left(k[[t_1, \dots, t_d]]\right)^{S^d} = k[[e_1, \dots, e_d]]$$

where e_j is the j-th elementary symmetric power series in the variables t_i.

The ring $k[[e_1, \dots, e_d]]$ is a formal power series ring in d variables. A fundamental theorem in commutative algebra states that a local ring is regular if and only if its completion is a formal power series ring over a field. Since $\hat{\mathcal{O}}_{X^{(d)},z} \cong k[[e_1, \dots, e_d]]$, the local ring $\mathcal{O}_{X^{(d)},z}$ is a regular local ring of dimension d. This implies that $X^{(d)}$ is smooth over k of dimension d, completing the proof. □

Throughout this chapter let C be a projective smooth geometrically connected curve over k. Let $\mathfrak{m} = \sum n_i P_i$ be a fixed modulus. Let $U = C \setminus \mathfrak{m}$. For any modulus $\mathfrak{n} \subset \mathfrak{m}$, we define $Z_\mathfrak{n}$ as the closed subscheme of $C^{(\deg \mathfrak{n})} = \mathrm{Sym}_k^{\deg \mathfrak{n}}(C)$ defined by $\mathfrak{n}$ as a point of $C^{(\deg \mathfrak{n})}$ We then define $X_\mathfrak{n}$ as the blowup of $C^{(\deg \mathfrak{n})}$ at $Z_\mathfrak{n}$, and we denote by $E_\mathfrak{n} = Z_\mathfrak{n} \times_{C^{(d)}} X_\mathfrak{n}$ the exceptional divisor of this blowup, it is irreducible of codimension 1. We denote by $\eta_\mathfrak{n}$ the generic point of $E_\mathfrak{n}$.

Diagrammatically:

$$\begin{tikzcd} \overline{\{\eta_\mathfrak{n}\}} = E_\mathfrak{n} \arrow[r, hook] \arrow[d] & X_\mathfrak{n} \arrow[d, "\pi_\mathfrak{n}"] \\ Z_\mathfrak{n} \arrow[r, hook] & C^{(\deg \mathfrak{n})} \end{tikzcd} \tag{1.8}$$

Our main goal in this chapter is to prove Theorem 3:

Theorem 52. *Let G be a finite abelian group. And let $\mathcal{P} \to U$ be a G-torsor in $U_{\acute{E}t}$ with ramification bounded by $\mathfrak{m} = \sum n_i P_i$. Let $\mathfrak{n} = n_i P_i \subset \mathfrak{m}$ be a sub modulus of $\mathfrak{m}$. Then $\mathcal{P}^{[\deg \mathfrak{n}]}$ is tamely ramified at $\eta_\mathfrak{n}$.*

Our approach proceeds in three stages. First, we establish the result for the cyclic case $G = \mathbb{Z}/p\mathbb{Z}$ and for groups G where $\gcd(|G|, p) = 1$. Next, we extend the proof to cyclic p-groups $G = \mathbb{Z}/p^r\mathbb{Z}$. Finally, for a general finite abelian group G, we conclude by applying the structure theorem for finite abelian groups and Proposition 15 to decompose the torsor into these constituent cases.

1.2 Ramification of G-Torsors

Generally speaking, assume we are given a locally Noetherian scheme X, a dense open $U \subset X$, a finite étale morphsm $f : Y \to U$ and a prime divisor $Z \subset X$ such that $Z \cap U = \emptyset$ and the local ring $\mathcal{O}_{X,\xi}$ of X at the generic point ξ of Z is a discrete valuation ring. Setting $K_\xi = \mathrm{Frac}(\mathcal{O}_{X,\xi})$ we obtain a cartesian square

$$\begin{tikzcd} \mathrm{Spec}(K_\xi) \arrow[r] \arrow[d] & U \arrow[d] \\ \mathrm{Spec}(\mathcal{O}_{X,\xi}) \arrow[r] & X \end{tikzcd}$$

of schemes. In particular, we see that $Y \times_U \mathrm{Spec}(K_\xi)$ is the spectrum of a finite separable algebra L_ξ/K_ξ.

Definition 53. We say:

1. Y is *unramified* over X at (Z, ξ) resp. Y is *tamely ramified* over X at (Z, ξ) if L_ξ/K_ξ is unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$ (Definition 27)
2. Y is unramified over X in codimension 1, resp. Y is tamely ramified over X in codimension 1 if L_ξ/K_ξ is unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$ for every (Z, ξ) as above.

More precisely, we decompose L_ξ into a product of finite separable field extensions of K_ξ and we require each of these to be unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$.

Now back to our orginial assumptions, C is projective smooth geometrically connected curve over a field k and $U = C \setminus \mathfrak{m}$. Every (Z, ξ) is now a closed point P and $\mathcal{O}_{C,P}$ is a discrete valuation ring. Assume $f : Y \to U$ is actually G-torsor $\mathcal{P} \to U$ in $U_{\text{Ét}}$ for G finite abelian group.

By passing to the completion $\widehat{\mathcal{O}}_{C,P}$, we obtain the complete local field $\widehat{K}_P$. Restricting the torsor $\mathcal{P}$ to the punctured formal neighborhood $\mathrm{Spec}(\widehat{K}_P)$ allows us to study the ramification in a strictly local context. This restriction yields a G-torsor over a field, which necessarily decomposes into a disjoint union of spectra of finite separable field extensions

$$\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)} \cong \bigsqcup \mathrm{Spec}(F)$$

such that:

1. These extensions $F/\widehat{K}_P$ are all isomorphic and Galois.
2. The Galois group $H = \mathrm{Gal}(F/\widehat{K}_P)$ identifies with the stabilizer of any connected component of the pullback $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$, which is a subgroup of G.
3. The number of such components is then given by the index $[G : H]$.

Definition 54. We say that the extension $F/\widehat{K}_P$ *has ramification bounded by* r if the ramification group H^r is trivial. We say $\mathcal{P}$ *has ramification bounded by* r *at* P if the corresponding local field extensions $F/\widehat{K}_P$ satisfy this condition.

Definition 55. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. We say the G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if, for each point P_i, the local restriction $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_{P_i})}$ has ramification bounded by the coefficient n_i.

This local property remains invariant under base change to the separable closure. Let $\bar{s} = \mathrm{Spec}(\bar{k}) \to \mathrm{Spec}(k)$ be a geometric point. The higher ramification groups for $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$ are isomorphic to those of the geometric restriction $\mathcal{P}_{\bar{k}}$ at any point $\bar{x}$ lying over P. Thus an equivalent defintion:

Definition 56. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. The G-torsor $\mathcal{P} \to U$ has **ramification bounded by $\mathfrak{m}$** if for every geometric point $\bar{x}$ of $\mathfrak{m}$ (with image $\bar{s}$ in $\mathrm{Spec}(k)$), the restriction of $\mathcal{P}$ to

$$\mathrm{Spec}(\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}) \times_{C_{\bar{k}}} U_{\bar{k}}$$

has ramification bounded by the multiplicity of $\mathfrak{m}_{\bar{s}}$ at $\bar{x}$ in the sense of Definition 54.

These definitions are equivalent because $\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}} \cong \widehat{\mathcal{O}}_{C,P} \otimes_k \bar{k}$. Furthermore, the notions of tame ramification and unramifiedness derived here coincide with the codimension-1 criteria in Definition 53.

There is another equiviliant formulation through the language of characters. The group G, being a finite abelian group over k, is étale. Consequently, the correspondence between G-torsors and the fundamental group allows us to describe G-torsors via characters. Specifically, there is an isomorphism: (Corollary 14)

$$\mathrm{Hom}_{\text{cont.}}(\pi_1^{\text{ét}}(U, \overline{x}), G) \xrightarrow{\cong} \mathrm{Tors}(U_{\text{ét}}, G)$$

In our local setting, where we consider the restriction to the complete local field $\widehat{K}_P$, the fundamental group identifies with the absolute Galois group $G_{\widehat{K}_P} := \mathrm{Gal}(\widehat{K}_P^{\text{sep}}/\widehat{K}_P)$. Thus, the restricted G-torsor $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$ corresponds to a unique continuous homomorphism:

$$\rho : G_{\widehat{K}_P} \to G$$

Under this correspondence, the torsor $\mathcal{P}$ has ramification bounded by r at P if and only if the homomorphism ρ trivializes the r-th ramification group in the upper numbering:

$$\rho(G^r_{\widehat{K}_P}) = \{1\}$$

Basic Properties of Ramification of G-Torsors

We now prove some elementary lemmas regarding the ramification of G-torsors.

Lemma 57. *Let G be a finite abelian group and X be a locally Noetherian scheme over a field k. Let $U \subset X$ be a dense open subset and let (Z, ξ) be a prime divisor in the complement $X \setminus U$. Assume $\mathcal{P}_1$ and $\mathcal{P}_2$ are two G-torsors on $U_{\acute{e}t}$, Such that $\mathcal{P}_1$ has ramification bounded by r_1 at (Z, ξ), and $\mathcal{P}_2$ has ramification bounded by r_2 at (Z, ξ). Then their product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ has ramification bounded by $max(r_1, r_2)$ at (Z, ξ).*

Proof. Let $A = \widehat{\mathcal{O}_{X,\xi}}$ and let $K = Frac(A)$. Let $\rho_1, \rho_2 : G_K \to G$ be the associated continious homomorphisms corresponding to the G-torsors $\mathcal{P}_1|_{\mathrm{Spec}(K)}$, $\mathcal{P}_2|_{\mathrm{Spec}(K)}$. We finish by noticing that the associated character of $(\mathcal{P}_1 \otimes^G \mathcal{P}_2)|_{\mathrm{Spec}(K)}$ is $\rho = \rho_1 + \rho_2$. □

Lemma 58. *Let $f : X \to Y$ be a morphism of locally Noetherian schemes. Let $U_X \subset X$ and $U_Y \subset Y$ be dense open subschemes such that $f^{-1}(U_Y) \subset U_X$.*

Let (Z_X, ξ_X) and (Z_Y, ξ_Y) be prime divisors of X and Y such that $Z_X \cap U_X = \emptyset$ and $Z_Y \cap U_Y = \emptyset$. Suppose $f(\xi_X) = \xi_Y$ and that f is étale at ξ_X. Let $\mathcal{P}$ be a G-torsor over U_Y, and let $f^{-1}\mathcal{P}$ be its pullback to $f^{-1}(U_Y) \subset U_X$. Then, the ramification of $f^{-1}\mathcal{P}$ at ξ_X is bounded by r if and only if the ramification of $\mathcal{P}$ at ξ_Y is bounded by r.

Proof. The boundedness of ramification is determined by the behavior of the torsor over the completion of the local rings at the generic points. Let $A = \mathcal{O}_{Y,\xi}$ and $B = \mathcal{O}_{X,\xi_X}$ be the discrete valuation rings at the generic points, with fraction fields K and L respectively. Since f is étale at ξ_X, the map $A \to B$ is a flat, unramified (hence weakly unramified) local homomorphism. Consequently, the extension of completions $\widehat{L}/\widehat{K}$ is a finite unramified extension of complete discretely valued fields. We finish by noticing that the upper numbering filtration on the absolute Galois group is compatible with unramified base change.[3] $\square$

Let C be a smooth, proper (projective) curve over a field k, and let $U \subset C$ be a dense open subset. Let G be a finite abelian group. and let $\pi_U : U' \to U$ be a G-torsor in $U_{\text{Ét}}$. Let C' be the normal proper model of U' over k (i.e. the unique smooth proper curve which has U' as a dense open subset). Let $\pi : C' \to C$ be the induced morphism. (which is finite generically étale over U). Let $U'^{(d)}$ be the standard d-th symmetric product of U' as a scheme, and let $U'^{[d]}$ be the torsor-theoetric symmetric product, by Corollary 23 we have a canonical morphism $q_U : U'^{(d)} \to U'^{[d]}$. Note that $C'^{(d)}$ is smooth and finite over $C^{(d)}$, hence it is the normalization of $C^{(d)}$ inside $K(C'^{(d)})$. $U'^{(d)}$ is a dense open subset of $C'^{(d)}$, hence $C'^{(d)}$ is a normal proper model of $U'^{(d)}$. Let $C'^{[d]}$ be the normalization of $C^{(d)}$ in the function field $K(U'^{[d]})$, then q_U extends to a map $q : C'^{(d)} \to C'^{[d]}$, and we have the following diagrams:

$$\begin{tikzcd} U' \arrow[r, hook] \arrow[d, "\pi_U"'] & C' \arrow[d, "\pi"] \\ U \arrow[r, hook] & C \end{tikzcd} \qquad \begin{tikzcd} U'^{(d)} \arrow[r, hook] \arrow[d, "q_U"'] & C'^{(d)} \arrow[d, "q"] \\ U'^{[d]} \arrow[r, hook] \arrow[d] & C'^{[d]} \arrow[d] \\ U^{(d)} \arrow[r, hook] & C^{(d)} \end{tikzcd}$$

Lemma 59. *In the context of the preceding discussion, the morphism $U'^{(d)} \to U'^{[d]} \hookrightarrow C'^{[d]}$ is unramified in codimension* 1 *(see Definition 53).*

Proof. Let $C' \setminus U' = \{x_1, \dots, x_m\}$. For each $x \in C' \setminus U'$, define the locus

$$Z_x = \{D \in C'^{(d)} \mid x \in \mathrm{Supp}(D)\}.$$

Each Z_x is a prime divisor in the symmetric product $C'^{(d)}$, and the boundary $C'^{(d)} \setminus U'^{(d)}$ is precisely the finite union $\bigcup_{i=1}^m Z_{x_i}$.

Let η be the generic point of Z_x. It suffices to show that the local ring extension $\mathcal{O}_{C'^{[d]},q(\eta)} \hookrightarrow \mathcal{O}_{C'^{(d)},\eta}$ is weakly unramified (Definition 26).

Let $p : C'^d \to C'^{(d)}$ be the natural quotient. Consider the closed subvariety

$$H = \{x\} \times C' \times \cdots \times C' \subset C'^d.$$

[3]Specifically, let $G_K = \mathrm{Gal}(K^{sep}/K)$ and $G_L = \mathrm{Gal}(L^{sep}/L)$. For an unramified extension, the Herbrand function is the identity, which implies that for any $r \geq 0$:

$$G_L^r = G_K^r \cap G_L$$

The subvariety H is irreducible, and its image under p is exactly Z_x. Consequently, p maps the generic point η_H of H to the generic point η of Z_x.

$$\begin{tikzcd}
\eta_H \arrow[r] \arrow[d] & H \arrow[r] \arrow[d] & C'^d \arrow[d, "p"] \\
\eta \arrow[r] \arrow[d] & Z_x \arrow[r] \arrow[d] & C^{(d)} \arrow[d, "q"] \\
q(\eta) \arrow[r] & q(Z_x) \arrow[r] & C'^{[d]}
\end{tikzcd}$$

It suffices to show that the local ring extension $\mathcal{O}_{C'^{[d]},q(\eta)} \hookrightarrow \mathcal{O}_{C'^d,\eta_H}$ is weakly unramified. Let $R = \mathcal{O}_{C'^d,\eta_H}$. And let $A = \mathcal{O}_{C'^{[d]},q(\eta)}$.

Recall that $C'^{[d]}$ is the quotient of C'^d by the finite group $\Xi = K_G \rtimes S_d$ (see Corollary 23), and the morphism $q \circ p : C'^d \to C'^{[d]}$ is *branched Galois Cover* (finite, surjective map of normal integral schemes $X \to Y$, with function field extension $K(X)/K(Y)$ Galois).

Hence the decomoposition group of η_H in Ξ and the inertia group of η_H in Ξ are defined:

$$D_{\eta_H} = \{g \in \Xi \mid g(\eta_H) = \eta_H\}$$

$$I_{\eta_H} = \{g \in D_{\eta_H} \mid g^*(u) \equiv u \pmod{\mathfrak{m}_R} \text{ for all } u \in R\}$$

Claim: The inertia group I_{η_H} is trivial.

Let $K = K(C')$ be the function field of C', then $K(C'^d) = K^{\otimes_k d} \cong Frac(K_1 \otimes_k K_2 \otimes_k \cdots \otimes_k K_d)$ Where K_i is the copy of K corresponding to the i-th factor of C'^d. The closed subscheme H corresponds to the prime ideal $\mathfrak{p}_H \subset \mathcal{O}_{C'^d}$, and $R = \mathcal{O}_{C'^d,\mathfrak{p}_H}$. The residue field of R is exactly $K(H) \cong Frac(k(x) \otimes_k K_2 \otimes_k \cdots \otimes_k K_d)$ where $k(x)$ is the residue field of x.

The action of $\gamma = (g_1, \ldots, g_d, \sigma) \in \Xi$, via the pullback (γ^*) on $K(C'^d)$ does two things (in that order):

1. It permutes the tensor factors according to σ. A function strictly in K_i is mapped to a function in $K_{\sigma(i)}$.
2. It applies the field automorphism $g_i^* \in \mathrm{Aut}(K/k)$ to the respective factors.

Let $\gamma \in I_{\eta_H} \subset \Xi$, Then the induced automorphism $\bar{\gamma}^*$ on $K(H)$ must be the identity map. This implies that σ must fix i for every $i \geq 2$ (The K_i are linearly independent), and hence $\sigma = id$. The trivial action of $\bar{\gamma}^*$ restricts to a trivial action of g_i^* on K_i for every $i \geq 2$, and hence $g_i = 1$ for every $i \geq 2$ (since the action of G on $K(C')$ is faithful). Finally, since $g_1 g_2 \cdots g_d = 1$, we also have $g_1 = 1$. Since the inertia group I_{η_H} is trivial, the local ring extension $A \hookrightarrow R$ is weakly unramified. (See for example [Stacks, Tag 09E3], [Stacks, Tag 0BTF])

□

1.3 Proof of Theorem 3

We argue that it is enough to prove Theorem 52 for $C = \mathbb{P}^1_k$, $\mathfrak{n} = d \cdot 0$, and $\mathcal{P} \to \mathbb{G}_m = \mathbb{P}^1_k \setminus \{0, \infty\}$ a G-torsor given by an explicit global equation. Informally, this is because *the ramification of*

$\mathcal{P}^{[d]}$ *at* $\eta_{\mathfrak{n}}$ *is a formal–local invariant of* $\mathcal{P}$ *at* P: it depends only on the completed local ring $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$ together with the restriction of $\mathcal{P}$ to the punctured formal disc $\operatorname{Spec}(k((x)))$, since all of the constructions involved — self products, quotients by finite groups, and blowups — commute with the flat base change to the completion. We emphasize that this is not a general principle that can simply be quoted: it is a statement about the particular constructions at hand, and it is proved by checking each construction separately. This is the content of Lemma 60 and Corollary 61 below. The second ingredient of the reduction is Lemma 62: every local torsor class appearing in Theorem 52 is realized by an explicit torsor over $\mathbb{G}_m \subset \mathbb{P}^1_k$.

Throughout this section we assume that k is algebraically closed; in particular every closed point of C is k-rational, and $\mathfrak{n} = dP$ with $P \in C(k)$ and $d = \deg \mathfrak{n}$.

Remark. The assumption that k is algebraically closed is harmless in our situation, for the following reasons.

1. **(Base change to $\bar{k}$.)** Assume k is perfect, so that $\bar{k} = k^{sep}$. The formation of $C^{(d)}$ (a quotient by a finite group), of the blowup $X_{\mathfrak{n}}$ along $Z_{\mathfrak{n}}$, and of $\mathcal{P}^{[d]}$ (Corollary 23) commutes with the flat base change $\operatorname{Spec}(\bar{k}) \to \operatorname{Spec}(k)$. Since $E_{\mathfrak{n}} \cong \mathbb{P}_k^{\deg \mathfrak{n} - 1}$ is geometrically irreducible, $(E_{\mathfrak{n}})_{\bar{k}}$ is irreducible and its generic point $\bar{\eta}$ lies over $\eta_{\mathfrak{n}}$; moreover $\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}} \to \mathcal{O}_{(X_{\mathfrak{n}})_{\bar{k}},\bar{\eta}}$ is a weakly unramified extension of discrete valuation rings with separable residue field extension. The proof of Lemma 58 uses only these two properties of the extension (there the extension was moreover finite étale), so it applies verbatim: the ramification of $(\mathcal{P}_{\bar{k}})^{[d]} = (\mathcal{P}^{[d]})_{\bar{k}}$ at $\bar{\eta}$ is bounded by r if and only if the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is. Hence unramifiedness and tameness at $\eta_{\mathfrak{n}}$ may be checked after base change to $\bar{k}$.
2. **(Rationality of P — a caveat.)** If P is a closed point with $\kappa(P) \neq k$, then after base change to $\bar{k}$ the point $Z_{\mathfrak{n}}$ corresponds to the divisor $\sum_{\bar{P} \mapsto P} n\bar{P}$ on $C_{\bar{k}}$, whose support consists of $[\kappa(P) : k]$ distinct points, whereas the computation below treats a modulus supported at a single rational point. We therefore prove Theorem 52 under the standing assumption that k is algebraically closed; this suffices for our purposes, since in the proof of Theorem 75 the field is assumed algebraically closed before the present theorem is invoked.

We now formulate the formal–local invariance precisely. Fix $P \in C(k)$ lying in the support of $\mathfrak{m}$, set $\mathfrak{n} = dP$, and fix a uniformizer x of $\mathcal{O}_{C,P}$; it induces an isomorphism $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$. The scheme $\operatorname{Spec}(k[[x]])$ has two points — the closed point, mapping to P, and the generic point, mapping to the generic point of C — so the punctured formal disc $\operatorname{Spec}(k((x)))$ maps to U, and we denote by

$$\mathcal{P}_x := \mathcal{P} \times_U \operatorname{Spec}(k((x)))$$

the corresponding restriction of $\mathcal{P}$; it is a G-torsor over $\operatorname{Spec}(k((x)))$. Let $e_1, \ldots, e_d \in k[[x_1, \ldots, x_d]]$ be the elementary symmetric polynomials in $x_1, \ldots, x_d$, and set

$$V = \operatorname{Spec}\Big(k[[e_1, \ldots, e_d]][e_d^{-1}]\Big), \qquad W = \operatorname{Spec}\Big(k[[x_1, \ldots, x_d]][(x_1 \cdots x_d)^{-1}]\Big).$$

The inclusion $k[[e_1, \ldots, e_d]] \subset k[[x_1, \ldots, x_d]]$ induces a finite flat morphism $\hat{r} : W \to V$ (note that inverting $e_d = x_1 \cdots x_d$ inverts each x_i), and for each i we let

$$q_i : W \to \operatorname{Spec}(k((x))), \qquad x \mapsto x_i.$$

The group S_d acts on W over V by permuting $x_1, \ldots, x_d$, and $q_{\sigma(i)} = q_i \circ \sigma$.

Lemma 60 (Formal–local invariance). *In the above notation:*

1. *The choice of x induces an isomorphism $\widehat{\mathcal{O}_{C^{(d)},Z_{\mathfrak{n}}}} \cong k[[e_1,\dots,e_d]]$ carrying the maximal ideal to $(e_1,\dots,e_d)$, such that the induced morphism $c : \operatorname{Spec}(k[[e_1,\dots,e_d]]) \to C^{(d)}$ is flat and satisfies $c^{-1}(U^{(d)}) = V$; we write $c_V : V \to U^{(d)}$ for the restriction.*

2. *There is an induced isomorphism $\widehat{\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}}} \cong \kappa[[e_d]]$, where $\kappa = k(\frac{e_1}{e_d},\dots,\frac{e_{d-1}}{e_d})$. In particular $\widehat{K}_{\eta_{\mathfrak{n}}} := \operatorname{Frac}(\widehat{\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}}}) \cong \kappa((e_d))$, and the canonical morphism $\operatorname{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}}) \to U^{(d)}$ factors as $\operatorname{Spec}(\kappa((e_d))) \to V \xrightarrow{c_V} U^{(d)}$, where the first map does not depend on $(C,\mathcal{P})$.*

3. *There is a canonical isomorphism of G-torsors over V*
$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \;\cong\; \Xi \backslash \Big(q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x \Big).$$

Consequently, the G-torsor $\mathcal{P}^{[d]}|_{\operatorname{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ — and hence whether the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is bounded by r (resp. tame, resp. unramified) — depends only on d and on the isomorphism class of the G-torsor $\mathcal{P}_x$ over $k((x))$, and not otherwise on the pair $(C,\mathcal{P})$.

Proof. **(1)** Since P is a k-rational point of the smooth curve C, the uniformizer x induces $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$. As completion at rational points takes products of k-varieties to completed tensor products, we get $\widehat{\mathcal{O}_{C^d,(P,\dots,P)}} \cong k[[x]]\widehat{\otimes}_k \cdots \widehat{\otimes}_k k[[x]] = k[[x_1,\dots,x_d]]$. Write $A = \mathcal{O}_{C^{(d)},Z_{\mathfrak{n}}}$ and $B = \mathcal{O}_{C^d,(P,\dots,P)}$. Since k is algebraically closed, a closed point $(Q_1,\dots,Q_d) \in C^d$ maps to $Z_{\mathfrak{n}} = dP$ under the finite projection $r : C^d \to C^{(d)}$ if and only if $Q_1 = \cdots = Q_d = P$; hence $(P,\dots,P)$ is the unique point of C^d over $Z_{\mathfrak{n}}$, $B = (r_*\mathcal{O}_{C^d})_{Z_{\mathfrak{n}}}$ is a finite A-module, and $A = B^{S_d}$ (formation of invariants commutes with localization at invariant primes). Since $\mathfrak{m}_B$ is the unique maximal ideal of B over $\mathfrak{m}_A$, we have $\mathfrak{m}_B^N \subset \mathfrak{m}_A B \subset \mathfrak{m}_B$ for some N, so the $\mathfrak{m}_A$-adic and $\mathfrak{m}_B$-adic topologies on B agree, and therefore $B \otimes_A \hat{A} \cong \hat{B} \cong k[[x_1,\dots,x_d]]$. Now $A = B^{S_d}$ is the kernel of $B \to \prod_{\sigma \in S_d} B$, $b \mapsto (\sigma(b) - b)_\sigma$; as $\hat{A}$ is flat over A, tensoring with $\hat{A}$ preserves this kernel, so

$$\hat{A} \cong (B \otimes_A \hat{A})^{S_d} \cong k[[x_1,\dots,x_d]]^{S_d}.$$

Finally, $k[[x_1,\dots,x_d]]^{S_d} = k[[e_1,\dots,e_d]]$: the ring $k[x_1,\dots,x_d]$ is a finite free module over $k[e_1,\dots,e_d]$, the same topological argument identifies $k[[x_1,\dots,x_d]]$ with $k[x_1,\dots,x_d] \otimes_{k[e_1,\dots,e_d]} k[[e_1,\dots,e_d]]$, and taking S_d-invariants — again a kernel, preserved by the flat base change $k[e_1,\dots,e_d] \to k[[e_1,\dots,e_d]]$ — yields $k[[e_1,\dots,e_d]]$. Under these identifications the maximal ideal of $\hat{A}$ is $(e_1,\dots,e_d)$, and c is flat as the composition of the flat morphisms $\operatorname{Spec}(\hat{A}) \to \operatorname{Spec}(A) \to C^{(d)}$.

It remains to check that $c^{-1}(C^{(d)} \setminus U^{(d)}) = V(e_d)$ as sets. We have $C^{(d)} \setminus U^{(d)} = \bigcup_{Q \in \operatorname{Supp}(\mathfrak{m})} Z_Q$ with $Z_Q = \{D : Q \in \operatorname{Supp}(D)\}$, as in the proof of Lemma 59. For $Q \neq P$ the divisor Z_Q does not contain the point $Z_{\mathfrak{n}} = dP$, so its ideal is not contained in $\mathfrak{m}_A$, and $c^{-1}(Z_Q) = \emptyset$. For $Q = P$, pull back along the finite surjective morphism $\operatorname{Spec}(\hat{B}) \to \operatorname{Spec}(\hat{A})$: the preimage of Z_P in C^d is $\bigcup_i p_i^{-1}(P)$, whose trace on $\operatorname{Spec}(\hat{B}) = \operatorname{Spec}(k[[x_1,\dots,x_d]])$ is $V(x_1) \cup \cdots \cup V(x_d) = V(x_1 \cdots x_d) = V(e_d)$; by surjectivity, $c^{-1}(Z_P) = V(e_d)$ as claimed. Hence $c^{-1}(U^{(d)}) = V$, and likewise $\operatorname{Spec}(\hat{B}) \times_{C^d} U^d = W$.

(2) Blowing up commutes with flat base change, and the center $Z_{\mathfrak{n}}$ pulls back under c to the closed point $V(e_1,\dots,e_d)$; hence

$$\hat{Y} := \operatorname{Bl}_{(e_1,\dots,e_d)}\big(\operatorname{Spec} k[[e_1,\dots,e_d]]\big) \cong X_{\mathfrak{n}} \times_{C^{(d)}} \operatorname{Spec}(\hat{A}),$$

and the exceptional divisor $\hat{E} \subset \hat{Y}$ is the base change of $E_{\mathfrak{n}}$ along $\operatorname{Spec}(\hat{A}/\hat{\mathfrak{m}}) = \operatorname{Spec}(\kappa(Z_{\mathfrak{n}}))$, i.e. the projection $\mathrm{pr} : \hat{Y} \to X_{\mathfrak{n}}$ restricts to an isomorphism $\hat{E} \cong E_{\mathfrak{n}} \cong \mathbb{P}^{d-1}_k$. Let $\hat{\eta}$ be the generic point of $\hat{E}$. Then $\mathrm{pr}(\hat{\eta}) = \eta_{\mathfrak{n}}$, and the induced local homomorphism of discrete valuation rings $\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}} \to \mathcal{O}_{\hat{Y},\hat{\eta}}$ carries a uniformizer to a uniformizer (the ideal of $E_{\mathfrak{n}}$ pulls back to the ideal of $\hat{E}$) and induces an isomorphism of residue fields (both are the function field of $\mathbb{P}^{d-1}_k$); hence it induces an isomorphism on completions. The completion of $\mathcal{O}_{\hat{Y},\hat{\eta}}$ is computed exactly as for the blowup of affine space at the origin, which we carry out in detail below: on the chart $u_d \neq 0$ we have $e_i = \frac{u_i}{u_d} e_d$, the local ring at $\hat{\eta}$ is $\left(\hat{A}[\frac{u_1}{u_d}, \ldots, \frac{u_{d-1}}{u_d}]\right)_{(e_d)}$, its residue field is $\kappa = k(\frac{u_1}{u_d}, \ldots, \frac{u_{d-1}}{u_d}) = k(\frac{e_1}{e_d}, \ldots, \frac{e_{d-1}}{e_d})$, and its completion is $\kappa[[e_d]]$. Finally, e_d is invertible in $\widehat{K}_{\eta_{\mathfrak{n}}} = \kappa((e_d))$, so the composite $\operatorname{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}}) \to \operatorname{Spec}(\mathcal{O}_{\hat{Y},\hat{\eta}}) \to \hat{Y} \to \operatorname{Spec}(\hat{A})$ factors through V, giving the asserted factorization of $\operatorname{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}}) \to U^{(d)}$ through c_V.

(3) Let $h : \mathcal{P}^{\times_k d} \to U^{(d)}$ denote the canonical morphism; it is finite, being the composition of the finite étale morphism $\mathcal{P}^{\times_k d} \to U^d$ with the finite morphism $U^d \to U^{(d)}$. By Corollary 23, $\mathcal{P}^{[d]} = \Xi \backslash \mathcal{P}^{\times_k d} = \underline{\operatorname{Spec}}_{U^{(d)}}\left((h_* \mathcal{O}_{\mathcal{P}^{\times_k d}})^{\Xi}\right)$. Pushforward along the affine morphism h commutes with the flat base change $c_V : V \to U^{(d)}$, and the formation of Ξ-invariants is a kernel, hence also commutes with flat base change; therefore

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \cong \Xi \backslash \left(\mathcal{P}^{\times_k d} \times_{U^{(d)}} V\right).$$

By step (1), $U^d \times_{U^{(d)}} V \cong W$, so $\mathcal{P}^{\times_k d} \times_{U^{(d)}} V \cong \mathcal{P}^{\times_k d} \times_{U^d} W$. For each i, the composition $W \to \operatorname{Spec}(k[[x_1, \ldots, x_d]]) \to \operatorname{Spec}(k[[x]]) \to C$ — the middle map being the completion of the projection $p_i : C^d \to C$, i.e. $x \mapsto x_i$ — factors through $\operatorname{Spec}(k((x))) \to U$, since x_i is invertible on W; this composite is precisely q_i, and it commutes with $p_i : U^d \to U$. Since $\mathcal{P}^{\times_k d} = p_1^{-1}\mathcal{P} \times_{U^d} \cdots \times_{U^d} p_d^{-1}\mathcal{P}$, base changing to W gives an isomorphism

$$\mathcal{P}^{\times_k d} \times_{U^d} W \cong q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x,$$

which is equivariant for the actions of G^d (in particular of K_G) and of S_d — the latter acting on W by permuting the x_i, compatibly with $q_{\sigma(i)} = q_i \circ \sigma$. Passing to Ξ-quotients yields the displayed isomorphism.

For the final statement: the schemes V and W, the maps q_i and $\hat{r}$, the Ξ-action, and the point $\operatorname{Spec}(\kappa((e_d))) \to V$ of (2) are all constructed from d alone; by (3), the torsor $\mathcal{P}^{[d]}|_{\operatorname{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ is the pullback along $\operatorname{Spec}(\kappa((e_d))) \to V$ of a torsor constructed from $\mathcal{P}_x$ and these data. Since the ramification at $\eta_{\mathfrak{n}}$ is by definition read off from $\mathcal{P}^{[d]}|_{\operatorname{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ together with the discrete valuation ring $\kappa[[e_d]]$ (Definition 53, Definition 54), it depends only on $(\mathcal{P}_x, d)$. $\square$

Corollary 61 (Reduction to $\mathbb{P}^1$)**.** *Let C, C' be smooth projective geometrically connected curves over the algebraically closed field k, let $\mathcal{P} \to U = C \setminus \mathfrak{m}$ and $\mathcal{P}' \to U' = C' \setminus \mathfrak{m}'$ be G-torsors, let $P \in \operatorname{Supp}(\mathfrak{m})(k)$ and $P' \in \operatorname{Supp}(\mathfrak{m}')(k)$, and let x, x' be uniformizers at P, P' respectively. Assume that $\mathcal{P}_x \cong \mathcal{P}'_{x'}$ as G-torsors, compatibly with the isomorphism $k((x)) \cong k((x'))$, $x \mapsto x'$. Then for every $d \geq 1$ and every r, the ramification of $\mathcal{P}^{[d]}$ at η_{dP} is bounded by r if and only if the ramification of $\mathcal{P}'^{[d]}$ at $\eta_{dP'}$ is; in particular, the one is unramified (resp. tamely ramified) at η_{dP} if and only if the other is at $\eta_{dP'}$.*

Proof. Immediate from Lemma 60: both $\mathcal{P}^{[d]}|_{\operatorname{Spec}(\widehat{K}_{\eta_{dP}})}$ and $\mathcal{P}'^{[d]}|_{\operatorname{Spec}(\widehat{K}_{\eta_{dP'}})}$ are identified, compatibly with the discrete valuation ring $\kappa[[e_d]]$, with the pullback along $\operatorname{Spec}(\kappa((e_d))) \to V$ of $\Xi \backslash (q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x)$. $\square$

Lemma 62 (Realization over $\mathbb{G}_m$). *Let k be algebraically closed of characteristic $p > 0$, and let $\mathcal{Q}$ be a G-torsor over* $\operatorname{Spec}(k((x)))$.

1. *If $\mathcal{Q}$ is tamely ramified, then there exist an integer e prime to p, an injective homomorphism $\iota : \mathbb{Z}/e\mathbb{Z} \hookrightarrow G$, and an integer a prime to e, such that*

$$\mathcal{P}' := \iota_*\Big(\operatorname{Spec}\Big(k[x, x^{-1}][T]/(T^e - x^a)\Big)\Big)$$

is a G-torsor over $\mathbb{G}_m$, tamely ramified at 0 and ∞, with $\mathcal{P}'_x \cong \mathcal{Q}$. Here $T^e = x^a$ is regarded as a $\mathbb{Z}/e\mathbb{Z}$-torsor via a fixed primitive e-th root of unity $\zeta_e \in k$, and $\iota_*\mathcal{Q}' := (\mathcal{Q}' \times G)/(\mathbb{Z}/e\mathbb{Z})$ denotes the induced G-torsor.

2. *If $G = \mathbb{Z}/p\mathbb{Z}$ and $\mathcal{Q}$ has ramification bounded by d (Definition 54), then there exists $f \in x^{-1}k[x^{-1}]$, either $f = 0$ or $f = cx^{-m} + a_{-m+1}x^{-m+1} + \cdots + a_{-1}x^{-1}$ with $c \neq 0$, $p \nmid m$ and $m < d$, such that*

$$\mathcal{P}' := \operatorname{Spec}\Big(k[x, x^{-1}][X]/(X^p - X - f)\Big)$$

is a $\mathbb{Z}/p\mathbb{Z}$-torsor over $\mathbb{G}_m$ with $\mathcal{P}'_x \cong \mathcal{Q}$. Moreover $\mathcal{P}'$ extends to a torsor over $\mathbb{P}^1_k \setminus \{0\}$ — in particular it is unramified at ∞ — and it has ramification bounded by d at 0.

In both cases $\mathcal{P}'_x$ denotes the restriction of $\mathcal{P}'$ to $\operatorname{Spec}(k((x)))$, *as before.*

Proof. **(1)** Since k is algebraically closed, the residue field of $k((x))$ is algebraically closed, so $k((x))$ admits no non-trivial unramified extensions, and its tamely ramified extensions are exactly the Kummer extensions $k((x^{1/e}))$ with $p \nmid e$ (Theorem 30). Hence the tame quotient G^t of $\operatorname{Gal}(k((x))^{sep}/k((x)))$ is procyclic, $G^t \cong \varprojlim_{p \nmid e} \mu_e(k) \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$; fix a topological generator γ. Let $\rho : \operatorname{Gal}(k((x))^{sep}/k((x))) \to G$ be the continuous homomorphism classifying $\mathcal{Q}$. Tameness means that ρ kills the wild inertia subgroup, so ρ factors through G^t and is determined by $g := \rho(\gamma)$, whose order e is prime to p. Let $\iota : \mathbb{Z}/e\mathbb{Z} \xrightarrow{\sim} \langle g \rangle \subset G$ be given by $1 \mapsto g$, so that $\rho = \iota \circ \chi$ for a surjective character $\chi : G^t \to \mathbb{Z}/e\mathbb{Z}$. On the other hand, for $a \in \mathbb{Z}$ the equation $T^e = x^a$ defines a $\mu_e \cong \mathbb{Z}/e\mathbb{Z}$-torsor over $\operatorname{Spec}(k((x)))$, and by Kummer theory the classes of x^a, $a \in \mathbb{Z}/e\mathbb{Z}$, exhaust $\operatorname{Hom}_{\text{cont.}}(G^t, \mathbb{Z}/e\mathbb{Z}) \cong \mathbb{Z}/e\mathbb{Z}$, the class of x^a being surjective precisely when $\gcd(a, e) = 1$. Choose a with the class of x^a equal to χ; then $\mathcal{Q} \cong \iota_*(T^e = x^a)$ over $k((x))$. The same equation over $k[x, x^{-1}]$ defines a $\mathbb{Z}/e\mathbb{Z}$-torsor over $\mathbb{G}_m$ (étale since e is invertible in k), so $\mathcal{P}' := \iota_*\big(\operatorname{Spec}(k[x, x^{-1}][T]/(T^e - x^a))\big)$ is a G-torsor over $\mathbb{G}_m$ with $\mathcal{P}'_x \cong \mathcal{Q}$; it is tamely ramified at 0 and at ∞ since its local characters there factor through the tame quotients.

(2) If $\mathcal{Q}$ is trivial, take $f = 0$. Otherwise, since $\mathbb{Z}/p\mathbb{Z}$ is simple and $k((x))$ admits no non-trivial unramified extensions, $\mathcal{Q} = \operatorname{Spec}(F)$ for a cyclic extension $F/k((x))$ of degree p together with an isomorphism $\operatorname{Gal}(F/k((x))) \cong \mathbb{Z}/p\mathbb{Z}$. By Artin–Schreier theory such data correspond to non-zero classes in $k((x))/\wp(k((x)))$, where $\wp(u) = u^p - u$, via $g \mapsto k((x))[X]/(X^p - X - g)$. We claim that every class admits a representative f as in the statement:

- $\wp$ is surjective on $k[[x]]$: given $h = \sum_{i \geq 0} b_i x^i$, solve $\wp(u) = h$ with $u = \sum_{i \geq 0} u_i x^i$ by choosing u_0 with $u_0^p - u_0 = b_0$ (possible as k is algebraically closed) and setting recursively $u_i = u_{i/p}^p - b_i$, with the convention $u_{i/p} = 0$ when $p \nmid i$. Hence, we may choose a representative in $x^{-1}k[x^{-1}]$.

- If the representative has pole order m divisible by p, with leading term cx^{-m}, then subtracting $\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p}$ strictly decreases the pole order. Iterating, we reach a representative f which is either 0 (impossible, as the class is non-zero) or has pole order m prime to p.

By Theorem 31, the unique ramification jump of $F = k((x))[X]/(X^p - X - f)$ is at m, so the ramification of $\mathcal{Q}$ is bounded by d if and only if $m < d$. Finally, $f \in k[x, x^{-1}]$, so the same equation defines a finite étale $\mathbb{Z}/p\mathbb{Z}$-cover $\mathcal{P}' \to \mathbb{G}_m$ (a torsor via $X \mapsto X + 1$) with $\mathcal{P}'_x \cong \mathcal{Q}$; and writing $y = x^{-1}$, we have $f \in y\,k[y]$, so $X^p - X - f$ defines a finite étale cover of $\mathbb{A}^1_k = \operatorname{Spec}(k[y]) = \mathbb{P}^1_k \setminus \{0\}$ extending $\mathcal{P}'$, whence $\mathcal{P}'$ is unramified at ∞. □

Combining Corollary 61 with Lemma 62, in order to prove Theorem 52 for a torsor which is tamely ramified at P, or for a $\mathbb{Z}/p\mathbb{Z}$-torsor with ramification bounded by d at P, we may — and do — assume from now on that

$$C = \mathbb{P}^1_k, \qquad \mathfrak{m} = d \cdot 0 + \infty, \qquad \mathfrak{n} = d \cdot 0, \qquad U = \mathbb{G}_m,$$

and that $\mathcal{P} \to \mathbb{G}_m$ is one of the explicit torsors of Lemma 62.

We start by analyzing the blowup $\tilde{X} = \mathrm{Bl}_0(X)$ of $X = \mathbb{A}^n_k$ where $0 \in X$ be the origin. Recall that $\tilde{X} \subset X \times_k \mathbb{P}^{n-1}_k$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \cdots : u_n]$ are the homogeneous coordinates of $\mathbb{P}^{n-1}_k$. The exceptional divisor $E \subset \tilde{X}$ is the fiber over the origin, $E = \{(0, [u_1 : \cdots : u_n])\}$, which is of codimension 1 in $\tilde{X}$. Let $\eta \in E$ be the generic point of E, and let $R = \mathcal{O}_{\tilde{X},\eta}$ be the associated local ring. This ring R is a discrete valuation ring (DVR) with fraction field $K = K(X) = k(x_1, \dots, x_n)$.

On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1}x_1$. The coordinate ring is:

$$\mathcal{O}_{\tilde{X}}(U_1) = k\left[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1}\right]$$

In this chart, the generic point η corresponds to the prime ideal $\mathfrak{p}_1 = (x_1)$. Thus, the local ring is $R = k[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1}]_{(x_1)}$. The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \dots, \frac{u_n}{u_1})$, and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_1]]$$

In this local ring, x_1 is a uniformizer. Note that any x_i (for $i > 1$) can also serve as a uniformizer, as $x_i = (\frac{u_i}{u_1})x_1$ and $\frac{u_i}{u_1}$ is a unit in R.

For any monomial $M = x_1^{a_1} \dots x_n^{a_n} \in K$, we can write:

$$M = x_1^{\sum a_i} \left(\frac{u_2}{u_1}\right)^{a_2} \cdots \left(\frac{u_n}{u_1}\right)^{a_n}$$

Since the term in parentheses is a unit in R, the valuation ν_E associated with E satisfies:

$$\nu_E(M) = \sum a_i = \deg M$$

Consequently, for any polynomial $f = f_d + f_{d+1} + \cdots + f_l$, where f_i is the homogeneous part of degree i, we have $\nu_E(f) = d$ (the order of vanishing at the origin).

Our first result is:

Theorem 63. *Let $\mathcal{P} \to \mathbb{G}_m \subset \mathbb{P}^1_k$ be a G torsor which is either*

1. *tamely ramified at* 0
2. *totally ramified at* 0 *with* $G = \mathbb{Z}/p\mathbb{Z}$ *and ramificaiton bounded by* d.

Then the ramification of the G-torsor $\mathcal{P}^{[d]} \to \mathbb{G}_m^{(d)}$ at $\eta_{\mathfrak{m}}$ the generic point of $E_{\mathfrak{m}} \subset X_{\mathfrak{m}}$ is

1. *tamely ramified if $\mathcal{P}$ was tamely ramified*
2. *unramified if $\mathcal{P}$ was totally ramified with $G = \mathbb{Z}/p\mathbb{Z}$ and ramification bounded by d*

Proof. The local ring at the generic point of the excptional divisor of the blowup of the affine space at 0 point is $R = k[x_d, \frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d}]_{(x_d)}$. Where for every $i < d$, we have $x_i = \frac{u_i}{u_d} x_d$ are all uniformizers. The residue field is $\kappa(\eta) = k(\frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d})$ and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_d]]$$

In our situation, when we take symmetric product of the affine space, the situation is similiar with different coordinates if we let $e_1, \dots, e_d$ be the symmetric polynomials in $x_1, \dots x_d$ then: The local ring is $R = k[e_d, \frac{u_2}{u_d}, \dots, \frac{u_{d-1}}{u_d}]_{(e_d)}$. $e_i = \frac{u_i}{u_d} e_d$ are all uniformizers. The residue field being: $\kappa(\eta) = k(\frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d})$ and the completion: $\hat{R} = \kappa(\eta)[[s_d]]$ Note that from $e_i = \frac{u_i}{u_d} e_d$ We get $\frac{e_i}{e_d} = \frac{u_i}{u_d}$ in the fracion field. hence

$$\hat{K} = k(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{u_d})((e_d)) \tag{1.9}$$

We compute directly the extesntion of complete valued fields over the complete valued field at the generic point. Note that by Corollary 61 and Lemma 62 we can assume $\mathcal{P} = \operatorname{Spec} k[x, x^{-1}][X]/(X^n - a)$ for $a \in k[x, x^{-1}]$ or $\mathcal{P} = \operatorname{Spec} k[x, x^{-1}][X]/(X^p + X - f(x, x^{-1}))$ where $f(x, x^{-1}) = cx^{-m} + a_{-m+1}x^{-m+1} + \cdots a_{-1}x^{-1} + a_0 = cx^{-m} + f_{-m+1}(x^{-1})$ where $m < d$.

We deal with each case separately.

Artin-Schreier Extensions:

Set $R = k[x, x^{-1}]$, and $S = \operatorname{Spec} k[x, x^{-1}][X]/(X^p + X - f(x^{-1}))$ we have

$$\begin{aligned}
R^{\otimes_k d} &= k[x_1, x_1^{-1}, \dots, x_d, x_d^{-1}] \\
S^{\otimes_k d} &= k[x_1, x_1^{-1}, \dots, x_d, x_d^{-1}][X_1, \dots, X_d]/(X_1^p - X_1 - f(x_1), \dots, X_d^p - X_d - f(x_d)) \\
&= R^{\otimes_k d}[X_1, \dots, X_d]/(X_1^p - X_1 - f(x_1), \dots, X_d^p - X_d - f(x_d))
\end{aligned}$$

Next, we want to understand the ring corresponding to $p_1^{-1}(\mathcal{P}) \otimes \cdots \otimes p_d^{-1}(\mathcal{P})$ on C^d - the d'th-product of the $G = \mathbb{Z}/p\mathbb{Z}$-torsors $p_1^{-1}(\mathcal{P}), \dots, p_d^{-1}(\mathcal{P})$ on U^d. It corresponds to qoutient: $\left(p_1^{-1}(\mathcal{P}) \times \cdots \times p_d^{-1}(\mathcal{P})\right)/G^{d-1}$ where the action of G^{d-1} on the product is:

$$(g_1, \dots, g_{d-1}) \cdot (p_1, p_2, \dots, p_{d-1}, p_d) = (g_1(p_1), g_1^{-1}g_2(p_2), \dots, g_{d-2}^{-1}g_{d-1}(p_{d-1}), g_{d-1}^{-1}(p_d))$$

The affine ring corrsponding to the torsor product is $\left(S^{\otimes_k d}\right)^{G^{d-1}}$

Recall that the action of $g \in G = \mathbb{Z}/p\mathbb{Z}$ on X is $g(X) = X + g$ (g correspond to a number $0 \le g \le p-1$). So, the action of $(g_1, \dots, g_{d-1})$ on the generators $(X_1, X_2, \dots, X_{d-1}, X_d)$ Is $X_1 \mapsto X_1 + g_1$, $X_i \mapsto X_i - g_{i-1} + g_i$ for $1 < i < d$ and $X_d \mapsto X_d - g_{d-1}$. So we see that $Y = X_1 + \dots + X_d$ is invariant. Moreover $Y^P - Y - \sum_{i=1}^d f(x_i) = 0$ is irreducible degree p equation for Y, Since we are quotienting a rank p^d extesntion by a group of order p^{d-1} the resulting invaraint subring must have rank p over $R^{\otimes_k d}$, So we conclude:

$$\left(S^{\otimes_k d}\right)^{G^{d-1}} \cong R^{\otimes_k d}[Y]/(Y^p - Y - \sum_{i=1}^d f(x_i))$$

The group S_d acts on $R^{\otimes_k d} = k[x_1^{\pm 1}, x_2^{\pm 1}, \dots, x_d^{\pm 1}]$ by permuting the variables $\{x_i\}_{i=1}^d$. And since $Y = \sum_i^d X_i$ it leaves Y invaraint. The invaraint subring $(R^{\otimes_k d})^{S_d}$ is simply $k[e_1, e_2, \dots, e_d, e_d^{-1}]$ where $\{e_i\}$ are the symmetric polynomials in $x_1, \dots, x_d$:

$$e_k = \sum_{1 \le j_1 < j_2 < \dots < j_k \le d} x_{j_1} x_{j_2} \dots x_{j_k}$$

To find $\left(R^{\otimes_k d}[Y]/(Y^p - Y - \sum_{i=1}^d f(x_i))\right)^{S_d}$ its enough to express $\sum_{i=1}^d f(x_i)$ in $e_1, \dots, e_d$, this can be done with the newton polynomials, moreover, we claim the following:

Lemma 64. *Let $f(x) = cx^{-m} + a_{-m+1}x^{-m+1} + \cdots a_{-1}x^{-1} + a_0$, define $\alpha(x_1, ..., x_d) = \sum_1^d f(x_i)$, and deonte by $e_1, ..., e_d$ the elementary symmetric polynomials in $x_1, ..., x_d$. If $m < d$ then $\alpha(x_1, ..., x_d) \in k(e_1/e_d, e_2/e_d, ..., e_{d-1}/e_d)$*

Proof. Changing variables $y_i = x_i^{-1}$ for each $i \in \{1, \dots, d\}$ We get

$$\alpha = \sum_{i=1}^d f(x_i) = \sum_{i=1}^d \Big(cy_i^m + a_{-m+1}y_i^{m-1} + \dots + a_{-1}y_i + a_0\Big)$$

Rearranging the sums, we get

$$\alpha = c\sum_{i=1}^d y_i^m + a_{-m+1}\sum_{i=1}^d y_i^{m-1} + \dots + a_{-1}\sum_{i=1}^d y_i + da_0$$

Let $p_k(y_1, \dots, y_d) = \sum_{i=1}^d y_i^k$ be the k-th power sum symmetric polynomial. The expression for α is a linear combination of these power sums:

$$\alpha = cp_m(y) + a_{-m+1}p_{m-1}(y) + \dots + a_{-1}p_1(y) + da_0$$

According to the *Fundamental Theorem of Symmetric Polynomials*, any symmetric polynomial in $y_1, \dots, y_d$ can be expressed as a polynomial in the elementary symmetric polynomials $e_k(y_1, \dots, y_d)$. Since $m < d$, α is a polynomial in $e_1(y), e_2(y), \dots, e_m(y)$. ($y = (y_1, \dots, y_d)$) The elementary symmetric polynomials in $y_i = 1/x_i$ are related to the elementary symmetric polynomials in x_i as follows:

$$e_k(y_1, \dots, y_d) = \sum_{1 \le i_1 < \dots < i_k \le d} \frac{1}{x_{i_1} \dots x_{i_k}} = \frac{\sum_{1 \le j_1 < \dots < j_{d-k} \le d} x_{j_1} \dots x_{j_{d-k}}}{x_1 x_2 \dots x_d}$$

Thus,

$$e_k(y_1, \dots, y_d) = \frac{e_{d-k}(x_1, \dots, x_d)}{e_d(x_1, \dots, x_d)}$$

which concludes the proof. $\square$

Finally, restricting $\mathcal{P}^{[d]}$ to spec $\hat{K}$ we get by (1.9) and Theorem 31 the result. (that $\mathcal{P}$ is unramified at the generic point of the exceptional divisor of the blowup).

Kummer Extensions: Few things are different in that case,

Set $R = k[x, x^{-1}]$, and $S = R[X]/(X^n - f)$ where $f = f(x, 1/x) \in R$ In this case we have $chark = p$ and $gcd(p, n) = 1$.

We have

$$\begin{aligned} R^{\otimes_k d} &= k[x_1, x_1^{-1}, \dots, x_d, x_d^{-1}] \\ S^{\otimes_k d} &= k[x_1, x_1^{-1}, \dots, x_d, x_d^{-1}][X_1, \dots, X_d]/(X_1^n - f_1, \dots, X_d^n - f_d) \\ &= R^{\otimes_k d}[X_1, \dots, X_d]/(X_1^n - f_1, \dots, X_d^n - f_d) \end{aligned}$$

Where $f_i = f(x_i, x_i^{-1})$

Next, we want to figure out $\left(S^{\otimes_k d}\right)^{G^{d-1}}$

Recall that the action of $g \in G = \mathbb{Z}/n\mathbb{Z}$ on X is $g(X) = \zeta^g X$ (g correspond to a number $0 \le g \le n-1$). So, the action of $(g_1, \dots, g_{d-1})$ on the generators $(X_1, X_2, \dots, X_{d-1}, X_d)$ Is $X_1 \mapsto \zeta^{g_1} X_1$, $X_i \mapsto \zeta^{g_i - g_{i-1}} X_i$ for $1 < i < d$ and $X_d \mapsto \zeta^{-g_{d-1}} X_d$.

So we see that $Y = X_1 X_2 \dots X_d$ is invariant. And $Y^n - \prod_{i=1}^d f_i$ is irreducible degree n equation for Y, So, like before, we conclude:

$$\left(S^{\otimes_k d}\right)^{G^{d-1}} \cong R^{\otimes_k d}[Y]/(Y^n - \prod_{i=1}^d f_i)$$

The group S_d acts on $R^{\otimes_k d}[Y]/(Y^n - \prod_{i=1}^d f_i)$ by permuting the indices. on the variables x_i, On $Y = \prod_1^d X_i$ it is invaraint. The invaraint subring $(R^{\otimes_k d})^{S_d}$ is simply $k[e_1, e_2, \dots, e_d, e_d^{-1}]$ like before.

The polynomial $F = \prod_{i=1}^d f_i$ is symmetric in $\{x_i\}_1^d$ so it can be expressed as a polynomial $\tilde{F}(e_1, \dots, e_d)$ in the elementary symmetric variables. Hence the qoutient ring is:

$$\left(\frac{k[x_1^{\pm 1}, \dots, t_d^{\pm 1}][Y]}{(Y^n - \prod_{i=1}^d f_i)}\right)^{S_d} \cong \frac{k[e_1, \dots, e_d, e_d^{-1}][Y]}{(Y^n - \tilde{F}(e_1, \dots, e_d))}$$

So we see again, that restricting $\mathcal{P}^{[d]}$ to spec $\hat{K}$ we get by (1.9) and Theorem 30, that $\mathcal{P}$ is tamely ramified at the generic point of the exceptional divisor of the blowup. $\square$

So, we conclude

Corollary 65. *In the general settings of a curve C, and $\mathcal{P} \to U = C \setminus \mathfrak{m}$ a G-torsor with ramification bounded by $\mathfrak{n} = nP \subset \mathfrak{m}$ at P, we have:*

1. *If $G = \mathbb{Z}/p\mathbb{Z}$ then $\mathcal{P}^{[\deg \mathfrak{n}]}$ is unramified at $\eta_\mathfrak{n}$*
2. *If $\mathcal{P}$ is tamely ramified at $\mathfrak{n}$ then $\mathcal{P}^{[\deg \mathfrak{n}]}$ is tamely ramified at $\eta_\mathfrak{n}$*

Proof. By the remark opening this section we may assume k is algebraically closed, so that $P \in C(k)$ and $\deg \mathfrak{n} = n$. The restriction $\mathcal{P}_x$ of $\mathcal{P}$ to the punctured formal disc at P is tamely ramified, resp. a $\mathbb{Z}/p\mathbb{Z}$-torsor with ramification bounded by n, so by Lemma 62 there is a torsor $\mathcal{P}' \to \mathbb{G}_m$ of the corresponding explicit form with $\mathcal{P}'_x \cong \mathcal{P}_x$. By Corollary 61, the ramification of $\mathcal{P}^{[\deg \mathfrak{n}]}$ at $\eta_\mathfrak{n}$ agrees with that of $\mathcal{P}'^{[\deg \mathfrak{n}]}$ at $\eta_{n \cdot 0}$, which was computed in Theorem 63. $\square$

Next, we generalize to the case $G = \mathbb{Z}/p^r\mathbb{Z}$.

Theorem 66. *Assume $G = \mathbb{Z}/p^r\mathbb{Z}$. Then $\mathcal{P}^{[\deg(\mathfrak{n})]}$ is unramified at $\eta_{\mathfrak{n}}$.*

The proof occupies the remainder of this subsection. In contrast with what the phrase "we generalize" may suggest, the argument is *not* an induction on r with Corollary 65(1) as its base case: it is a direct argument, treating all r at once, in which each of the three steps of the Artin–Schreier case of Theorem 63 — realization of the local torsor by an explicit equation over $\mathbb{G}_m$, identification of the symmetric-power cover at $\eta_{\mathfrak{n}}$, and the degree estimate — is replaced by its Witt-vector analogue, using the theory collected in Section 1.1.6. In particular the case $r = 1$ is *reproved* (as the special case of Witt vectors of length one) rather than used. For an explanation of why the seemingly natural induction on r fails — in short: the hypothesis bounds the ramification of $\mathcal{P}$ in *upper* numbering, while the break of the top degree-p layer $\mathcal{P} \to \mathcal{P}'$ is the largest *lower*-numbering break, which can be of size about $p^{r-1}d$ — see Chapter A.

Standing reduction

As in the proof of Corollary 65, we may assume k is algebraically closed. We may further assume that $\mathcal{P} \to U$ is *totally ramified* at P with inertia equal to all of G: writing $R = \widehat{\mathcal{O}_{C,P}}$, $K = \mathrm{Frac}(R)$, and letting $H = \rho(G^0) \subseteq G$ be the image of the inertia subgroup under the homomorphism $\rho\colon \mathrm{Gal}(K^{sep}/K) \to G$ classifying $\mathcal{P}|_{\mathrm{Spec}\,K}$, decompose $\mathcal{P} \to \mathcal{P}_{G/H} \to U$ as in Proposition 15, where $\mathcal{P} \to \mathcal{P}_{G/H}$ is an H-torsor and $\mathcal{P}_{G/H} \to U$ a G/H-torsor. Then $\mathcal{P}_{G/H}$ is unramified at P, hence extends over P; taking any point $Q \in \mathcal{P}_{G/H}$ over P yields $\widehat{\mathcal{O}_{\mathcal{P}_{G/H},Q}} \cong \widehat{\mathcal{O}_{C,P}}$, and since k is algebraically closed the torsor $\mathcal{P}_{G/H}$ is locally trivial at Q. This replaces $(G, \mathcal{P} \to U)$ by $(H, \mathcal{P} \to \mathcal{P}_{G/H})$ with $H \cong \mathbb{Z}/p^s\mathbb{Z}$ for some $s \le r$; as the theorem is proved for all r simultaneously, we may rename and assume $H = G$.

After this reduction, the restriction $\mathcal{Q} := \mathcal{P}_x$ of $\mathcal{P}$ to the punctured formal disc $\mathrm{Spec}\,k((x))$ at P (with x a uniformizer at P) is a *connected* $\mathbb{Z}/p^r\mathbb{Z}$-torsor: $\mathcal{Q} = \mathrm{Spec}\,L$ for a cyclic, totally ramified extension L/K of degree p^r, with ramification bounded by $d := \deg \mathfrak{n}$ (Definition 54); equivalently, the largest upper-numbering break u of L/K satisfies $u < d$ *strictly*.

Realization over $\mathbb{G}_m$ for Witt vectors

The first step generalizes Lemma 62(2) from $\mathbb{Z}/p\mathbb{Z}$ to $\mathbb{Z}/p^r\mathbb{Z}$: every local torsor class as above is realized by an explicit Artin–Schreier–Witt equation over $\mathbb{G}_m$ whose Witt datum is a vector of *polynomials in x^{-1}* with controlled pole orders. Recall from Definition 44 the notion of a reduced vector; we add:

Definition 67. A vector $\vec{g} = (g_0, \ldots, g_{r-1}) \in W_r(k((x)))$ is a **reduced polynomial vector** if it is reduced (Definition 44) and moreover every $g_j \in x^{-1}k[x^{-1}]$; then $m(g_j) = \deg_{x^{-1}} g_j$, and $m(g_j) = 0$ if and only if $g_j = 0$.

Lemma 68 (surjectivity of $\wp$ on integral Witt vectors)**.** $\wp\colon W_r(k[[x]]) \to W_r(k[[x]])$ *is surjective.*

Proof. Induction on r. For $r = 1$ this is the first bullet in the proof of Lemma 62(2): given $h = \sum_{i \ge 0} b_i x^i$, solve $u^p - u = h$ coefficientwise, using that k is algebraically closed for the constant term. For $r > 1$, let $\vec{h} \in W_r(k[[x]])$. By induction choose $\vec{u}' \in W_{r-1}(k[[x]])$ with $\wp\vec{u}' = t(\vec{h})$, and lift it to $\tilde{u} := (\vec{u}', 0) \in W_r(k[[x]])$. By Lemma 39(i), $t\big(\wp\tilde{u} \ominus \vec{h}\big) = \wp\vec{u}' \ominus t(\vec{h}) = 0$, so $\wp\tilde{u} \ominus \vec{h} = V^{r-1}(g)$

for some $g \in k[[x]]$. Choose $w \in k[[x]]$ with $\wp w = -g$ (case $r = 1$ again). Then by Lemma 39(ii) and additivity of V^{r-1},

$$\wp\big(\tilde{u} \oplus V^{r-1}(w)\big) = \wp\tilde{u} \oplus V^{r-1}(\wp w) = \big(\vec{h} \oplus V^{r-1}(g)\big) \oplus V^{r-1}(-g) = \vec{h}. \qquad \square$$

Proposition 69 (reduced polynomial representatives)**.** *Every class in $W_r(k((x)))/\wp W_r(k((x)))$ has a reduced polynomial representative: for every $\vec{f} \in W_r(k((x)))$ there exists $\vec{u} \in W_r(k((x)))$ such that $\vec{g} := \vec{f} \ominus \wp\vec{u}$ has all components in $x^{-1}k[x^{-1}]$ with pole orders either 0 or prime to p.*

Proof. Induction on r.

Base case $r = 1$. This is the argument of Lemma 62(2): first, writing $f = f^- + f^+$ with $f^- \in x^{-1}k[x^{-1}]$ and $f^+ \in k[[x]]$, surjectivity of $\wp$ on $k[[x]]$ removes f^+; then, as long as the leading term is cx^{-m} with $p \mid m$, $m > 0$, subtracting $\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p}$ strictly decreases the pole order while staying inside $x^{-1}k[x^{-1}]$; this terminates at 0 or at pole order prime to p.

Inductive step. Let $\vec{f} \in W_r(k((x)))$. By induction there is $\vec{u}' \in W_{r-1}(k((x)))$ such that $\vec{g}' := t(\vec{f}) \ominus \wp\vec{u}'$ is a reduced polynomial vector in W_{r-1}. Lift $\vec{u}'$ to $\tilde{u} := (\vec{u}', 0) \in W_r(k((x)))$ and set $\vec{h} := \vec{f} \ominus \wp\tilde{u}$. By Lemma 39(i), $t(\vec{h}) = \vec{g}'$, i.e.

$$\vec{h} = (g'_0, \ldots, g'_{r-2}, h_{r-1}), \qquad h_{r-1} \in k((x)),$$

with the first $r-1$ components already reduced polynomials. It remains to repair the last component *without touching the others*; by Lemma 39(iii), adjustments of the form $\ominus\,\wp\big(V^{r-1}(z)\big) = \ominus\, V^{r-1}(\wp z)$ change only the last component, by $-\wp(z)$. So:

- Write $h_{r-1} = h^- + h^+$ with $h^- \in x^{-1}k[x^{-1}]$, $h^+ \in k[[x]]$, choose $w \in k[[x]]$ with $\wp w = h^+$ (Lemma 68 with $r = 1$), and replace $\vec{h}$ by $\vec{h} \ominus \wp(V^{r-1}w)$: the last component becomes h^-.
- While the last component has leading term cx^{-m} with $p \mid m$, $m > 0$, replace $\vec{h}$ by $\vec{h} \ominus \wp\big(V^{r-1}(c^{1/p}x^{-m/p})\big)$: the last component changes by $-\wp(c^{1/p}x^{-m/p}) = -cx^{-m} + c^{1/p}x^{-m/p}$, so its pole order strictly decreases and it stays in $x^{-1}k[x^{-1}]$. This terminates at 0 or at pole order prime to p.

The result is a reduced polynomial vector $\wp$-equivalent to $\vec{f}$. $\square$

Remark. The individual components of a Witt vector cannot be truncated to their principal parts one at a time — Witt addition is not componentwise ((1.5)), so a componentwise operation changes the Artin–Schreier–Witt class. The point of Proposition 69 is that all the adjustments are performed by subtracting genuine elements of $\wp W_r(k((x)))$, inside the Witt ring.

Lemma 70 (Realization over $\mathbb{G}_m$ for W_r)**.** *Let k be algebraically closed of characteristic $p > 0$ and let $\mathcal{Q}$ be a* connected *$\mathbb{Z}/p^r\mathbb{Z}$-torsor over* $\operatorname{Spec}(k((x)))$ *with ramification bounded by d (Definition 54). Then there is a reduced polynomial vector $\vec{f} = (f_0, \ldots, f_{r-1})$, $f_j \in x^{-1}k[x^{-1}]$, such that, with $m_j := \deg_{x^{-1}} f_j$:*

1. *$p^{r-1-j}\, m_j < d$ for all $0 \le j \le r-1$; equivalently $m_j < d\, p^{j-(r-1)}$;*
2. *$\mathcal{P}' := \operatorname{Spec}\Big(k[x, x^{-1}][\vec{X}]/\big(\wp\vec{X} \ominus \vec{f}\big)\Big)$ is a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\mathbb{G}_m$ with $\mathcal{P}'_x \cong \mathcal{Q}$;*
3. *$\mathcal{P}'$ extends to a torsor over $\mathbb{P}^1_k \setminus \{0\}$ — in particular it is unramified at ∞ — and has ramification bounded by d at 0.*

Proof. $\mathcal{Q} = \operatorname{Spec} L$ with L/K cyclic totally ramified of degree p^r and largest upper break $u < d$ (Theorem 41). Let $[\vec{f_0}] \in W_r(K)/\wp$ be its Artin–Schreier–Witt class and let $\vec{f}$ be a reduced polynomial representative (Proposition 69).

First, $f_0 \neq 0$: otherwise Corollary 42 would bound the order of the class by p^{r-1}, contradicting $[L:K] = p^r$. Hence $m_0 \geq 1$ and $p \nmid m_0$, and the Schmid–Witt break formula (Theorem 45) applies: $\max_j p^{r-1-j} m_j = u < d$, which is 1.

For 2: by Lemma 40 (with $A = k[x, x^{-1}] \ni f_j$), $\mathcal{P}'$ is a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\mathbb{G}_m$; its restriction to $\operatorname{Spec} k((x))$ is the Artin–Schreier–Witt torsor of the class of $\vec{f}$, i.e. $\mathcal{Q}$ (Theorem 41).

For 3: in the coordinate $y = x^{-1}$ at ∞ we have $f_j \in y\,k[y]$, so $\vec{f} \in W_r(k[y])$ and Lemma 40 (with $A = k[y]$) extends $\mathcal{P}'$ to a torsor over $\mathbb{A}^1_k = \operatorname{Spec} k[y] = \mathbb{P}^1_k \setminus \{0\}$; a torsor over the full local ring at ∞ is unramified there. The ramification bound at 0 holds because $\mathcal{P}'_x \cong \mathcal{Q}$. $\square$

Remark. Connectedness of $\mathcal{Q}$ is guaranteed in the application by the standing reduction (inertia equal to all of G). Note also the following sanity check: for every j with $p^{r-1-j} \geq d$, the bound 1 forces $m_j = 0$, i.e. $f_j = 0$ — all sufficiently low Witt components of a reduced representative vanish outright. This makes visible how strong the hypothesis $u < d$ is at the lower levels.

The symmetric-power cover at $\eta_{\mathfrak{n}}$

We now generalize the invariant-ring computation in the Artin–Schreier case of Theorem 63 to Witt coordinates. Fix $\vec{f} \in W_r(R)$, $R = k[x, x^{-1}]$, and let

$$S \;=\; R[\vec{X}]/\big(\wp\vec{X} \ominus \vec{f}\big)$$

be the coordinate ring of the torsor $\mathcal{P}' \to \mathbb{G}_m$ of Lemma 70. Write $R^{\otimes_k d} = k[x_1^{\pm 1}, \ldots, x_d^{\pm 1}]$ and

$$S^{\otimes_k d} = R^{\otimes_k d}[\vec{X}_1, \ldots, \vec{X}_d]/\big(\wp\vec{X}_1 \ominus \vec{f}(x_1), \ldots, \wp\vec{X}_d \ominus \vec{f}(x_d)\big),$$

where $\vec{X}_i = (X_{i,0}, \ldots, X_{i,r-1})$ and $\vec{f}(x_i)$ is the image of $\vec{f}$ under $x \mapsto x_i$. Exactly as in the proof of Theorem 63, the d-fold product torsor $p_1^{-1}\mathcal{P}' \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P}'$ on U^d is the quotient of $\operatorname{Spec} S^{\otimes_k d}$ by $K_G \cong G^{d-1}$, embedded in G^d as the kernel of the sum map via $(g_1, \ldots, g_{d-1}) \mapsto (g_1, g_2 - g_1, \ldots, -g_{d-1})$, and acting (in Witt coordinates, writing $\underline{g}$ for the image of $g \in G$ under $\mathbb{Z}/p^r\mathbb{Z} \cong W_r(\mathbb{F}_p)$, Example 37) by

$$(g_1, \ldots, g_{d-1})\colon \quad \vec{X}_1 \mapsto \vec{X}_1 \oplus \underline{g}_1, \qquad \vec{X}_i \mapsto \vec{X}_i \oplus \underline{g}_i \ominus \underline{g}_{i-1}\ (1 < i < d), \qquad \vec{X}_d \mapsto \vec{X}_d \ominus \underline{g}_{d-1}.$$

Proposition 71 (identification of the product torsor)**.** *Set*

$$\vec{Y} := \bigoplus_{i=1}^{d} \vec{X}_i \in W_r\big(S^{\otimes_k d}\big), \qquad \vec{\alpha} := \bigoplus_{i=1}^{d} \vec{f}(x_i) \in W_r\big(R^{\otimes_k d}\big),$$

where $\bigoplus$ denotes the d-fold Witt vector sum $\oplus$ (not a direct sum); in particular $\vec{Y}$ is a single Witt vector of length r. Then:

1. *Every component Y_n of $\vec{Y}$ is G^{d-1}-invariant, and*

$$\big(S^{\otimes_k d}\big)^{G^{d-1}} \;\cong\; R^{\otimes_k d}[\vec{Y}]/\big(\wp\vec{Y} \ominus \vec{\alpha}\big)$$

   as G-torsors over $R^{\otimes_k d}$ (the residual $G = G^d/G^{d-1}$ acting by $\vec{Y} \mapsto \vec{Y} \oplus \underline{g}$).

2. *Every component α_n of $\vec{\alpha}$ is a symmetric Laurent polynomial, i.e. lies in $\left(R^{\otimes_k d}\right)^{S_d} = k[e_1, \ldots, e_d, e_d^{-1}]$, and*
$$\left(\left(S^{\otimes_k d}\right)^{G^{d-1}}\right)^{S_d} \;\cong\; k[e_1, \ldots, e_d, e_d^{-1}][\vec{Y}]/\left(\wp\vec{Y} \ominus \vec{\alpha}\right).$$

Consequently, restricting to the complete local field $\widehat{K}_\eta = \kappa(\eta)((e_d))$, $\kappa(\eta) = k(e_1/e_d, \ldots, e_{d-1}/e_d)$, of (1.9),
$$\mathcal{P}'^{[d]}\big|_{\operatorname{Spec} \widehat{K}_\eta} \;\cong\; \operatorname{Spec}\, \widehat{K}_\eta[\vec{Z}]/\left(\wp\vec{Z} \ominus \vec{\alpha}\right).$$

Proof. 1 An element $\sigma = (g_1, \ldots, g_{d-1}) \in G^{d-1}$ acts as a ring automorphism of $S^{\otimes_k d}$: this is its *original* action, defined on the generators $X_{i,n}$ by the twisted formulas above (which mix components through the carry polynomials of (1.5)). By functoriality of W_r (Section 1.1.6), the induced map $W_r(\sigma)$ on $W_r(S^{\otimes_k d})$ is componentwise and commutes with $\oplus, \ominus$; on the generators it reproduces the twisted action, $W_r(\sigma)(\vec{X}_i) = \vec{X}_i \oplus \underline{g}_i \ominus \underline{g}_{i-1}$ (with $\underline{g}_0 = \underline{g}_d = \vec{0}$). Hence, using commutativity and associativity of $\oplus$, the Witt sum transforms by the telescoping constant
$$W_r(\sigma)(\vec{Y}) \;=\; \bigoplus_{i=1}^{d} W_r(\sigma)(\vec{X}_i) \;=\; \vec{Y} \oplus \underline{g}_1 \oplus (\underline{g}_2 \ominus \underline{g}_1) \oplus \cdots \oplus (\ominus\, \underline{g}_{d-1}) \;=\; \vec{Y}.$$

Since $W_r(\sigma)$ is componentwise, this equality of r-tuples says exactly $\sigma(Y_n) = Y_n$ for every n, i.e. each $Y_n \in S^{\otimes_k d}$ is invariant. Since $\wp = F - \mathrm{id}$ is additive (Definition 38 ff.),
$$\wp\vec{Y} = \bigoplus_{i=1}^{d} \wp\vec{X}_i = \bigoplus_{i=1}^{d} \vec{f}(x_i) = \vec{\alpha}.$$

Hence there is a well-defined $R^{\otimes_k d}$-algebra homomorphism
$$\phi\colon\; T := R^{\otimes_k d}[\vec{Z}]/\left(\wp\vec{Z} \ominus \vec{\alpha}\right) \;\longrightarrow\; \left(S^{\otimes_k d}\right)^{G^{d-1}}, \qquad \vec{Z} \mapsto \vec{Y}.$$

By Lemma 40, $\operatorname{Spec} T$ is a G-torsor over U^d. The target is the product torsor (the contracted product of the $p_i^{-1}\mathcal{P}'$ along the sum map $G^d \to G$), a G-torsor over U^d by Proposition 22 and the surrounding discussion; its residual G-action is induced by $(g, 0, \ldots, 0) \in G^d$, i.e. $\vec{X}_1 \mapsto \vec{X}_1 \oplus \underline{g}$, under which $\vec{Y} \mapsto \vec{Y} \oplus \underline{g}$. So $\operatorname{Spec} \phi$ is a G-equivariant morphism of G-torsors over U^d, hence an isomorphism by Proposition 5.

2 S_d acts on $S^{\otimes_k d}$ by permuting the tensor factors, i.e. $x_i \mapsto x_{\sigma(i)}$, $\vec{X}_i \mapsto \vec{X}_{\sigma(i)}$. As in 1, each $\sigma \in S_d$ is a ring automorphism, so $W_r(\sigma)$ is componentwise and commutes with $\oplus$. Since $\oplus$ is commutative and associative, permuting the summands of a Witt sum leaves it unchanged; thus $\vec{Y}$ is S_d-invariant (componentwise, as before) and $W_r(\sigma)(\vec{\alpha}) = \bigoplus_i \vec{f}(x_{\sigma(i)}) = \vec{\alpha}$, so each $\alpha_n \in R^{\otimes_k d}$ is a symmetric Laurent polynomial, hence lies in $k[e_1, \ldots, e_d, e_d^{-1}]$ (the S_d-invariants of $R^{\otimes_k d}$, as in the proof of Theorem 63). By Lemma 40, T is free over $R^{\otimes_k d}$ with basis the monomials $\prod_n Y_n^{a_n}$, $0 \le a_n < p$, each of which is S_d-invariant; writing an element in this basis, it is S_d-invariant if and only if all its coefficients are. Therefore
$$T^{S_d} = \left(R^{\otimes_k d}\right)^{S_d}[\vec{Y}]/\left(\wp\vec{Y} \ominus \vec{\alpha}\right) = k[e_1, \ldots, e_d, e_d^{-1}][\vec{Y}]/\left(\wp\vec{Y} \ominus \vec{\alpha}\right).$$

The final statement follows exactly as in the $r = 1$ case: by definition $\mathcal{P}'^{[d]}$ is the quotient by $\Xi = K_G \rtimes S_d$ computed above (Corollary 23), and its restriction to $\operatorname{Spec} \widehat{K}_\eta$ along $k[e_1, \ldots, e_d, e_d^{-1}] \to \widehat{K}_\eta$ is the Artin–Schreier–Witt cover of the image of $\vec{\alpha}$ in $W_r(\widehat{K}_\eta)$. ☐

Remark. The proof via Proposition 5 replaces the rank count of the $r = 1$ argument ($p^{rd}/p^{r(d-1)} = p^r$ together with irreducibility of the invariant's minimal equation); the rank count alone would still require showing that $\vec{Y}$ generates the invariant ring, which is exactly what equivariance gives for free. Note also that a pure restriction argument on H^1's ("the class of the product torsor is the sum of the pulled-back characters") identifies the cover over $k(x_1, \ldots, x_d)$ but does *not* by itself descend it to the symmetric base $k(e_1, \ldots, e_d)$; the invariant-theoretic computation above is what effects the descent.

The degree count

Throughout this paragraph, $\vec{f}$ is the reduced polynomial vector of Lemma 70, so $m_j = \deg_{x^{-1}} f_j$ satisfies $m_j < d\,p^{j-(r-1)}$, and $y_i := 1/x_i$, so that $f_j(x_i) \in y_i\, k[y_i]$ is a polynomial in y_i of degree m_j.

Lemma 72 (symmetric regularity)**.** *Let $\Phi \in k[y_1, \ldots, y_d]^{S_d}$ be symmetric with every monomial of total degree $< d$. Then $\Phi \in \kappa(\eta) = k(e_1/e_d, \ldots, e_{d-1}/e_d)$; in particular Φ is regular at $\eta_{\mathfrak{n}}$.*

Proof. Φ is a k-combination of monomial symmetric functions $m_\lambda(y)$ with $|\lambda| < d$. Each $m_\lambda(y)$ is an isobaric polynomial in the elementary symmetric functions $e_\mu(y) = \prod_i e_{\mu_i}(y)$ with $\mu \vdash |\lambda|$; every part $\mu_i \leq |\lambda| < d$, so only $e_1(y), \ldots, e_{d-1}(y)$ occur — never $e_d(y)$. Now $e_k(y) = e_{d-k}(x)/e_d(x)$, which for $1 \leq k \leq d-1$ lies in $\kappa(\eta)$. Hence $\Phi \in \kappa(\eta) \subset \widehat{\mathcal{O}}_\eta = \kappa(\eta)[[e_d]]$. $\square$

Theorem 73 (regularity of all Witt components)**.** *With $\vec{f}$ as in Lemma 70, every component α_n $(0 \leq n \leq r-1)$ of $\vec{\alpha} = \bigoplus_{i=1}^d \vec{f}(x_i)$ lies in $\kappa(\eta)$; in particular $\vec{\alpha} \in W_r\big(\widehat{\mathcal{O}}_\eta\big)$.*

Proof. Fix $n \leq r-1$ and a monomial $\prod_{i,j}\big(f_j(x_i)\big)^{e_{ij}}$ of α_n, viewed as a universal isobaric polynomial in the $f_j(x_i)$ of weight p^n (Lemma 43, with $f_j(x_i)$ of weight p^j); set $E_j := \sum_i e_{ij}$, so $\sum_j E_j p^j = p^n$. Expanding each $f_j(x_i)$ as a polynomial in y_i of degree m_j, every resulting monomial in the y_i has total degree

$$\leq \sum_{i,j} e_{ij}\, m_j \;=\; \sum_j E_j\, m_j \;<\; \sum_j E_j\, d\,p^{j-(r-1)} \;=\; d\,p^{-(r-1)} \sum_j E_j p^j \;=\; d\,p^{n-(r-1)} \;\leq\; d,$$

strictly: some $E_j > 0$ (as $\sum_j E_j p^j = p^n > 0$), and the pole bound $m_j < d\,p^{j-(r-1)}$ of Lemma 701 is strict even when $m_j = 0$. Thus every monomial of α_n — a symmetric polynomial in $y_1, \ldots, y_d$ by Proposition 712 — has total y-degree $< d$, and Lemma 72 gives $\alpha_n \in \kappa(\eta)$. $\square$

Remark. The bound is exact: the weight $p^{j-(r-1)}$ is paired against the isobaricity constraint $\sum_j E_j p^j = p^n$, so the estimate $d\,p^{n-(r-1)} \leq d$ is independent of how the Witt carries distribute the exponents E_j; the hypothesis "break $< d$ at P" is used at *every* Witt level at once. The sanity check of Lemma 70 records the extreme case: components with $p^{r-1-j} \geq d$ vanish outright in the reduced representative.

Proof of the theorem

Proof of Theorem 66. By the standing reduction above we may assume k algebraically closed and $\mathcal{P} \to U$ totally ramified at P with inertia all of G, so that $\mathcal{Q} := \mathcal{P}_x$ is a connected $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\operatorname{Spec} k((x))$ with ramification bounded by $d = \deg \mathfrak{n}$.

Lemma 70 produces a torsor $\mathcal{P}' \to \mathbb{G}_m$, defined by a reduced polynomial vector $\vec{f}$ with $p^{r-1-j}m_j < d$, with $\mathcal{P}'_x \cong \mathcal{Q}$. By Corollary 61 (valid for arbitrary finite abelian G), the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ agrees with that of $\mathcal{P}'^{[d]}$ at $\eta_{d\cdot 0}$; so we may assume $C = \mathbb{P}^1_k$, $\mathfrak{m} = d \cdot 0 + \infty$, $\mathfrak{n} = d \cdot 0$, $\mathcal{P} = \mathcal{P}'$.

By Proposition 71,

$$\mathcal{P}'^{[d]}\big|_{\mathrm{Spec}\,\widehat{K}_\eta} \cong \mathrm{Spec}\ \widehat{K}_\eta[\vec{Z}]/(\wp\vec{Z} \ominus \vec{\alpha}), \qquad \vec{\alpha} = \bigoplus_{i=1}^{d} \vec{f}(x_i),$$

with $\widehat{K}_\eta = \kappa(\eta)((e_d))$ the complete local field at $\eta_{\mathfrak{n}}$ ((1.9)) and $\widehat{\mathcal{O}}_\eta = \kappa(\eta)[[e_d]]$ its valuation ring.

By Theorem 73, $\vec{\alpha} \in W_r(\kappa(\eta)) \subset W_r(\widehat{\mathcal{O}}_\eta)$. Hence, by Lemma 40 applied over $A = \widehat{\mathcal{O}}_\eta$, the Artin–Schreier–Witt cover $\wp\vec{Z} = \vec{\alpha}$ is finite étale over $\widehat{\mathcal{O}}_\eta$, and $\mathcal{P}'^{[d]}|_{\mathrm{Spec}\,\widehat{K}_\eta}$ is its generic fibre: every field factor of $\widehat{K}_\eta[\vec{Z}]/(\wp\vec{Z} \ominus \vec{\alpha})$ is the fraction field of a finite étale $\widehat{\mathcal{O}}_\eta$-algebra, i.e. an unramified extension of $\widehat{K}_\eta$ with respect to the discrete valuation ring $\widehat{\mathcal{O}}_\eta$. By Definition 53, $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$. □

Remark. The argument proves all r simultaneously; it reproves the case $r = 1$ (Corollary 65(1)) as the special case of a vector of length one and never invokes it.

Finally, **Proof of Theorem theorem 52**: Idea: decompose G be the structure theorem, and finish by the above.

1.4 Extending To Product of Blowups

We conclude this sesction by proving sume auxilary results that would be useful in Chapter 2.

Let X and Y be smooth schemes over a field k, and let $x \in X$ and $y \in Y$ be closed points. We denote the blowups of these schemes at the given points by $\pi_X : \mathrm{Bl}_x(X) \to X$ and $\pi_Y : \mathrm{Bl}_y(Y) \to Y$. Furthermore, let $\pi_{X\times Y} : \mathrm{Bl}_{(x,y)}(X \times_k Y) \to X \times_k Y$ be the blowup of the product scheme at the point (x, y). We denote by E_X, E_Y, and $E_{X\times Y}$ the respective exceptional divisors, and let η_X, η_Y, and $\eta_{X\times Y}$ be their generic points.

In this section, we establish the following result concerning the stability of ramification bounds under the external product of torsors.

Proposition 74. *Let G be a finite abelian group. Suppose $\mathcal{G}_X$ and $\mathcal{G}_Y$ are G-torsors defined on open subsets $U_X \subset \mathrm{Bl}_x(X)$ and $U_Y \subset \mathrm{Bl}_y(Y)$ that are disjoint from the exceptional divisors. If the ramification of $\mathcal{G}_X$ at η_X and $\mathcal{G}_Y$ at η_Y is bounded by r, then the external product torsors*

$$\mathcal{G}_{X\times Y} := pr_1^{-1}\mathcal{G}_X \otimes pr_2^{-1}\mathcal{G}_Y$$

has ramification bounded by r at the generic point $\eta_{X\times Y}$ of the exceptional divisor in the product blowup.

The proposition is purely local in nature, it suffices to consider the case where X and Y are affine. More precisely, by the smoothness of X and Y, we may restrict our attention to open neighborhoods of x and y that are etale over affine spaces. The rest of this section treats that case.

The Affine Case

Let $X = \mathbb{A}^n_k$ be the affine n-space over a field k, and let $0 \in X$ be the origin. Let $\tilde{X} = \mathrm{Bl}_0(X)$ be the blowup of X at the origin. Recall that $\tilde{X} \subset X \times_k \mathbb{P}^{n-1}_k$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \cdots : u_n]$ are the homogeneous coordinates of $\mathbb{P}^{n-1}_k$. The exceptional divisor $E \subset \tilde{X}$ is the fiber over the origin, $E = \{(0, [u_1 : \cdots : u_n])\}$, which is of codimension 1 in $\tilde{X}$. Let $\eta \in E$ be the generic point of E, and let $R = \mathcal{O}_{\tilde{X},\eta}$ be the associated local ring. This ring R is a discrete valuation ring (DVR) with fraction field $K = K(X) = k(x_1, \ldots, x_n)$.

On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1} x_1$. The coordinate ring is:

$$\mathcal{O}_{\tilde{X}}(U_1) = k\left[x_1, \frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1}\right]$$

In this chart, the generic point η corresponds to the prime ideal $\mathfrak{p}_1 = (x_1)$. Thus, the local ring is $R = k[x_1, \frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1}]_{(x_1)}$. The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1})$, and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_1]]$$

In this local ring, x_1 is a uniformizer. Note that any x_i (for $i > 1$) can also serve as a uniformizer, as $x_i = (\frac{u_i}{u_1})x_1$ and $\frac{u_i}{u_1}$ is a unit in R.

For any monomial $M = x_1^{a_1} \ldots x_n^{a_n} \in K$, we can write:

$$M = x_1^{\sum a_i} \left(\frac{u_2}{u_1}\right)^{a_2} \cdots \left(\frac{u_n}{u_1}\right)^{a_n}$$

Since the term in parentheses is a unit in R, the valuation ν_E associated with E satisfies:

$$\nu_E(M) = \sum a_i = \deg M$$

Consequently, for any polynomial $f = f_d + f_{d+1} + \cdots + f_l$, where f_i is the homogeneous part of degree i, we have $\nu_E(f) = d$ (the order of vanishing at the origin).

The Product Case

Now, let $X = \mathbb{A}^n$ and $Y = \mathbb{A}^m$ with origins $x = 0$ and $y = 0$. As before, $\mathrm{Bl}_0(X) \subset X \times \mathbb{P}^{n-1}$ and $\mathrm{Bl}_0(Y) \subset Y \times \mathbb{P}^{m-1}$ have exceptional divisors E_X and E_Y respectively. Consider the product $X \times_k Y \cong \mathbb{A}^{n+m}_k$. The blowup of the product at the origin $(0,0)$, denoted $\mathrm{Bl}_{(0,0)}(X \times_k Y)$, is a subscheme of $(X \times Y) \times \mathbb{P}^{n+m-1}$ defined by:

$$\begin{cases} x_i w_j = x_j w_i & 1 \leq i, j \leq n \\ y_k w_{n+l} = y_l w_{n+k} & 1 \leq k, l \leq m \\ x_i w_{n+j} = y_j w_i & 1 \leq i \leq n,\ 1 \leq j \leq m \end{cases}$$

where $[w_1 : \cdots : w_{n+m}]$ are the homogeneous coordinates of $\mathbb{P}^{n+m-1}$. The exceptional divisor $E_{X \times Y}$ is isomorphic to $\mathbb{P}^{n+m-1}$.

Comparison of Blowups

Both $\mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$ and $\mathrm{Bl}_{(0,0)}(X \times_k Y)$ are birational to $X \times_k Y$. They share a common dense open set $\tilde{U}$ defined by the condition that neither the X-coordinates nor the Y-coordinates vanish simultaneously in the projective space:

$$\tilde{U} = \{((x,y), [w_1 : \cdots : w_{n+m}]) \mid (w_1, \ldots, w_n) \neq 0 \text{ and } (w_{n+1}, \ldots, w_{n+m}) \neq 0\}$$

This yields a diagram of open immersions:

$$\begin{tikzcd} & \tilde{U} \arrow[dl, "f_1"'] \arrow[dr, hook, "f_2"] & \\ \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y) & & \mathrm{Bl}_{(0,0)}(X \times_k Y) \end{tikzcd}$$

where f_1 maps the coordinates to the respective projectivizations $[w_1 : \cdots : w_n]$ and $[w_{n+1} : \cdots : w_{n+m}]$. Also note that the generic point $\eta_{X \times Y}$ of $E_{X \times_k Y}$ is inside $\tilde{U}$

Extensions of DVRs

Let S be the local ring of the generic point $\eta_{X \times Y}$ of $E_{X \times Y}$ in $\tilde{U}$. On the chart where $w_1 \neq 0$ and $w_{n+1} \neq 0$, we have $x_1 = (\frac{w_1}{w_{n+1}})y_1$. Since $\frac{w_1}{w_{n+1}}$ is a unit in this chart, x_1 and y_1 are equivalent as uniformizers. We have:

$$S = k\left[x_1, \frac{w_2}{w_1} \cdots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1} \cdots \frac{w_{n+m}}{w_1}\right]_{(x_1)} = k\left[\frac{w_1}{w_{n+1}}, \frac{w_2}{w_{n+1}} \cdots \frac{w_n}{w_{n+1}}, y_1 \cdots \frac{w_{n+m}}{w_{n+1}}\right]_{(y_1)}$$

$$k(\eta_{X \times Y}) = k\left(\frac{w_2}{w_1} \cdots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1} \cdots \frac{w_{n+m}}{w_1}\right)$$

Let R_X be the local ring of the exceptional divisor E_X in $\mathrm{Bl}_0(X)$. The pullback of $E_X \times_k \mathrm{Bl}_0(Y)$ along f_1 induces an extension of DVRs $R_X \hookrightarrow S$. Which is:

1. Weakly Unramified: x_1 is a uniformizer in both R_X and S, so the ramification index is $e = 1$.

2. Residually Transcendental: The residue field extension $\kappa(\eta_X) \subset \kappa(\eta)$ is:

$$k\left(\frac{w_2}{w_1}, \ldots, \frac{w_n}{w_1}\right) \subset k\left(\frac{w_2}{w_1}, \ldots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1}, \ldots, \frac{w_{n+m}}{w_1}\right)$$

   Hence separable.

Since this extension is generated by transcendental elements, it is separable and formally smooth at the maximal ideal ([Stacks, Tag 09E7]).

Ramification of G-Torsors

Let G be a finite abelian group. Let $\mathcal{Q}$ be a G-torsor on an open $U \subset \mathrm{Bl}_0(X)$ disjoint from E_X. Let $V \subset \mathrm{Bl}_0(Y)$ be an open subscheme, and let $\pi_X : \mathrm{Bl}_0(X) \times_k V \to \mathrm{Bl}_0(X)$ be the projection onto the first factor. By restricting this projection to $U \times_k V$, we obtain the pullback G-torsor:

$$\pi_X^{-1}(\mathcal{Q}) \cong \mathcal{Q} \times_k V$$

which is defined on the open subset $U \times_k V \subset \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$. The extension of local rings $R_X \to S$ is weakly unramified (the ramification index $e = 1$) and residually transcendental with seprable residue field exntesion. Under these conditions the ramification filtration is preserved. Therefore, the pullback $\pi_X^{-1}(\mathcal{Q})$ has ramification bounded by r at the generic point $\eta_{X \times Y}$ of the exceptional divisor $E_{X \times Y}$ if and only if the original torsor $\mathcal{Q}$ has ramification bounded by r at the generic point η_X of E_X

And we finish by Lemma 57.

Chapter 2

Geometric Class Field Theory

The main theorem of this chapter is

Theorem 75. *Let Λ be a finite ring of cardinality invertible in k, and let $\mathcal{F}$ be an étale sheaf of Λ-modules, locally free of rank 1 on U, with ramification bounded by $\mathfrak{m}$. Considering $U^{(d)}$ as an open subscheme of the blowup $\tilde{C}_{\mathfrak{m}}^{(d)}$ of $C^{(d)}$, we have that for sufficiently large integer d, $\mathcal{F}^{[d]}$ is tamely ramified on $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)} = E_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)}$.*

From which we will deduce Theorem 2.

We start by recalling the necessarily material, and then move to proving this theorem.

2.1 Etale Fundamental Groups and Tame Fundamental Groups

We recall the definition and basic properties of the etale fundamental group, following stacks project [Stacks, Tag 0BQ6]

Proposition 76 ([Stacks, Tag 0C0J]). *Let $f : X \to S$ be a flat proper morphism of finite presentation whose geometric fibres are connected and reduced. Assume S is connected and let $\overline{s}$ be a geometric point of S. Then there is an exact sequence*

$$\pi_1(X_{\overline{s}}) \to \pi_1(X) \to \pi_1(S) \to 1$$

of fundamental groups.

Corollary 77. *Let $f : X \to S$ be a proper smooth morphism of finite presentation whose geometric fibres are connected. Assume S is connected and let $\overline{s}$ be a geometric point of S. Then there is an exact sequence*

$$\pi_1(X_{\overline{s}}) \to \pi_1(X) \to \pi_1(S) \to 1$$

of fundamental groups.

2.2 Generalized Picard Scheme

We recall the notion of generalized Jacobian varieties and study their fundamental properties. The material presented here is primarily adapted from [Gui19] and [Tak19]. For further background on the general theory of abelian varieties and Jacobians, the reader may also consult [Mil08]. Let S be a scheme and let C be a projective smooth S-scheme whose geometric fibers are connected and of dimension 1. Let $\mathfrak{m}$ be a modulus on C, defined as an effective Cartier divisor of C/S (i.e., a closed subscheme of C which is finite flat of finite presentation over S). We denote the projection $C \times_S T \to T$ by pr for any S-scheme T.

The Functor of Points

Let d be an integer. For an S-scheme T, we consider the set of data $(\mathcal{L}, \psi)$ where:

- $\mathcal{L}$ is an invertible sheaf of degree d on C_T.
- $\psi : \mathcal{O}_{\mathfrak{m}_T} \xrightarrow{\sim} \mathcal{L}|_{\mathfrak{m}_T}$ is a trivialization of $\mathcal{L}$ along the modulus.

Two such pairs $(\mathcal{L}, \psi)$ and $(\mathcal{L}', \psi')$ are said to be isomorphic if there exists an isomorphism of invertible sheaves $f : \mathcal{L} \to \mathcal{L}'$ such that the following diagram commutes:

$$\begin{tikzcd} & \mathcal{O}_{\mathfrak{m}_T} \arrow[dl, "\psi'"'] \arrow[dr, "\psi"] & \\ \mathcal{L}'|_{\mathfrak{m}_T} \arrow[rr, "f|_{\mathfrak{m}_T}"] & & \mathcal{L}|_{\mathfrak{m}_T} \end{tikzcd}$$

We define the presheaf $\mathrm{Pic}^{d,\mathrm{pre}}_{C,\mathfrak{m}}$ on Sch/S by assigning to T the set of isomorphism classes of such pairs. Let $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ denote the étale sheafification of this presheaf.

Representability and Structure

The fundamental properties of this functor are as follows:

1. $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ is represented by an S-scheme. (Note: If $\mathfrak{m}$ is faithfully flat over S, the presheaf is already a étale sheaf).
2. $\mathrm{Pic}^{0}_{C,\mathfrak{m}}$ is a smooth commutative group S-scheme with geometrically connected fibers, referred to as the *generalized Jacobian variety* of C with modulus $\mathfrak{m}$.
3. For any d, $\mathrm{Pic}^{d}_{C,\mathfrak{m}}$ is a $\mathrm{Pic}^{0}_{C,\mathfrak{m}}$-torsor.

In the case where $\mathfrak{m} = 0$, we recover the standard Jacobian variety, denoted simply as Pic^{d}_{C}.

Relation to the Standard Jacobian

We now examine the behavior of the generalized Picard scheme under the variation of the modulus. By viewing the structure along the modulus as an additional rigidification, we obtain natural transition maps corresponding to the inclusion of moduli.

Let $\mathfrak{m}_1$ and $\mathfrak{m}_2$ be moduli such that $\mathfrak{m}_1 \subset \mathfrak{m}_2$. There exists a natural map

$$\mathrm{Pic}^d_{C,\mathfrak{m}_2} \to \mathrm{Pic}^d_{C,\mathfrak{m}_1}$$

obtained by restricting the isomorphism ψ. Since $\mathfrak{m}_2$ is a finite S-scheme, this map is a surjection as a morphism of étale sheaves. In particular, for any modulus $\mathfrak{m}$, there is a natural surjective morphism of étale sheaves:

$$\mathrm{Pic}^d_{C,\mathfrak{m}} \to \mathrm{Pic}^d_C.$$

Local Freeness and Base Change

Let $\mathfrak{m}$ be a modulus which is everywhere strictly positive. Let g denote the genus of C, which is a locally constant function on S. We restrict our attention to degrees d satisfying the condition:

$$d \geq \max\{2g - 1 + \deg \mathfrak{m}, \deg \mathfrak{m}\}. \tag{2.1}$$

Assuming S is quasi-compact, such a d always exists.

Fix an integer d satisfying the condition above. Let T be an S-scheme and let $\mathcal{L}$ be an invertible sheaf of degree d on C_T. One can show that the pushforwards $\mathrm{pr}_*\mathcal{L}$ and $\mathrm{pr}_*\mathcal{L}(-\mathfrak{m})$ are locally free sheaves and their formations commute with any base change. Explicitly, for any morphism of S-schemes $f : T' \to T$, the base change morphisms are isomorphisms:

$$f^*\mathrm{pr}_*\mathcal{L} \xrightarrow{\sim} \mathrm{pr}_* f^*\mathcal{L}$$

and

$$f^*\mathrm{pr}_*(\mathcal{L}(-\mathfrak{m})) \xrightarrow{\sim} \mathrm{pr}_* f^*(\mathcal{L}(-\mathfrak{m})).$$

In particular, following [Gui19], if $\mathcal{L}$ is invertible $\mathcal{O}_C$-module with degree d satisfying 2.1 on each fiber of f then, $\mathrm{pr}_*\mathcal{L}$ is a locally free $\mathcal{O}_S$-module of rank $d - g + 1$.

For further background and verification of these constructions, we refer the reader to Milne's notes on abelian Varieties ([Mil08]).

2.3 The Abel-Jacobi Morphism and its Fibers

Let $U = C\backslash\mathfrak{m}$ be the complement of the modulus in C. The effective cartier divisors of degree d which are prime to $\mathfrak{m}$ are parameterized by the symmetric power $\mathrm{Sym}^d_S(U) = U^{(d)}$ over S (See [Gui19] Proposition 4.12, [Mil08] Theorem 3.13). For any such divisor $D \in U^{(d)}$, the associated line bundle $\mathcal{O}_C(D)$ admits a canonical trivialization along $\mathfrak{m}$. Specifically, the canonical section 1_D is regular and non-vanishing on $\mathfrak{m}$ because $\mathrm{supp}(D) \cap \mathrm{supp}(\mathfrak{m}) = \emptyset$. This section restricts to a nowhere-vanishing section on the subscheme $\mathfrak{m}$, thereby determining a trivialization $\psi_D^{-1} : \mathcal{O}_C(D)|_{\mathfrak{m}} \xrightarrow{\sim} \mathcal{O}_{\mathfrak{m}}$. This is done functorially in families, yielding a morphism from the symmetric power to the generalized Picard scheme (over S):

$$\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}, \quad D \mapsto [(\mathcal{O}_C(D), \psi_D)], \tag{2.2}$$

When $\mathfrak{m} = 0$, $d \geq max\{2g-1, 0\}$ and C admits a section over S, $C^{(d)}$ is a projective space bundle over Pic^d_C, It is proper, surjective with geometrically connected fibers.

Guignard ([Gui19] Theorem 4.14) proves that for $\mathfrak{m} > 0$ and d satisfying (2.1), the Abel-Jacobi morphism Φ_d is surjective smooth of relative dimension $d - \deg \mathfrak{m} - g + 1$, with geometrically connected fibers.

When $S = \operatorname{Spec}(k)$, the geometric-fibers of Φ_d are well understood:

Theorem 78. *Assuming $S = \operatorname{Spec}(k)$ and $d \geq max\{2g - 1 + \deg \mathfrak{m}, \deg \mathfrak{m}\}$. Then, the geometric-fibers of the Abel-Jacobi morphism*

$$\Phi_d : U^{(d)} \to \operatorname{Pic}^d_{C,\mathfrak{m}}$$

over any point are isomorphic to

$$\begin{cases} \mathbb{A}^{d-\deg \mathfrak{m}-g+1}_{k^{sep}} & \text{if } m > 0 \\ \mathbb{P}^{d-g}_{k^{sep}} & \text{if } m = 0 \end{cases}$$

In both cases Φ_d is a fibration in affine spaces or projective spaces, depending on whether $\mathfrak{m}$ is non-zero or zero.

Proof. see [Ten15] Propositions 3.13-3.14, or [Tót11] Prop 2.1.4:

□

2.4 Compactification of Blowup of Symmetric Powers of a Curve

We recall that our objective is to descend the local system $\mathcal{F}^{[d]}$ from $U^{(d)}$ to $\operatorname{Pic}^d_{C,\mathfrak{m}}$ along the Abel-Jacobi map Φ_d:

$$\begin{tikzcd} \mathcal{F}^{[d]} \arrow[d] & \\ U^{(d)} \arrow[r, "\Phi_d"] & \operatorname{Pic}^d_{C,\mathfrak{m}} \end{tikzcd}$$

(Here, the purple arrow emphasizes that the morphism is of sheaves on the étale site).

However, we encounter an obstruction: in the case we are considering ($\mathfrak{m} > 0$), the fibers of Φ_d are affine spaces (of the same degree) rather than the better-behaved projective spaces. This hint that a solution to this problem is to compactify the morphism to yield projective fibers.

This section describes the result of the compactification constructed by [Tak19] via the method of blowup.

Let $\mathfrak{m} = \sum_{i=1}^{n} k_P P$ with $\deg P = d_P$ be a modulus on C, and let d satisfy (2.1). Takeuchi ([Tak19]) defines $Z_0 = Z_0(\mathfrak{m}, d)$ as the closed subscheme of $C^{(d)}$ defined by the map $C^{(d-\deg \mathfrak{m})} \to C^{(d)}$ adding $\mathfrak{m}$. He also defines $X_{\mathfrak{m},d}$ as the blowup of $C^{(d)}$ along Z_0. Let $E_0 = E_{\mathfrak{m},d} = Z_0(\mathfrak{m}, d) \times_{C^{(d)}} X_{\mathfrak{m},d}$ be the exceptional divisor of the blowup. It is irreducible of codimension 1, and we let $\eta_0 = \eta_{\mathfrak{m},d}$ be its generic point.

Diagrammatically:

$$\begin{tikzcd} \overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\ Z_0 \arrow[r, "c.i"] & C^{(d)} \end{tikzcd}$$

Incoporating $U^{(d)}$, the local system $\mathcal{F}^{[d]}$ and the Abel-Jacobi map, we have:

$$\begin{tikzcd}
 & & \mathcal{F}^{[d]} \arrow[d] & \\
\overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] & U^{(d)} \arrow[l, hook'] \arrow[r, "\Phi_d"] \arrow[dl, hook] & \mathrm{Pic}^d_{C,\mathfrak{m}} \\
Z_0 \arrow[r, "c.i"] & C^{(d)} & &
\end{tikzcd}$$

In Section 3 of [Tak19] Takeuchi constructs, for d satisfying (2.1) a compactification denoted by $\tilde{C}_{\mathfrak{m}}^{(d)}$ and proves the following:

Theorem 79 (Takeuchi)**.** *The scheme $\tilde{C}_{\mathfrak{m}}^{(d)}$ is an open subscheme of $X_{\mathfrak{m},d}$ containing $U^{(d)}$. The morphism $\Phi_d : U^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ extends to a morphism $\tilde{\Phi}_d : \tilde{C}_{\mathfrak{m}}^{(d)} \to \mathrm{Pic}^d_{C,\mathfrak{m}}$ which makes $\tilde{C}_{\mathfrak{m}}^{(d)}$ a projective space bundle over $\mathrm{Pic}^d_{C,\mathfrak{m}}$. Furthermore, the complement of $U^{(d)}$ in $\tilde{C}_{\mathfrak{m}}^{(d)}$ is isomorphic to the fiber product $E_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)}$.*

Diagrammatically we have:

$$\begin{tikzcd}
 & & \mathcal{F}^{[d]} \arrow[d] \\
E_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)} \arrow[r] \arrow[d] & \tilde{C}_{\mathfrak{m}}^{(d)} \arrow[d, hook] \arrow[dr, "\tilde{\Phi}_d"] & U^{(d)} \arrow[l, hook'] \arrow[d, "\Phi_d"] \\
\overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] & \mathrm{Pic}^d_{C,\mathfrak{m}} \\
Z_0 \arrow[r, "c.i"] & C^{(d)} &
\end{tikzcd}$$

Also note that $E_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)} = Z_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)}$

2.5 Proof of Theorem 75

In what follows we reduce Theorem 75 to Theorem 52

2.5.1 Reduction Lemmas

The following lemma is adapted from [Tak19] (Lemma 4.1)

Lemma 80. *Let C be a projective, smooth, and geometrically connected curve over a perfect field k. Let $\mathfrak{m} = \sum_{i=1}^{r} k_i P_i$ be an effective divisor where $P_1, \dots, P_r$ are distinct closed points. Let $U = C \setminus \mathfrak{m}$ and let $d \geq \deg \mathfrak{m}$.*

Suppose $\mathfrak{n}_1, \dots, \mathfrak{n}_l$ are pairwise coprime submoduli of $\mathfrak{m}$ such that $\mathfrak{m} = \sum_{j=1}^{l} \mathfrak{n}_j$. Consider the summation morphism:

$$\pi : C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d - \deg \mathfrak{m})} \longrightarrow C^{(d)}$$

defined by $(D_1, \dots, D_l, D_{extra}) \mapsto \sum_{j=1}^{l} D_j + D_{extra}$.

Then π *is étale at the generic point of the closed subvariety*

$$V = \{\mathfrak{n}_1\} \times_k \cdots \times_k \{\mathfrak{n}_l\} \times_k C^{(d-\deg \mathfrak{m})}$$

inside the domain $C^{(\deg \mathfrak{n}_1)} \times_k \cdots \times_k C^{(\deg \mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$.

Proof. We may assume that k is algebraically closed (hence $\deg P_i = 1$ for all i). By miracle flatness π is flat, it is quasi-finite and projective as a map between projecive spaces. so we condlude π is finite and flat. It is enough to show that there exists a closed point Q of $\mathfrak{n}_1 + \dots \mathfrak{n}_l + C^{(d-\deg \mathfrak{m})} \subset C^{(d)}$ over which there are $\deg \pi$ points on $C^{(\mathfrak{n}_1)} \times_k \cdots \times_k C^{(\mathfrak{n}_l)} \times_k C^{(d-\deg \mathfrak{m})}$. (Because it will be unramified at this point and thus also at the generic point of V.) Choose Q as a point corresponding to a divisor $\mathfrak{n}_1 + \cdots + \mathfrak{n}_l + P_{r+1} + \cdots + P_{r+d-\deg \mathfrak{m}}$, where $P_1, \dots, P_{r+d-\deg \mathfrak{m}}$ are distinct points of $U(k)$. ☐

Corollary 81. *The morphism* $C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} \xrightarrow{\pi} C^{(d)}$ *is finite flat everywhere, and étale at the generic point of the closed subvariety* $Z_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$.

Following from this, we look at the following diagram, coming from the flat base change $C^{(d-\deg \mathfrak{m})} \to \mathrm{Spec}(k)$ (Proposition 20) of (1.8):

$$\begin{tikzcd}
E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \arrow[r] \arrow[d] & X_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \arrow[d] & \mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})} \arrow[d] \\
Z_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \arrow[r] & C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} & U^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} \arrow[ul]
\end{tikzcd}$$

Note that $U^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$ is dense open subscheme of $X_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$, And $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$ is a prime divisor of $X_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$. Hence it is well defined question according to Definition 53 to ask whether $\mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$.

Lemma 82. *If* $\mathcal{F}^{[\deg \mathfrak{m}]}$ *is tamely ramified at* $\eta_{\mathfrak{m}}$, *then* $\mathcal{F}^{[\deg \mathfrak{m}]} \times_k C^{(d-\deg \mathfrak{m})}$ *is tamely ramified at the generic point* θ *of* $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})}$.

Proof. This follows from [Stacks, Tag 0EYD] ☐

Replacing $C^{(d-\deg \mathfrak{m})}$ with the dense open subscheme $U^{(d-\deg \mathfrak{m})} \subset C^{(d-\deg \mathfrak{m})}$, we get that the G-torsor $p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]}$ ($\mathcal{P}$ correspodns to $\mathcal{F}$ under Proposition 9) is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \subset U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$, where $p_1 : U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to U^{(\deg \mathfrak{m})}$ is the projection to the first factor.

Looking at the second projection $p_2 : X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \to U^{(d-\deg \mathfrak{m})}$, and the fact that $\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is étale on $U^{(d-\deg \mathfrak{m})}$ we get that $p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]} = X_{\mathfrak{m}} \times_k \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is étale on $X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$. Hence, its restriction to $U^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ is unramified at θ.

Thus, by the following lemma, we conclude that $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]} = p_1^{-1}\mathcal{P}^{[\deg \mathfrak{m}]} \otimes^G p_2^{-1}\mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at θ the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$.

Lemma 83. *Let* $X \to \mathrm{Spec}\, k$ *be a scheme over a field* k, *and let* $\mathcal{P}_1, \mathcal{P}_2$ *be two* G*-torsors on* U_{et}. *Let* ξ *be the generic point of a prime divisor* $D \subset X$. *If* $\mathcal{P}_1$ *is tamely ramified at* ξ, *and* $\mathcal{P}_2$ *is unramified at* ξ, *then the product* $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ *is tamely ramified at* ξ.

Proof. This follows from Lemma 57. □

Combining this with Corollary 81 we get

Corollary 84. *Let $\mathfrak{m}$ be a modulus as above, and let $\eta_{\mathfrak{m}}$ be the generic point of $E_{\mathfrak{m}}$. Let $\mathcal{P}$ be a G-torsor on U_{et} with ramification bounded by $\mathfrak{m}$. Assume $\mathcal{P}^{[\deg \mathfrak{m}]}$ is tamely ramified at $\eta_{\mathfrak{m}}$. Then $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at the generic point θ of $E_{\mathfrak{m}} \times_k C^{(d-\deg \mathfrak{m})} \subset C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})}$, and $\mathcal{P}^{[d]}$ is tamely ramified at the generic point $\eta_0 = \eta_{\mathfrak{m},d}$ of $E_0 = E_{\mathfrak{m},d}$*

Proof. The first assertion, that $\mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ is tamely ramified at θ, follows from the preceding discussion. Thus, it remains to show that $\mathcal{P}^{[d]}$ is tamely ramified at η_0.

Consider the blowup diagram defining $X_{\mathfrak{m},d}$:

$$\begin{tikzcd}
\overline{\{\eta_0\}} = E_0 \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
Z_0 \arrow[r, "c.i"] & C^{(d)}
\end{tikzcd}$$

By performing a base change along the flat addition map $+ : C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to C^{(d)}$, we obtain the following commutative diagram:

$$\begin{tikzcd}
& \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} E_0 \arrow[r] \arrow[d] & \overline{\{\eta_0\}} = E_0 \arrow[d] \\
& \left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d} \arrow[r] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "c.i"] & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "+"] & C^{(d)}
\end{tikzcd}$$

Since blowups commute with flat base change, and the inverse image of the center Z_0 under the map $+$ is $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$, the scheme $\left(C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}\right) \times_{C^{(d)}} X_{\mathfrak{m},d}$ is the blowup of $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})}$ along $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. This, in turn, is isomorphic to the base change of the blowup $X_{\mathfrak{m}}$ (of $C^{(\deg \mathfrak{m})}$ along $\mathfrak{m}$) via the (flat) projection $C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \to C^{(\deg \mathfrak{m})}$.

Assembling these facts, we obtain the following Cartesian square:

$$\begin{tikzcd}
& E_{\mathfrak{m}} \times U^{(d-\deg \mathfrak{m})} \arrow[r, "\tilde{+}"] \arrow[d] & \overline{\{\eta_0\}} = E_0 \arrow[d] \\
& X_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "\tilde{+}"] \arrow[d] & X_{\mathfrak{m},d} \arrow[d, "\pi"] \\
\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "c.i"] & C^{(\deg \mathfrak{m})} \times_k U^{(d-\deg \mathfrak{m})} \arrow[r, "+"] & C^{(d)}
\end{tikzcd}$$

The point θ defined in the Corollary is the generic point of $E_{\mathfrak{m}} \times_k U^{(d-\deg \mathfrak{m})}$. Let η be the generic point of $\mathfrak{m} \times_k U^{(d-\deg \mathfrak{m})}$. By Corollary 81, the map $+$ is étale at η. Consequently, the lifted map $\tilde{+}$ is étale at θ. Given the isomorphism $(+^{-1})(\mathcal{P}^{[d]}) \cong \mathcal{P}^{[\deg \mathfrak{m}]} \boxtimes \mathcal{P}^{[d-\deg \mathfrak{m}]}$ from Proposition 25, the tame ramification of the box product at θ descends to the tame ramification of $\mathcal{P}^{[d]}$ at η_0 by applying Lemma 58.

$\square$

Lemma 85. *Let $\mathfrak{n}_1, \mathfrak{n}_2 \subset \mathfrak{m}$ be two coprime sub moduli of $\mathfrak{m}$. Assume $\mathcal{P}^{[\deg \mathfrak{n}_1]}$, $\mathcal{P}^{[\deg \mathfrak{n}_2]}$ are at most tamely ramified at $\eta_{\mathfrak{n}_1}$, $\eta_{\mathfrak{n}_2}$ respectively. Then $\mathcal{P}^{[\deg \mathfrak{n}_1 + \deg \mathfrak{n}_2]}$ is at most tamely tamified at $\eta_{\mathfrak{n}_1+\mathfrak{n}_2}$.*

Proof. It follows from Proposition 74 and Proposition 25 $\square$

2.6 Proof of Theorem 75

Proof of Theorem 75. Let $\mathcal{F}$ be as in Theorem 75, $\mathfrak{m} = \sum_{i=1}^{n} k_P P$ with $\deg P = d_P$ Then by Theorem 3 for every $\mathfrak{n} \subset \mathfrak{m}$ of the form $\mathfrak{n} = k_P P$, $\mathcal{F}^{[\deg \mathfrak{n}]}$ is at most tamely ramified at $\eta_{\mathfrak{n}}$. By Lemma 85, $\mathcal{F}^{[\deg \mathfrak{m}]}$ is then at most tamely ramified at $\eta_{\mathfrak{m}}$. And thus by Corollary 84 $\mathcal{F}^{[d]}$ is tamely ramified at the generic point η_0 of E_0 $\square$

2.7 Proof of Theorem 2

We work over $S = \operatorname{Spec} k$, for k perfect.

By Proposition 9, its equivalent to prove:

Theorem 86. *Let $G = \Lambda^{\times}$ be a finite abelian group (Λ as before), and let $\mathcal{P}$ be a G-torsor on U, with ramification bounded by $\mathfrak{m}$. Then, for sufficiently large integer d, there exists a unique (up to isomorphism) G-torsor $\mathcal{Q}_d$ on $\operatorname{Pic}^d_{C,\mathfrak{m}}$, such that the pullback of $\mathcal{Q}_d$ by Φ_d is isomorphic to $\mathcal{P}^{[d]}$.*

Proof. We divide the proof into two cases, when $\mathfrak{m} = 0$ and when $\mathfrak{m} > 0$.

Case 1: $\mathfrak{m} = 0$. By Section 2.3, for d large enough, the Abel-Jacobi morphism $\Phi_d : C^{(d)} \to \operatorname{Pic}^d_C$ is proper surjective and smooth, with geometrically connected fibers, each isomorphic to $\mathbb{P}^{d-g}_{k^{sep}}$, Hence by Corollary 77 it induces an exact sequence of etale fundamental groups:

$$\pi_1^{et}(\mathbb{P}^{d-g}_{k^{sep}}) \to \pi_1^{et}(C^{(d)}) \to \pi_1^{et}(\operatorname{Pic}^d_C) \to 1$$

But $\mathbb{P}^{d-g}_{k^{sep}}$ is simply connected ([Ten15] Example 4.9, [Tót11] Example 1.4.12), hence its etale fundamental group is trivial, and we get an isomorphism of etale fundamental groups:

$$\pi_1^{et}(C^{(d)}) \cong \pi_1^{et}(\operatorname{Pic}^d_C)$$

Implying the theorem in this case.

Case 2: $\mathfrak{m} > 0$. In this case by Theorem 79, for d large enough, the Abel-Jacobi morphism $\Phi_d : U^{(d)} \to \operatorname{Pic}^d_{C,\mathfrak{m}}$ extends to a proper surjective and smooth map, with geometrically connected fibers isomorphic to projective spaces,

$$\tilde{\Phi}_d : \tilde{C}^{(d)}_{\mathfrak{m}} \to \operatorname{Pic}^d_{C,\mathfrak{m}}$$

Hence we get an isomorphism of etale fundamental groups:

$$\pi_1^{et}(\tilde{C}^{(d)}_{\mathfrak{m}}) \cong \pi_1^{et}(\operatorname{Pic}^d_{C,\mathfrak{m}})$$

By Theorem 75, $\mathcal{F}^{[d]}$ is tamely ramified on the boundary divisor $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$.

Thus, by Lemma 87 below, we have: $\mathcal{F}^{[d]}$ extends to a locally constant sheaf $\tilde{\mathcal{F}}^{[d]}$ on $\tilde{C}_{\mathfrak{m}}^{(d)}$, which by the isomorphism of etale fundamental groups above, corresponds to a unique locally constant sheaf $\mathcal{G}_d$ on $\mathrm{Pic}_{C,\mathfrak{m}}^d$, such that $\tilde{\Phi}_d^* \mathcal{G}_d \cong \tilde{\mathcal{F}}^{[d]}$. Restricting back to $U^{(d)}$, we get $\Phi_d^* \mathcal{G}_d \cong \mathcal{F}^{[d]}$, as required. $\square$

Lemma 87. *If $\mathcal{F}^{[d]}$ is a locally constant sheaf on $U^{(d)}$ which is tamely ramified along the boundary divisor $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$, then $\mathcal{F}^{[d]}$ extends to a locally constant sheaf $\tilde{\mathcal{F}}^{[d]}$ on $\tilde{C}_{\mathfrak{m}}^{(d)}$.*

Proof. The lemma we are referencing above can be proved in two routes:
Route 1 - Showing $\ker\{\pi_1^{ab}(U^{(d)}) \to \pi_1^{ab}(\mathrm{Pic}_{C,\mathfrak{m}}^d)\}$ is pro-p group.
Route 2 - Showing

$$\pi_1^{t,ab}(U^{(d)}) \to \pi_1^{t,ab}(\mathrm{Pic}_{C,\mathfrak{m}}^d)$$

is isomorphism to its image. here one needs to be precise.

$\square$

Appendix A

A Wrong Path

In this appendix we want to explain why Theorem 66 is diffcult to prove by induction on r, using the previously proven case of $G = \mathbb{Z}/p\mathbb{Z}$. (Corollary 65).

A.1 Setup and notation

Let $\mathcal{P} \to U$ be a G-torsor over $U = C \setminus \mathfrak{m}$, where G is a finite abelian group, $H \trianglelefteq G$, and C is a smooth projective curve. Fix $p_\infty \in \mathfrak{m}$ and an integer d. Assume the G torsor $\mathcal{P} \to U$ has ramification strictly bounded by d at p_∞. (i.e. $< d$) according to the definition in Definition 54 (i.e. the higher d'th-ramification group of the extension of DVRs at p_∞ is trivial).

Recall the four quotient schemes, defined in: Section 1.1.4

$$\mathcal{P}^{[d]_H}, \qquad \mathcal{P}^{[d]_G}, \qquad (\mathcal{P}_{G/H})^{(d)}, \qquad (\mathcal{P}_{G/H})^{[d]},$$

together with the canonical Cartesian square

$$\begin{tikzcd}
\mathcal{P}^{[d]_H} \arrow[r, "H"] \arrow[d] & (\mathcal{P}_{G/H})^{(d)} \arrow[d, "q"] \\
\mathcal{P}^{[d]_G} \arrow[r, "H"'] \arrow[dr, bend right, "G"'] & (\mathcal{P}_{G/H})^{[d]} \arrow[d, "G/H"] \\
& U^{(d)},
\end{tikzcd}$$

in which the arrows labelled H, G/H and G are torsors under the indicated finite constant groups (in particular, *étale*), and the upper square is Cartesian (Proposition 24).

The G/H-torsor $\mathcal{P}_{G/H} \to U$ compactifies to a finite map of smooth projective curves $f\colon C' \to C$. Pick $p'_\infty \in f^{-1}(p_\infty)$

$$f^{(d)}\colon C'^{(d)} \to C^{(d)}$$

for the induced map of d-fold symmetric powers; it sends $d \cdot p'_\infty \mapsto d \cdot p_\infty$. Let

$$\pi_C \colon X_C \to C^{(d)}, \qquad \pi_{C'} \colon X_{C'} \to C'^{(d)}$$

be the blowups at $d \cdot p_\infty$ and $d \cdot p'_\infty$ respectively, with exceptional divisors $E_C, E_{C'}$ and generic points $\eta_C, \eta_{C'}$. The DVRs of these generic points are

$$V_C := \mathcal{O}_{X_C, \eta_C} \subset K(C^{(d)}), \qquad V_{C'} := \mathcal{O}_{X_{C'}, \eta_{C'}} \subset K(C'^{(d)}).$$

A.2 The master diagram

The torsor tower over the open base $U^{(d)}$ embeds into the projective geometry of $C^{(d)}$ and $C'^{(d)}$, which in turn admit canonical blowups at the points at infinity. We assemble all of this into a single commutative diagram.

$$\begin{tikzcd}
\mathcal{P}^{[d]_H} \arrow[r, "H"] \arrow[d] & (\mathcal{P}_{G/H})^{(d)} \arrow[r, hook, "\iota'"] \arrow[d, "q"] & C'^{(d)} \arrow[dd, "f^{(d)}"] & X_{C'} \arrow[l, "\pi_{C'}"'] \arrow[dd, dashed, "f_X"] \\
\mathcal{P}^{[d]_G} \arrow[r, "H"'] \arrow[dr, bend right, "G"'] & (\mathcal{P}_{G/H})^{[d]} \arrow[d, "G/H"] & & \\
& U^{(d)} \arrow[r, hook, "\iota"'] & C^{(d)} & X_C \arrow[l, "\pi_C"]
\end{tikzcd}$$

Reading the diagram:

- The **left block** (columns 1–2) is the torsor tower of Proposition 24: H-torsors horizontally, the G/H-torsor on the lower right, and the composite G-torsor $\mathcal{P}^{[d]_G} \to U^{(d)}$.
- The **middle block** (column 3) records the compactifications. The map ι is the canonical open immersion $U^{(d)} \hookrightarrow C^{(d)}$. The map ι' is the open immersion $(\mathcal{P}_{G/H})^{(d)} \hookrightarrow C'^{(d)}$ obtained from compactifying $\mathcal{P}_{G/H} \subset C'$. The vertical $f^{(d)}$ is the finite morphism of symmetric powers; restricting along ι' recovers the canonical map $(\mathcal{P}_{G/H})^{(d)} \to U^{(d)}$, which is the composite of q with the G/H-torsor.
- The **right block** (column 4) records the blowups at the diagonal points at infinity. The dashed arrow $f_X \colon X_{C'} \dashrightarrow X_C$ is the rational lift of $f^{(d)}$, defined on the strict transform; on generic points of the exceptional divisors it sends $\eta_{C'} \mapsto \eta_C$ so we have induced extension of DVRs $V_C \hookrightarrow V_{C'}$

A.3 Unramifiedness of $\mathcal{P}^{[d]_G}$ at η_C

So, going back to the begining of this appendix, a natural way to prove Theorem 66 from Corollary 65 by induction on r, comes down to proving the following theorem.

Theorem 88. *Suppose*

(H1) $(\mathcal{P}_{G/H})^{[d]} \to U^{(d)}$ *is unramified at* η_C*; and*

(H2) $\mathcal{P}^{[d]_H} \to (\mathcal{P}_{G/H})^{(d)}$ *is unramified at* $\eta_{C'}$.

(H3) $G = \mathbb{Z}/p^r\mathbb{Z}$ *is cyclic of prime power order.*

Then $\mathcal{P}^{[d]_G} \to U^{(d)}$ *is unramified at* η_C.

And the next section is devoted to explaining why this is not an easy task, and in fact, we were not able to prove it.

A.4 Difficulties in Proving the Theorem

First, note that we can assume $\mathcal{P} \to U$ is connected, and that H is non-trivial.

(R1) $\mathcal{P} \to U$ *is connected.* Let $\rho\colon \pi_1(U) \to G$ be the monodromy of $\mathcal{P}$ and put $G_0 := \operatorname{Im}(\rho)$, $H_0 := H \cap G_0$. Pick a connected component $\mathcal{P}_0 \subseteq \mathcal{P}$; it is a connected G_0-torsor, and $\mathcal{P}$ is the disjoint union of $[G : G_0]$ copies of $\mathcal{P}_0$ permuted by G. Since the symmetric-power quotients of Section 1.1.4 commute with such disjoint unions, each of the four schemes in Proposition 24 for $(\mathcal{P}, G, H)$ decomposes as a disjoint union of copies of the corresponding scheme for $(\mathcal{P}_0, G_0, H_0)$. Unramifiedness at η_C (resp. $\eta_{C'}$) is detected componentwise, so (H1) and (H2) are inherited by $(\mathcal{P}_0, G_0, H_0)$; (H3) is inherited because subgroups of cyclic p-groups are cyclic p-groups. Replacing $(\mathcal{P}, G, H)$ by $(\mathcal{P}_0, G_0, H_0)$ we may assume $\mathcal{P}$ is connected and ρ is surjective.

(R2) $H \neq 0$. If $H = 0$ then $\mathcal{P}_{G/H} = \mathcal{P}$ and $(\mathcal{P}_{G/H})^{[d]} = \mathcal{P}^{[d]_G}$, so (H1) is literally the conclusion.

We can read a decomposition of fields directly from the canonical Cartesian square of Proposition 24. After the reduction (R1) all four corners are integral: the three corners $\mathcal{P}^{[d]_G}$, $(\mathcal{P}_{G/H})^{[d]}$ and $(\mathcal{P}_{G/H})^{(d)}$ are integral, and the fourth, $\mathcal{P}^{[d]_H} = (K_H \rtimes S_d)\backslash \mathcal{P}^{\times_S d}$, is integral as a quotient of the integral scheme $\mathcal{P}^{\times_S d}$ (Theorem 47). So we may take function fields; using the open immersions ι, ι' of the master diagram together with $K_C = \operatorname{Frac} V_C = K(C^{(d)}) = K(U^{(d)})$ and $K_{C'} = \operatorname{Frac} V_{C'} = K(C'^{(d)}) = K\big((\mathcal{P}_{G/H})^{(d)}\big)$, we identify

$$K(U^{(d)}) = K_C, \qquad K\big((\mathcal{P}_{G/H})^{[d]}\big) = F, \qquad K\big((\mathcal{P}_{G/H})^{(d)}\big) = K_{C'}, \qquad K\big(\mathcal{P}^{[d]_G}\big) = E,$$

with E/F Galois of group H, F/K_C Galois of group G/H, and the composite E/K_C Galois of group G. The fourth corner $\mathcal{P}^{[d]_H}$ is the fibre product $\mathcal{P}^{[d]_G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)}$; being integral, its single function field is the compositum $E \cdot K_{C'}$. In particular its generic fibre $E \otimes_F K_{C'}$ over $(\mathcal{P}_{G/H})^{(d)}$ is a field, so by **??** the extensions E and $K_{C'}$ are linearly disjoint over F; that is,

$$E \cap K_{C'} = F,$$

and $E \cdot K_{C'}/K_{C'}$ is Galois with group H. We denote:

1. $K_C := \operatorname{Frac} V_C$, $K_{C'} := \operatorname{Frac} V_{C'}$;
2. $F := K\big((\mathcal{P}_{G/H})^{[d]}\big)$, $E := K(\mathcal{P}^{[d]_G})$;

3. $V_F := V_{C'}|_F$;
4. a place v_E of E above V_F, and a place $v_\star$ of $E \cdot K_{C'}$ above both v_E and $V_{C'}$.

Thus we have:

$$\begin{tikzcd}
& E \cdot K_{C'} \arrow[dl, dash] \arrow[dr, dash, "H"] & \\
E \arrow[dr, dash, "H"'] & & K_{C'} \arrow[dl, dash] \\
& F \arrow[d, dash, "G/H"] & \\
& K_C &
\end{tikzcd}
\qquad
\begin{tikzcd}
& v_\star \arrow[dl, dash] \arrow[dr, dash] & \\
v_E \arrow[dr, dash] & & V_{C'} \arrow[dl, dash] \\
& V_F \arrow[d, dash] & \\
& V_C &
\end{tikzcd}$$

(lines denote field inclusion / place lying-over, read upwards; in the left diagram a label is the Galois group of the extension where it is Galois). In addition E/K_C is Galois with group G.

Read through this dictionary, the hypotheses of the theorem become statements about the ramification of the places above:

(H1) $(\mathcal{P}_{G/H})^{[d]} \to U^{(d)}$ is unramified at η_C means that the G/H-extension F/K_C is unramified at V_C. In particular V_F is unramified over V_C, with $e(V_F \mid V_C) = 1$.

(H2) $\mathcal{P}^{[d]_H} \to (\mathcal{P}_{G/H})^{(d)}$ is unramified at $\eta_{C'}$ means that the H-extension $E \cdot K_{C'}/K_{C'}$ is unramified at $V_{C'}$; equivalently the inertia group $I(v_\star \mid V_{C'}) \leq H$ is trivial.

(H3) $G = \mathbb{Z}/p^r\mathbb{Z}$ makes G, hence H and every inertia group, a cyclic p-group; as the residue characteristic is p, the ramification of E/F is potentially *wild*.

The conclusion we are after — that $\mathcal{P}^{[d]_G} \to U^{(d)}$ is unramified at η_C — is the statement that the G-extension E/K_C is unramified at V_C, i.e. that the inertia $I(v_E \mid V_C) \leq G$ is trivial. By (H1) the lower layer F/K_C is already unramified at V_C, so this inertia lies in $H = \mathrm{Gal}(E/F)$ and coincides with the inertia of the top layer,

$$I(v_E \mid V_C) \;=\; I(v_E \mid V_F) \;\leq\; H.$$

The theorem therefore reduces to the single assertion

$$E/F \text{ is unramified at } V_F, \qquad \text{i.e.} \qquad I(v_E \mid V_F) = 1.$$

Now $E \cdot K_{C'}/K_{C'}$ is the base change of E/F along $F \hookrightarrow K_{C'}$, and under the restriction isomorphism $\mathrm{Gal}(E \cdot K_{C'}/K_{C'}) \cong \mathrm{Gal}(E/F) = H$ the inertia injects

$$I(v_\star \mid V_{C'}) \;\hookrightarrow\; I(v_E \mid V_F).$$

Hypothesis (H2) says the source is trivial; the goal is that the target is trivial. These need *not* coincide: the two differ precisely by the ramification of $K_{C'}/F$ at $V_{C'} \mid V_F$ — the ramification that the symmetric-power base change q introduces at the point at infinity — which can be wild and can absorb the ramification of E/F. It is exactly this discrepancy that obstructs deducing the goal from (H1)–(H3).

Put in other words: The base change $K_{C'}/F$ is wildly ramified, so the inertia comparison is not onto .

The inertia injection, and why (H2) is not enough

The injection used above is an instance of the following general fact.

Lemma 89. *Let E/F be a finite Galois extension, linearly disjoint over F from a field extension K/F, with compositum $E \cdot K$; thus restriction is an isomorphism $r\colon \operatorname{Gal}(E \cdot K/K) \xrightarrow{\sim} \operatorname{Gal}(E/F)$, $\sigma \mapsto \sigma|_E$. Let w be a place of $E \cdot K$, and write $w_K := w|_K$, $w_E := w|_E$, $w_F := w|_F$ for its restrictions. Then r carries the inertia group of w over w_K into that of w_E over w_F:*

$$r\big(I(w \mid w_K)\big) \;\subseteq\; I(w_E \mid w_F),$$

so that $I(w \mid w_K) \hookrightarrow I(w_E \mid w_F)$ is injective. Moreover, completing at w identifies $I(w \mid w_K)$ with the inertia group of the local compositum $\widehat{E} \cdot \widehat{K}$ over $\widehat{K}$; in particular, if $\widehat{E} \subseteq \widehat{K}$ then $I(w \mid w_K) = 1$, no matter how ramified E/F is at w_F.

Proof. That r is an isomorphism is the translation theorem for the Galois extension E/F and the linearly disjoint K/F (**??**(1) with $M = F$). For the inclusion, let $\sigma \in I(w \mid w_K)$. Since E/F is Galois and σ fixes $F \subseteq K$ pointwise, $\sigma(E) = E$ and $\sigma|_E \in \operatorname{Gal}(E/F)$ fixes w_F. As σ lies in the decomposition group of w we have $w \circ \sigma = w$; restricting to E gives $w_E \circ \sigma|_E = w_E$, so $\sigma|_E \in D(w_E \mid w_F)$. Finally the residue extension $\kappa(w_E) \hookrightarrow \kappa(w)$ is equivariant and σ acts trivially on $\kappa(w)$, hence $\sigma|_E$ acts trivially on $\kappa(w_E)$, i.e. $\sigma|_E \in I(w_E \mid w_F)$. Injectivity is inherited from that of r. For the last assertion, the completion of $E \cdot K$ at w is the local compositum $\widehat{E}\widehat{K}$, with $\operatorname{Gal}(\widehat{E}\widehat{K}/\widehat{K}) = D(w \mid w_K)$ and inertia subgroup $I(w \mid w_K)$; if $\widehat{E} \subseteq \widehat{K}$ then $\widehat{E}\widehat{K} = \widehat{K}$ and this group is trivial. □

Taking $w = v_\star$ (so $w_K = V_{C'}$, $w_E = v_E$, $w_F = V_F$) recovers the injection $I(v_\star \mid V_{C'}) \hookrightarrow I(v_E \mid V_F)$. The implication "(H2) $\Rightarrow$ goal" would follow if this were always onto; the next example shows it need not be, already for $G = \mathbb{Z}/p^2\mathbb{Z}$ over $\mathbb{F}_p$.

Example 90. Take $k = \mathbb{F}_p$, $G = \mathbb{Z}/p^2\mathbb{Z}$, and $H = p\,\mathbb{Z}/p^2\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z}$ the unique subgroup of order p, so $G/H \cong \mathbb{Z}/p\mathbb{Z}$. We describe the local fields at the place $V_{C'} \mid V_F$ — the only place where the obstruction lives — writing $\widehat{(\cdot)}$ for completions.

The base and the G-tower. Let $\widehat{K_C} = \mathbb{F}_p((t))$, and present the local G-extension $\widehat{E}/\widehat{K_C}$ (an Artin–Schreier–Witt extension) by its two layers:

- lower layer $\widehat{F}/\widehat{K_C}$ (group G/H): *split*, so $\widehat{F} = \mathbb{F}_p((t))$. Then F/K_C is unramified at V_C — this is (H1) — and $\widehat{V_F} = \widehat{V_C}$;
- upper layer $\widehat{E}/\widehat{F}$ (group H): the Artin–Schreier extension

  $$\widehat{E} = \widehat{F}(y), \qquad y^p - y = t^{-1}.$$

  By Theorem 31(3) (with $m = 1$, prime to p) this is totally, wildly ramified of degree p, with $I(v_E \mid V_F) = H \cong \mathbb{Z}/p\mathbb{Z} \neq 1$.

Hence $I(v_E \mid V_C) = I(v_E \mid V_F) = H \neq 1$: the extension E/K_C is ramified at V_C, so *the conclusion of the theorem fails.*

The base change. The places of $K_{C'}$ over V_F are governed by the symmetric-power cover q. Suppose at $V_{C'}$ this cover reproduces the *same* Artin–Schreier extension,

$$\widehat{K_{C'}} = \widehat{F}(y), \quad y^p - y = t^{-1}, \qquad \text{i.e.} \qquad \widehat{K_{C'}} = \widehat{E}.$$

This is compatible with the global linear disjointness $E \cap K_{C'} = F$: two *distinct* degree-p extensions of the global field F may have isomorphic completions at a place. Now the local compositum is $\widehat{E \cdot K_{C'}} = \widehat{E} \cdot \widehat{K_{C'}} = \widehat{E} = \widehat{K_{C'}}$, so $E \cdot K_{C'}/K_{C'}$ is split, hence unramified, at $V_{C'}$:

$$I(v_\star \mid V_{C'}) = \mathrm{Gal}\big(\widehat{E}/\widehat{K_{C'}}\big) = 1.$$

Thus (H2) holds, and (H3) holds by construction.

Outcome. All of (H1)–(H3) hold, yet E/K_C is wildly ramified at V_C with inertia $H \cong \mathbb{Z}/p\mathbb{Z}$. The inertia injection of Lemma 89 is here

$$\underbrace{I(v_\star \mid V_{C'})}_{=1} \hookrightarrow \underbrace{I(v_E \mid V_F)}_{=\mathbb{Z}/p\mathbb{Z}},$$

the inclusion of the trivial group: (H2) carries no information about the ramification of E/F, which has been entirely *absorbed* by that of $K_{C'}/F$ — the wild ramification introduced by q at the point at infinity.

Remark. This is globally consistent with everything above. Since $\widehat{E} \otimes_{\widehat{F}} \widehat{K_{C'}} = \widehat{E} \otimes_{\widehat{F}} \widehat{E}$ is not a field, $\mathcal{P}^{[d]_H}$ (function field $E \cdot K_{C'}$) *splits* at $V_{C'}$ into several places and is therefore unramified at $\eta_{C'}$, as (H2) demands; meanwhile $\mathcal{P}^{[d]_G}$ stays ramified at η_C. The hypotheses simply do not see the local coincidence $\widehat{K_{C'}} = \widehat{E}$, and ruling out such coincidences for a given $(\mathcal{P}, d)$ is exactly the global input the inductive argument lacks — which is why this is a wrong path.

Absorption made explicit: two local fields

Example 90 posited the local coincidence $\widehat{K_{C'}} = \widehat{E}$ abstractly. For completeness we exhibit it in the smallest possible setting — two honest local fields and a one-line Artin–Schreier reduction — to make the absorption phenomenon fully visible.

Example 91 (absorption of a wild break, explicit fields)**.** Let $k = \bar{\mathbb{F}}_p$ and let $F = k((t))$. Define a second local field $K = k((\pi))$, and embed $F \hookrightarrow K$ by

$$t \;=\; \frac{\pi^p}{1-\pi^{p-1}} \;=\; \pi^p\big(1 + \pi^{p-1} + \pi^{2(p-1)} + \cdots\big), \qquad \text{so} \qquad v_\pi(t) = p.$$

Since $t = \pi^p \cdot (\text{unit})$ and $dt/d\pi = -\pi^{2p-2}(1-\pi^{p-1})^{-2} \neq 0$, the extension K/F is separable, *totally (wildly) ramified of degree* p; equivalently it is the Artin–Schreier extension $\sigma^p - \sigma = t^{-1}$ with $\sigma = \pi^{-1}$, of break 1 (Theorem 31(3)).

Now let E/F be the Artin–Schreier extension

$$E = F(y), \qquad y^p - y = t^{-1}.$$

Over F this is ramified: $v_t(t^{-1}) = -1$ is prime to p, so by Theorem 31(3) it is totally wildly ramified of degree p, break 1, with $I(v_E \mid v_F) = \mathbb{Z}/p\mathbb{Z} \neq 1$.

Yet E is absorbed by K. Computing t^{-1} inside K,

$$t^{-1} \;=\; \pi^{-p}\big(1-\pi^{p-1}\big) \;=\; \pi^{-p} - \pi^{-1} \;=\; \wp\big(\pi^{-1}\big), \qquad \wp(x) := x^p - x,$$

so $y = \pi^{-1} \in K$ already solves $y^p - y = t^{-1}$. Hence $E = F(\pi^{-1}) \subseteq K$ (indeed $E = K$, both being degree p over F), the compositum is $E \cdot K = K$, and

$$E \cdot K/K \text{ is trivial, hence unramified:} \qquad I(v_\star \mid v_K) = 1,$$

while E/F is ramified. The ramification of E/F has been *entirely absorbed* by the wild base change to K — exactly the failure of surjectivity in the inertia injection $I(v_\star \mid v_K) \hookrightarrow I(v_E \mid v_F)$, here $1 \hookrightarrow \mathbb{Z}/p\mathbb{Z}$.

Remark. The mechanism, and why F cannot do it. Read over K, the datum t^{-1} has pole order $v_\pi(t^{-1}) = p$, *divisible by p*; this is what licenses the Artin–Schreier reduction $y \mapsto y - \pi^{-1}$, which removes the pole completely because $\wp(\pi^{-1})$ matches t^{-1} to all orders. The substitution is available over K but *not* over F: the reducing element π^{-1} has valuation -1, whereas $v_\pi(F^\times) = p\,\mathbb{Z}$ contains no element of valuation -1 at all. Thus the wild base change does not "fight" the ramification of E/F; it *supplies the one element of intermediate valuation* that the Artin–Schreier reduction needs. The ramification index $e(K/F) = p$ is the gate that turns the prime-to-p break 1 into the divisible-by-p pole order p over K.

General r. For $G = \mathbb{Z}/p^r$ the faithful local model has $\widehat{K_{C'}}/\widehat{F}$ totally ramified of degree p^{r-1} (the residue field k being algebraically closed, there is no residual part), and absorption is the proper inclusion $\widehat{E} \subsetneq \widehat{K_{C'}}$ of the top-layer $\mathbb{Z}/p$-extension. The explicit fields are obtained by iterating the recipe above: with $E = k((\pi))$, $\pi = y^{-1}$ as here, set $\pi = \rho^p(1-\rho^{p-1})^{-1}$ to get a degree-p overfield $k((\rho)) \supsetneq E$, and so on $r-1$ times. The pole order of the top datum is multiplied to $p^{r-1} \cdot 1 \equiv 0 \pmod p$ and reduced away by the same one-line substitution. Over an algebraically closed residue field, "$E \cdot K/K$ unramified with E/F ramified" is *equivalent* to $\widehat{E} \subseteq \widehat{K_{C'}}$ — there are no nontrivial unramified extensions to hide in — so absorption is precisely a containment of local fields, never a softer phenomenon.

Bibliography

[Sch37] Hermann Ludwig Schmid. “Zur Arithmetik der zyklischen p-Körper”. In: *Journal für die reine und angewandte Mathematik* 176 (1937), pp. 161–167.

[SGA1] Alexander Grothendieck. *Revêtements étales et groupe fondamental (SGA 1)*. Vol. 224. Lecture notes in mathematics. Springer-Verlag, 1971.

[KS79] Kimiaki Kanesaka and Koji Sekiguchi. “Representation of Witt vectors by formal power series and its applications”. In: *Tokyo Journal of Mathematics* 2.2 (1979), pp. 349–370. DOI: 10.3836/tjm/1270473027.

[Ser79] Jean-Pierre Serre. *Local Fields*. Springer, 1979.

[Koc97] Helmut Koch. *Algebraic Number Theory*. Vol. 62. Encyclopaedia of Mathematical Sciences. Berlin, Heidelberg: Springer-Verlag, 1997. ISBN: 978-3-540-63003-6.

[Tho05] Lara Thomas. “Ramification groups in Artin-Schreier-Witt extensions”. In: *Journal de théorie des nombres de Bordeaux* 17.2 (2005), pp. 689–720. DOI: 10.5802/jtnb.514. URL: https://jtnb.centre-mersenne.org/articles/10.5802/jtnb.514/.

[Mil08] James S. Milne. *Abelian Varieties (v2.00)*. Available at www.jmilne.org/math/. 2008.

[Tót11] Péter Tóth. “Geometric Abelian Class Field Theory”. In: (May 2011). Master of Science thesis. URL: https://math.bu.edu/people/rmagner/Seminar/GCFTthesis.pdf.

[Rab14] Joseph Rabinoff. *The Theory of Witt Vectors*. 2014. arXiv: 1409.7445 [math.RA]. URL: https://arxiv.org/abs/1409.7445.

[Ten15] Avichai Tendler. *Geometric Class Field Theory*. 2015. arXiv: 1507.00104 [math.AG]. URL: https://arxiv.org/abs/1507.00104.

[Stacks] The Stacks Project Authors. *Stacks Project*. https://stacks.math.columbia.edu. 2018.

[Gui19] Quentin Guignard. “On the ramified class field theory of relative curves”. In: *Algebra & Number Theory* 13 (May 2019). Revised version submitted on 29 May 2019, pp. 1299–1326. DOI: 10.2140/ant.2019.13.1299. arXiv: 1804.02243 [math.AG].

[Tak19] Daichi Takeuchi. “Blow-ups and the class field theory for curves”. In: *Algebraic Number Theory* 13.6 (2019), pp. 1327–1351. DOI: 10.2140/ant.2019.13.1327. URL: https://doi.org/10.2140/ant.2019.13.1327.

[You20] Alex Youcis. “Notes on Torsors”. Available at https://alex-youcis.github.io/torsors.pdf. 2020. URL: https://alex-youcis.github.io/torsors.pdf.

[EK25] G. Griffith Elder and Kevin Keating. *Artin–Schreier–Witt extensions and ramification breaks*. 2025. arXiv: 2503.16830. URL: https://arxiv.org/abs/2503.16830.